

AIM Foundations

**Curvature-Driven Modelling, Classification, Dynamics, and Optimization of
Structured Railway Alignments**

Uwe Falz

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Preface

This manuscript is intended as a long-term scientific foundation for alignment-based information modelling. It is not merely software documentation, but a structured knowledge base covering geometry, transition theory, numerical reconstruction, engineering constraints, dynamics, standards, and optimization.

Executive Summary

Alignment-Based Information Modelling (AIM)

Railway alignments are traditionally described as geometric curves in space. This perspective is insufficient for modern engineering tasks, which require the integration of design, measurement, construction, and optimization.

Alignment is not a curve. It is a curvature-driven system whose geometry emerges from a structured process.

This work introduces *Alignment-Based Information Modelling (AIM)* as a unified framework for this perspective.

—

Core Idea

AIM represents alignments intrinsically by curvature:

$$\kappa(s).$$

This formulation:

- unifies geometry and dynamics,
- reduces model complexity,
- enables efficient computation and optimization.

—

From Representation to Process

AIM replaces static modelling by an operator pipeline:

$$\mathcal{P} = \mathcal{W}^{-1} \circ \mathcal{R} \circ \mathcal{M} \circ \mathcal{W} \circ \mathcal{C}.$$

This pipeline connects:

- coordinate systems,
- world geometry,

- intrinsic track coordinates,
- modelling,
- realization,
- and optimization.

Alignment modelling becomes a process.

Hybrid Modelling

AIM combines structure and flexibility:

$$\kappa(s) = \kappa_{\text{param}}(s; \mathbf{T}) + \delta\kappa(s).$$

This hybrid representation:

- preserves engineering intent,
 - integrates measurement data,
 - avoids overfitting.
-

Realization as a First-Class Concept

The designed alignment is not identical to the constructed track.

A realization operator

$$\mathcal{R}$$

maps the theoretical model to physical geometry, introducing smoothing and constraints.

Design must therefore be performed in realized space.

Optimization

Alignment design is formulated as a structured nonlinear least-squares problem.

Due to:

- locality in arc-length,
- sparse Jacobians,
- intrinsic block structure,

the problem is efficiently solvable using SQP methods.

World-to-Track Mapping

A central component is the transformation:

$$x \leftrightarrow (s, d).$$

This mapping:

- links measurement data to the model,
 - introduces nonlinear complexity,
 - motivates hybrid and LOD-based strategies.
-

Key Principles

- curvature-centric modelling,
 - separation of model and realization,
 - hybrid parametric–discrete representation,
 - operator-based architecture,
 - optimization in realized space,
 - exploitation of sparsity and structure.
-

Outcome

AIM provides a coherent framework for:

- reconstruction from imperfect data,
 - physically meaningful optimization,
 - integration of design, measurement, and construction,
 - scalable computational implementation.
-

Conclusion

Alignment modelling is not a purely geometric task.

It is the integration of:

- geometry,
- physics,
- data,
- and computation.

AIM formalizes this integration as a structured operator framework, establishing a new foundation for railway alignment engineering.

Core Thesis

Alignment is not a curve.

It is a curvature-driven, realization-aware generative system.

Remark 0.1. Classical railway geometry represents track layouts as curves in Euclidean space. While sufficient for visualization, this perspective is inadequate for design, construction, and dynamic analysis.

Remark 0.2. Within the AIM framework, the fundamental object is not the curve $\gamma(s)$, but the curvature law $\kappa(s)$, complemented by the cant function $\phi(s)$. All geometric quantities are derived by integration.

Remark 0.3. This shifts the modelling paradigm from descriptive geometry to generative structure: alignments are defined by the laws that produce curves, not by the curves themselves.

—

Key Principles

Curvature as Primary Variable The alignment is defined by curvature $\kappa(s)$. Position $\gamma(s)$ is a derived quantity. This inversion simplifies modelling and exposes the intrinsic structure of railway geometry.

Realization as Operator The physically constructed track differs from the theoretical alignment. This transformation is captured by a realization operator $\mathcal{R}$, which includes discretization, interpolation, and mechanical response. Design must therefore be formulated in realized space.

Hybrid Representation Real-world data and engineering design require both structure and flexibility. A hybrid model combines parametric representations with local corrections, enabling robust reconstruction from imperfect data.

Analytical Structure All relevant quantities arise from arc-length based relations. As a consequence, derivatives with respect to parameters are available in closed form. This eliminates the need for numerical differentiation.

Efficient Optimization Alignment optimization reduces to a structured nonlinear least-squares problem with sparse and banded Jacobians. This enables efficient solution via Gauss–Newton or SQP methods.

—

Implication

Remark 0.4. The AIM framework unifies geometry, dynamics, realization, and data processing within a single coherent model.

It replaces curve-based thinking by a system-based perspective in which alignment, realization, and optimization are intrinsically linked.

Remark 0.5. This provides a foundation for alignment-based information modelling, capable of bridging theoretical design and practical railway engineering.

Abstract

Railway alignments are traditionally represented as geometric curves in space. This perspective is insufficient for modern engineering tasks, which require the integration of design, measurement, construction, and optimization.

This work introduces *Alignment-Based Information Modelling (AIM)*, a unified framework that reformulates alignment modelling as a curvature-driven and realization-aware system. Instead of treating geometry as a primary object, AIM represents alignments intrinsically through curvature and cant, from which all geometric and dynamic quantities are derived.

A central contribution is the introduction of a realization operator, which models the transformation from theoretical alignment to physically constructed track. This leads to the formulation of alignment design in realized space rather than in purely geometric terms.

The framework combines parametric structure with discrete corrections in a hybrid representation, enabling robust reconstruction from imperfect data. Alignment optimization is formulated as a structured nonlinear least-squares problem with sparse analytical Jacobians, allowing efficient solution via SQP methods.

AIM provides a coherent foundation that unifies geometry, dynamics, data, and computation, establishing a new perspective on railway alignment engineering.

Introduction

1. A Shift in Perspective

Remark 0.1. Railway alignment is commonly described as a geometric curve $\gamma(s)$ embedded in space.

Remark 0.2. This view is sufficient for visualization, but it fails to capture the quantities that actually govern railway design: curvature, cant, dynamics, and realization.

Remark 0.3. In practice, railway tracks are not designed, constructed, or maintained as curves, but through operations acting on curvature and discrete control points.

Summary 0.4. Alignment is not a curve. It is a curvature-driven system whose geometry emerges from its generative structure.

2. Limitations of Current Practice

Remark 0.5. Despite a long tradition of railway engineering, alignment modelling remains fragmented across several domains:

- geometric construction of curves,
- rule-based engineering design,
- data exchange formats such as LandXML and IFC,
- numerical evaluation and simulation.

Remark 0.6. These aspects are typically handled in separate tools and workflows, leading to discontinuities between modelling, analysis, and design.

Remark 0.7. In particular:

- geometry is treated independently of dynamics,
- transition curves are selected from predefined catalogues,
- data formats focus on representation rather than computation,
- optimization is not part of the core modelling paradigm.

Remark 0.8. As a consequence, alignment design lacks a unified internal model that connects data, geometry, dynamics, and realization.

3. Problem Statement

Remark 0.9. The absence of a unified modelling framework leads to fundamental limitations:

- inconsistent interpretation of alignment data,
- separation of geometry and dynamics,
- limited support for optimization-based design,
- high effort for integrating heterogeneous data sources.

Remark 0.10. These issues become increasingly critical as infrastructure systems grow in complexity and demand higher levels of automation, precision, and traceability.

4. AIM: A Unified Framework

Remark 0.11. This work introduces **Alignment-Based Information Modelling (AIM)** as a framework addressing these limitations.

Remark 0.12. The central idea is to represent alignment as a structured, state-based object defined by curvature and cant:

$$(\kappa(s), \phi(s)),$$

from which all geometric and dynamic quantities are derived.

Remark 0.13. Within this framework:

- curvature is the primary modelling variable,
 - geometry is obtained by integration,
 - transition curves are interpreted as control functions,
 - alignment becomes a generative system rather than a static object.
-

5. Core Claim

Proposition 0.14. *Railway alignment modelling can be reformulated as a curvature-driven, realization-aware, and optimization-ready system.*

Remark 0.15. This reformulation enables a unified treatment of:

- geometry,
- dynamics,

- construction and realization,
 - data integration,
 - and optimization.
-

6. Consequences

Remark 0.16. Adopting this perspective leads to several key consequences:

- alignment design becomes an optimization problem,
 - realization is treated as an operator acting on the model,
 - data reconstruction is formulated as an inverse problem,
 - hybrid models bridge parametric structure and measured data,
 - analytical derivatives arise naturally from the formulation.
-

7. Scope of the Work

Remark 0.17. The present manuscript develops the AIM framework across the following dimensions:

- mathematical foundations of curvature-based modelling,
 - structured representation of alignment elements,
 - classification and interpretation of transition curves,
 - modelling of realization and construction processes,
 - formulation of optimization problems and solution methods,
 - integration of real-world data and coordinate systems,
 - application to railway engineering use cases.
-

8. Contribution

Remark 0.18. The main contributions of this work are:

- a unified state-based alignment model,
- the reinterpretation of transition curves as curvature control functions,
- the introduction of a realization-aware modelling perspective,
- the formulation of alignment design as a structured optimization problem,
- the development of a pipeline connecting data, model, and computation.

Remark 0.19. Together, these contributions establish a coherent framework linking theoretical modelling and practical engineering.

9. Structure of the Manuscript

Remark 0.20. The manuscript is organized into several parts, progressing from conceptual foundations to practical applications:

- motivation and conceptual shift,
 - mathematical and state-based foundations,
 - modelling and representation,
 - realization and system perspective,
 - optimization methods,
 - engineering constraints and standards,
 - applications and outlook.
-

10. Outlook

Remark 0.21. The AIM framework is intended as a foundation for future development.

Remark 0.22. It opens the path toward:

- optimization-driven design systems,
- deeper integration with BIM,

Introduction

- network-level alignment modelling,
- interactive and real-time engineering tools.

Remark 0.23. In this sense, the work contributes to a broader transition from static geometry toward computational infrastructure modelling.

Glossary

Core Concepts

Alignment A structured representation of railway track geometry defined primarily through curvature $\kappa(s)$, together with associated state variables such as position, orientation, profile, and cant. An alignment is not merely a curve, but a system linking geometry, dynamics, realization, and operational behaviour.

Arc-Length Domain The intrinsic one-dimensional domain s shared by all synchronized longitudinal alignment bands. Within AIM, arc-length acts as the primary reference structure for geometry, cant, profile, dynamics, and optimization.

Curvature $\kappa(s)$ The fundamental geometric quantity describing the rate of change of direction along the track. Within AIM, curvature is the primary design variable.

In planar formulations this coincides with the standard curvature. On curved surfaces, the physically relevant quantity becomes geodesic curvature.

Cant The rail height difference between both rails, represented as a longitudinal band along the alignment.

The corresponding roll angle

$$\phi(s) = \arctan\left(\frac{u(s)}{g}\right)$$

is derived using the applicable gauge g .

Pose2 A minimal planar alignment state consisting of:

$$\gamma(s), \quad \theta(s).$$

Pose2 forms the reconstruction kernel of planar alignment geometry.

Pose3 A runtime-derived spatial railway state combining:

- planar geometry,
- profile evaluation,
- cant evaluation,
- and spatial frame construction.

State A geometric or physical description of the railway system at a given arc-length position s .

Sparse Alignment Structure

cGeom The central curvature-driven geometric band of the sparse alignment model. It contains:

- fixed elements,
- transition elements,
- pose2-based reconstruction anchors.

Cant Band A longitudinal band storing cant information correlated with fixed alignment elements.

The cant band:

- has no independent stationing system,
- stores rail height differences,
- follows the same normalized curvature family as the associated cGeom element,
- may contain local station offsets.

Profile Band A longitudinal band representing vertical geometry independently from planar curvature geometry.

Fixed Element An alignment element with constant curvature.

Typical special cases are:

- straight segments,
- circular arcs.

Transition Element An alignment element with non-constant curvature evolution.

Transition elements describe controlled changes between curvature states.

Sparse Representation A compact, reconstruction-oriented representation of an alignment using semantically meaningful elements rather than dense sampled geometry.

The sparse representation is reconstruction-oriented rather than sample-oriented.

Runtime Pose3 A pose3 state evaluated dynamically from cGeom, cant, profile, and runtime operators rather than stored explicitly as persistent truth.

Modeling and Representation

Parametric Model A low-dimensional representation of curvature or cant defined by a small set of parameters $\mathbf{T}$, such as transition types or segment lengths.

Sample-Based Model A discrete representation of curvature or geometry defined at a set of points $\{s_i\}$, typically used for measurement data or numerical optimization.

Hybrid Model A combination of parametric structure and discrete corrections:

$$\kappa(s) = \kappa_{\text{param}}(s; \mathbf{T}) + \delta\kappa(s).$$

It allows structured modelling with local flexibility.

Normalized Curvature Law A curvature law defined on the normalized interval

$$u \in [0, 1],$$

separating shape from physical scale.

Operators and Runtime Evaluation

Sampling Operator $\mathcal{S}$ Maps a continuous function to discrete values:

$$\mathcal{S}(\kappa) = \{\kappa(s_i)\}.$$

Interpolation Operator $\mathcal{I}$ Reconstructs a continuous curve or function from discrete data.

Mechanical Operator $\mathcal{M}$ Represents the physical response of the track structure, including smoothing and deformation effects.

Realization Operator $\mathcal{R}$ Maps a theoretical alignment to its physically realized form:

$$\mathcal{R} = \mathcal{M} \circ \mathcal{I} \circ \mathcal{S}.$$

Adjustment Operator $\mathcal{T}$ A local operator representing one construction or maintenance step, for example tamping:

$$\gamma_{k+1} = \mathcal{T}(\gamma_k).$$

Runtime Evaluation Dynamic computation of geometric or physical states from sparse bands and mathematical operators rather than from explicitly stored geometry.

Spaces and Transformations

Track Space A coordinate system aligned with the track, typically parameterized by arc-length s , where geometry and dynamics are defined.

World Space A global coordinate system, often CRS-based, used for georeferencing and data integration.

World-to-Track Transformation The mapping from world coordinates to intrinsic alignment coordinates, typically requiring projection onto the alignment and evaluation of s , d , κ , and cant.

Kilometre Line A longitudinal reference structure used for stationing, independent of the physical geometric alignment.

Optimization

Residual A component of the objective function representing deviation from a desired property, such as smoothness, dynamic equilibrium, or data fit.

Jacobian The matrix of derivatives of residuals with respect to parameters.

Within AIM, Jacobians are preferably derived analytically from arc-length-based relations.

Gauss–Newton Method An optimization method using the approximation

$$H \approx J^T J$$

for solving nonlinear least-squares problems.

Sequential Quadratic Programming (SQP) An iterative optimization framework solving a sequence of quadratic subproblems based on linearized constraints.

Realization and Construction

Realized Alignment The physically constructed track geometry, which differs from the theoretical alignment due to discretization, construction, and mechanical effects.

Control Points Discrete positions $\{s_i\}$ at which geometry is specified or adjusted during construction or maintenance.

Realization Error The deviation between target and realized curvature or geometry.

Inverse Realization Problem The problem of finding a target alignment such that

$$\mathcal{R}(\kappa_{\text{target}}) \approx \kappa_{\text{desired}}.$$

Advanced Geometry

Geodesic Curvature κ_g The intrinsic curvature of a curve on a surface, measuring the change of direction within the surface.

For railway alignment on the Earth, this is the physically relevant curvature.

Normal Curvature κ_n The component of curvature induced by the curvature of the underlying surface, for example the Earth ellipsoid.

Ellipsoidal Alignment An alignment defined directly on a reference ellipsoid rather than in a projected coordinate system.

Surface Frame A local coordinate frame on the ellipsoid consisting of tangent, lateral, and surface-normal directions.

Cant Rotation Rotation of the track cross-section around the tangent direction, defining superelevation.

Effective Lateral Acceleration The physically relevant lateral acceleration acting on a vehicle:

$$a_{\text{eff}} = v^2 \kappa_g - \mathbf{g} \cdot \mathbf{b}_\phi.$$

Pipeline

Pipeline The sequence of transformations from raw data to optimized alignment:

$$\text{data} \rightarrow \text{model} \rightarrow \text{state} \rightarrow \text{realization} \rightarrow \text{optimization}.$$

Failure Modes Typical breakdown points within the pipeline, such as inconsistent data, invalid coordinate systems, unstable projection, or failed optimization.

Design Principles Guiding principles ensuring robustness, interpretability, and consistency across the pipeline.

Teil I.

Why Alignments Matter

Remark 0.24. This part introduces the conceptual motivation of the AIM framework. It explains why railway alignments cannot be treated as mere geometric curves, but must be understood as structured engineering objects.

Remark 0.25. The central claim is that alignment modelling must integrate geometry, dynamics, realization, and data interpretation within a single coherent framework.

Remark 0.26. The following chapters therefore develop the problem setting, the need for a new perspective, and the guiding vision behind alignment-based information modelling.

1. Motivation

1.1. The Hidden Complexity of Railway Alignments

Remark 1.1. At first glance, a railway alignment appears to be a simple geometric object: a curve connecting two points in space.

Remark 1.2. However, this view is misleading. An alignment is not merely a curve, but the result of a complex interaction between geometry, physics, engineering constraints, and operational requirements.

Remark 1.3. Even small changes in curvature or cant can have significant effects on:

- passenger comfort,
- vehicle stability,
- track wear,
- and operational safety.

1.2. Fragmentation of Current Practice

Remark 1.4. In current engineering practice, alignment design is typically divided into separate domains:

- geometric design (plan and profile),
- cant design,
- dynamic evaluation,
- and regulatory verification.

Remark 1.5. These aspects are often treated sequentially rather than jointly. As a consequence:

- design iterations are required,
- optimal solutions are rarely achieved,
- and knowledge remains fragmented across tools and disciplines.

1.3. Limitations of Existing Models

Remark 1.6. Traditional alignment models are primarily geometric. They describe the track as a sequence of elements such as:

- straight lines,
- circular arcs,
- and transition curves.

Remark 1.7. While this representation is sufficient for construction, it is insufficient for:

- capturing dynamic behaviour,
- integrating measurement data,
- or supporting automated optimization.

Remark 1.8. In particular, curvature is often treated as a derived quantity, although it is the fundamental driver of railway dynamics.

1.4. Need for a Unified Representation

Remark 1.9. A modern alignment model must integrate:

- geometry,
- curvature evolution,
- cant,
- and dynamic effects.

Remark 1.10. Such a model should not treat these aspects independently, but as components of a single coherent system.

Remark 1.11. This requires a shift from element-based descriptions to function-based representations.

1.5. Alignment as a State Trajectory

Remark 1.12. Instead of viewing an alignment as a static geometric object, it can be interpreted as a trajectory of a system evolving along arc length.

Remark 1.13. In this interpretation:

- curvature acts as a control variable,
- cant represents an additional state component,

1. Motivation

- and the resulting motion determines physical behaviour.

Remark 1.14. This perspective naturally connects alignment design with modern concepts from:

- differential geometry,
- numerical analysis,
- and optimal control theory.

1.6. Towards Alignment-Based Information Modelling

Remark 1.15. The observations above motivate the development of a new modelling paradigm: *alignment-based information modelling* (AIM).

Remark 1.16. The central idea of AIM is to treat alignment not as a collection of geometric primitives, but as a structured, functional object defined by curvature and its derived quantities.

Remark 1.17. Within this framework:

- transitions are primary modelling elements,
- dynamics are embedded in the representation,
- and optimization becomes a natural design tool.

1.7. Connection to Practical Engineering

Remark 1.18. Despite its mathematical formulation, the AIM approach is not purely theoretical.

Remark 1.19. It directly addresses practical challenges such as:

- handling imperfect measurement data,
- integrating multiple coordinate systems,
- modelling complex track configurations,
- and supporting modern BIM workflows.

1.8. Objective of this Work

Remark 1.20. The objective of this manuscript is to provide a coherent foundation for alignment-based information modelling.

Remark 1.21. This includes:

1. Motivation

- a mathematical description of alignment structures,
- a unified state-based representation,
- methods for numerical reconstruction,
- integration of engineering constraints,
- and approaches to optimization.

Summary 1.22. Railway alignments are not simple geometric curves, but complex systems governed by curvature, cant, and dynamic effects.

A unified, curvature-driven modelling approach is required to capture this complexity and to enable modern, optimization-based design methods.

This work develops the foundations of such an approach.

2. Problem Statement

2.1. From Motivation to Formalization

Remark 2.1. The previous chapter has argued that railway alignment design is intrinsically a coupled problem involving geometry, dynamics, and engineering constraints.

Remark 2.2. We now formulate this insight as a precise problem statement.

2.2. Core Deficiency of Existing Approaches

Remark 2.3. Current alignment models are predominantly:

- element-based,
- geometry-centered,
- and weakly connected to dynamic behaviour.

Remark 2.4. As a consequence, they exhibit the following structural deficiencies:

- curvature is not treated as a primary design variable,
- cant is handled in a separate design process,
- dynamic quantities are evaluated only after geometric design,
- optimization is indirect and iterative rather than intrinsic.

2.3. Formal Gap

Remark 2.5. Let

$$\gamma : [0, L] \rightarrow \mathbb{R}^2$$

denote a planar alignment.

Remark 2.6. In classical formulations, the curve γ is the primary object, while curvature is derived as

$$\kappa(s) = \theta'(s).$$

Remark 2.7. However, from a dynamic perspective, the situation is reversed:

- curvature determines acceleration,

2. Problem Statement

- curvature variation determines jerk,
- and therefore curvature governs physical behaviour.

Remark 2.8. This reveals a fundamental mismatch between:

- geometric modelling (curve-first),
- and physical interpretation (curvature-first).

2.4. Missing Coupling of State Variables

Remark 2.9. Let

$$\kappa(s), \quad \phi(s)$$

denote curvature and cant.

Remark 2.10. In practice, these variables are strongly coupled through

$$a_{\text{eff}}(s) = v^2 \kappa(s) - g \sin \phi(s).$$

Remark 2.11. Despite this coupling:

- κ is typically designed geometrically,
- ϕ is designed in a separate process,
- and their interaction is only checked afterwards.

Remark 2.12. This separation prevents a unified optimization of the system.

2.5. Data and Reality Gap

Remark 2.13. Modern railway projects rely heavily on real-world data, including:

- measurement polylines,
- legacy alignment descriptions,
- and heterogeneous coordinate systems.

Remark 2.14. Existing models struggle to:

- integrate noisy and incomplete data,
- reconcile different coordinate reference systems,
- and maintain a consistent representation across scales.

Remark 2.15. This leads to a disconnect between:

- theoretical alignment models,
- and real-world engineering data.

2.6. Lack of a Unified Optimization Framework

Remark 2.16. Alignment design problems naturally take the form:

$$\min J(\kappa, \phi)$$

subject to geometric and engineering constraints.

Remark 2.17. However, in current practice:

- the objective function is implicit or fragmented,
- constraints are applied in separate stages,
- and optimization is not formulated as a single problem.

Remark 2.18. This prevents the use of systematic numerical optimization methods.

2.7. Target Formulation

Remark 2.19. We aim to formulate alignment design as a unified problem:

$$\text{find } (\kappa, \phi)$$

such that:

- geometric consistency is satisfied,
- engineering constraints are fulfilled,
- and a global objective functional is minimized.

Remark 2.20. This requires:

- a curvature-based representation,
- a coupled state model,
- and a consistent treatment of data and constraints.

2.8. Scope of the Problem

Remark 2.21. The problem addressed in this work includes:

- modelling of alignments via curvature functions,
- integration of cant as a state variable,
- reconstruction of geometry from curvature,

2. Problem Statement

- incorporation of engineering rules as constraints,
- and formulation of optimization problems.

Remark 2.22. The problem explicitly excludes:

- detailed multibody vehicle dynamics,
- full infrastructure modelling,
- and construction-specific considerations.

2.9. Research Questions

Remark 2.23. The following central questions arise:

- How can alignment be represented in a curvature-driven manner?
- How can curvature and cant be modelled as a coupled state?
- How can real-world data be integrated into such a model?
- How can engineering rules be expressed as mathematical constraints?
- How can alignment design be formulated as an optimization problem?

2.10. Working Hypothesis

Remark 2.24. We hypothesize that:

- a curvature-based state representation,
- combined with a unified optimization framework,

provides a more consistent and powerful foundation for railway alignment design.

Remark 2.25. In particular, we assume that:

- transition curves emerge naturally from optimization,
- geometry and dynamics can be unified,
- and real-world data can be systematically integrated.

2.11. Alignment

2.11.1. Motivation

Remark 2.26. In classical geometry, a railway track is often represented as a curve $\gamma(s)$ in space.

Remark 2.27. This representation is insufficient for engineering purposes, as it does not capture the intrinsic quantities governing construction, dynamics, and realization.

Remark 2.28. In particular, curvature and cant are not derived properties, but primary design variables.

—

2.11.2. Definition

Definition 2.29 (Alignment). An **alignment** is a structured object defined over arc-length $s \in [0, L]$, consisting of the following components:

- a curvature function

$$\kappa : [0, L] \rightarrow \mathbb{R},$$

- a cant function

$$\phi : [0, L] \rightarrow \mathbb{R},$$

- an initial pose

$$(\gamma_0, \theta_0),$$

together with the induced state variables

$$\theta(s) = \theta_0 + \int_0^s \kappa(\sigma) \, d\sigma,$$

$$\gamma(s) = \gamma_0 + \int_0^s (\cos \theta(\sigma), \sin \theta(\sigma)) \, d\sigma.$$

—

2.11.3. Interpretation

Remark 2.30. An alignment is not defined by its position $\gamma(s)$, but by its curvature $\kappa(s)$.

Remark 2.31. The curve $\gamma(s)$ is a derived quantity obtained by integration.

Remark 2.32. Thus, curvature is the primary design variable, while position is a consequence.

—

2. Problem Statement

2.11.4. State Structure

Definition 2.33 (Alignment state). The **state** of an alignment at position s is given by

$$(\gamma(s), \theta(s), \kappa(s), \phi(s)).$$

Remark 2.34. This state couples geometry, orientation, and dynamics in a unified representation.

2.11.5. Structural Nature

Proposition 2.35. *An alignment is a low-dimensional structured object, even though its realization may involve high-dimensional data.*

Remark 2.36. This structure enables:

- efficient storage,
 - robust optimization,
 - meaningful interpretation.
-

2.11.6. Relation to Realization

Remark 2.37. The alignment represents the intended geometry.

Remark 2.38. The physically constructed track is given by the realization operator:

$$\gamma_{\text{real}} = \mathcal{R}(\kappa, \phi).$$

Remark 2.39. Thus, the alignment exists in design space, while the track exists in physical space.

2.11.7. Relation to Data

Remark 2.40. Measured data typically provide sampled positions or polylines.

Remark 2.41. An alignment must be reconstructed from such data via inverse problems.

2.11.8. Key Insight

Proposition 2.42. *An alignment is not a curve, but a curvature-driven generative model of a curve.*

2. Problem Statement

2.11.9. Summary

Summary 2.43. The alignment is the central object of railway geometry.

It is defined by curvature and cant, from which all geometric and dynamic quantities are derived.

This perspective shifts the focus from describing curves to defining generative structures, forming the foundation of alignment-based information modelling.

2.12. Summary

Summary 2.44. Current alignment design methods suffer from fragmentation between geometry, dynamics, and data.

This work addresses the problem of developing a unified, curvature-driven framework that integrates these aspects into a single coherent model.

The goal is to enable consistent modelling, analysis, and optimization of railway alignments.

3. Vision

Why Railway is Different

Designing a railway alignment is fundamentally different from designing a road.

A road vehicle can steer.

A train cannot.

A road alignment may tolerate small geometric imperfections. A railway alignment cannot.

For a train:

- *curvature directly determines lateral acceleration,*
- *curvature variation determines jerk,*
- *and both directly affect safety, comfort, and wear.*

As a consequence:

- *geometry is not just shape,*
- *curvature is not just a derived quantity,*
- *and transitions are not optional smoothing elements.*

They are the system.

3.1. From Problem to Paradigm

Remark 3.1. The previous chapters have identified a fundamental limitation of current alignment modelling approaches: the fragmentation of geometry, dynamics, and data.

Remark 3.2. This motivates not only incremental improvements, but a shift in perspective.

3.2. Core Idea

Remark 3.3. The central idea of this work is to treat railway alignment as a *state trajectory* governed by curvature and its associated quantities.

3. Vision

Remark 3.4. Instead of describing alignment as a sequence of geometric elements, we propose to describe it as a function-based object:

$$(\kappa(s), \phi(s)).$$

Remark 3.5. In this view:

- curvature defines geometry,
- cant defines dynamic balancing,
- and their interaction determines physical behaviour.

3.3. Alignment as a Controlled System

Remark 3.6. The alignment is interpreted as a controlled system evolving along arc length:

$$\begin{aligned}\gamma'(s) &= (\cos \theta(s), \sin \theta(s)), \\ \theta'(s) &= \kappa(s), \\ \phi'(s) &= \omega(s).\end{aligned}$$

Remark 3.7. Here:

- $\kappa(s)$ acts as a geometric control,
- $\omega(s)$ governs cant evolution,
- and the resulting state determines both geometry and dynamics.

Remark 3.8. This formulation establishes a direct link between alignment design and optimal control theory.

3.4. Transitions as Primary Objects

Remark 3.9. In classical models, transitions are treated as auxiliary elements connecting geometric primitives.

Remark 3.10. In the proposed framework, transitions become the primary modelling objects, as they define the evolution of curvature.

Remark 3.11. This shifts the focus:

- from static geometry
- to controlled change of state.

3.5. Integration of Reality

Remark 3.12. A central requirement of the proposed approach is compatibility with real-world data.

Remark 3.13. This includes:

- measurement data with uncertainty,
- heterogeneous coordinate systems (CRS),
- legacy alignment descriptions,
- and complex track configurations.

Remark 3.14. The model must therefore support:

- robust reconstruction from imperfect data,
- explicit handling of coordinate transformations,
- and consistent representation across scales.

3.6. Unified Optimization Framework

Remark 3.15. Within this paradigm, alignment design is formulated as an optimization problem:

$$\min J(\kappa, \phi)$$

subject to:

- state dynamics,
- engineering constraints,
- and boundary conditions.

Remark 3.16. This allows:

- direct incorporation of comfort criteria,
- integration of dynamic effects,
- and systematic exploration of design alternatives.

3.7. Towards Alignment-Based Information Modelling

Remark 3.17. The proposed framework forms the basis of *alignment-based information modelling* (AIM).

Remark 3.18. AIM is characterized by:

- curvature-driven representation,
- state-based modelling,
- integration of geometry and dynamics,
- and compatibility with data-driven workflows.

Remark 3.19. In this sense, alignment is not a static object, but a structured, computable entity.

3.8. Implications

Remark 3.20. Adopting this perspective has several implications:

- design becomes a problem of state control,
- transitions emerge from functional requirements,
- optimization replaces iterative manual adjustment,
- and modelling becomes directly compatible with computation.

3.9. Long-Term Perspective

Remark 3.21. The proposed framework is intended as a foundation for future developments, including:

- integration with BIM systems,
- extension to network-level modelling,
- incorporation of vehicle dynamics,
- and real-time optimization.

Remark 3.22. It is not a finished solution, but a structured starting point.

3.10. Summary

Summary 3.23. This work proposes a paradigm shift from element-based alignment modelling to a curvature-driven, state-based representation.

By treating alignment as a controlled system and integrating geometry, dynamics, and data, it establishes a unified foundation for modelling and optimization.

This perspective forms the basis of alignment-based information modelling (AIM).

Teil II.

Mathematical Foundations

Remark 3.24. This part establishes the mathematical foundation of the AIM framework. Its purpose is not to develop abstract mathematics for its own sake, but to identify the intrinsic structures underlying railway alignment modelling.

Remark 3.25. In particular, the focus lies on arc-length based descriptions, curvature-driven geometry, and the distinction between primary and derived quantities.

Remark 3.26. These foundations provide the conceptual and formal basis for all later developments in modelling, realization, and optimization.

4. Axiomatic Foundation

4.1. Primitives

Definition 4.1. $s \in \mathbb{R}$ is the arc-length parameter.

Definition 4.2. $\gamma : I \rightarrow \mathbb{R}^2$ is a curve.

Definition 4.3. $\theta : I \rightarrow \mathbb{R}$ is the direction angle.

Definition 4.4. $\kappa : I \rightarrow \mathbb{R}$ is the curvature.

4.2. Axioms

Axiom 4.5.

$$\|\gamma'(s)\| = 1$$

Axiom 4.6.

$$\theta'(s) = \kappa(s)$$

Axiom 4.7.

$$\gamma'(s) = (\cos \theta(s), \sin \theta(s))$$

4.3. Consequence

Proposition 4.8. *Curvature uniquely defines the curve up to initial conditions.*

5. Notation and Terminology

5.1. Motivation

Remark 5.1. Railway alignment engineering combines concepts from geometry, surveying, dynamics, construction, realization, and operational railway systems.

Remark 5.2. Many inconsistencies in classical railway systems arise not from incorrect mathematics, but from ambiguous terminology and mixed conceptual layers.

Remark 5.3. The AIM framework therefore introduces a canonical notation and terminology system.

Remark 5.4. The purpose of this chapter is:

- to define canonical symbols,
- to stabilize terminology,
- to reduce ambiguity,
- and to separate fundamentally different conceptual layers.

5.2. General Principles

Principle 5.5 (Intrinsic-first principle). Whenever possible, railway geometry is formulated intrinsically using arc-length-based quantities.

Principle 5.6 (Runtime distinction principle). A distinction is made between:

- stored representations,
- runtime-evaluated states,
- and physically realized states.

Principle 5.7 (Target vs realization principle). Target geometry and realized geometry are treated as distinct objects.

Principle 5.8 (Canonical layer principle). AIM separates:

- intrinsic geometry,
- runtime interpretation,

5. Notation and Terminology

- realization,
- operational behaviour,
- and CRS-dependent embedding

into distinct conceptual layers.

5.3. Canonical Coordinates

5.3.1. Arc-Length

Definition 5.9 (Arc-length).

$$s$$

denotes intrinsic arc-length along the alignment.

Remark 5.10. Arc-length is the primary intrinsic coordinate within AIM.

5.3.2. Track Coordinates

Definition 5.11 (Track coordinates). Track-space coordinates are:

$$(s, d, h),$$

where:

- s : longitudinal coordinate,
- d : lateral offset,
- h : vertical offset.

5.3.3. World Coordinates

Definition 5.12 (World coordinates). World coordinates are denoted by:

$$x \in \mathbb{R}^n.$$

Remark 5.13. World coordinates are CRS-dependent.

5.4. Geometry Symbols

5.4.1. Alignment Curve

Definition 5.14 (Alignment curve).

$$\gamma(s)$$

denotes the intrinsic alignment geometry.

5. Notation and Terminology

Remark 5.15. The alignment curve represents the geometric backbone of the railway alignment.

5.4.2. Orientation

Definition 5.16 (Orientation angle).

$$\theta(s)$$

denotes horizontal alignment orientation.

5.4.3. Curvature

Definition 5.17 (Curvature).

$$\kappa(s)$$

denotes curvature as function of arc-length.

Remark 5.18. Curvature is the primary geometric quantity within AIM.

5.4.4. Cant

Definition 5.19 (Cant).

$$\phi(s)$$

denotes cant rotation.

Remark 5.20. Within AIM, cant is interpreted geometrically as frame rotation, independent of its engineering realization as rail height difference.

5.4.5. Cant Evolution

Definition 5.21 (Cant evolution).

$$\omega(s) = \phi'(s)$$

denotes cant-rate evolution.

5.5. Frame Symbols

5.5.1. Tangent Direction

Definition 5.22 (Tangent vector).

$$\mathbf{t}(s)$$

denotes the local tangent direction.

5.5.2. Lateral Direction

Definition 5.23 (Lateral vector).

$$\mathbf{n}(s)$$

denotes the local lateral direction.

5.5.3. Binormal Direction

Definition 5.24 (Binormal vector).

$$\mathbf{b}(s)$$

denotes the local binormal direction.

5.5.4. Local Frame

Definition 5.25 (Local frame).

$$F(s) = (\mathbf{t}(s), \mathbf{n}(s), \mathbf{b}(s))$$

denotes the local pose3 frame.

5.6. Pose and State Terminology

5.6.1. Pose

Definition 5.26 (Pose). A pose denotes a local intrinsic railway state attached to arc-length position s .

5.6.2. pose2

Definition 5.27 (pose2). The pose2 state is:

$$\text{pose2}(s) = (\gamma(s), \theta(s)).$$

Remark 5.28. pose2 represents intrinsic planar railway geometry.

5.6.3. pose3

Definition 5.29 (pose3). The pose3 state is:

$$\text{pose3}(s) = (\gamma(s), \theta(s), \phi(s)).$$

Remark 5.30. pose3 extends pose2 by including cant and spatial railway orientation.

5.6.4. Controls

Definition 5.31 (Control quantities). The canonical control quantities are:

$$\kappa(s), \quad \omega(s).$$

Remark 5.32. Curvature controls directional evolution, whereas cant-rate controls cant evolution.

5.6.5. State

Definition 5.33 (State). A state denotes a runtime-evaluated operational railway condition.

Remark 5.34. Within AIM, states are not merely stored parameter containers.

Remark 5.35. Runtime refers to operational evaluation of alignment-related quantities, independent of implementation details.

5.7. Target and Realization Notation

5.7.1. Target Quantities

Definition 5.36 (Target notation). Target quantities use the index:

$$(\cdot)_{\text{target}}.$$

5.7.2. Realized Quantities

Definition 5.37 (Realized notation). Realized quantities use the index:

$$(\cdot)_{\text{real}}.$$

5.7.3. Measured Quantities

Definition 5.38 (Measured notation). Measured quantities use the index:

$$(\cdot)_{\text{meas}}.$$

5.8. Operator Notation

5.8.1. Realization Operator

Definition 5.39 (Realization operator).

$$\mathcal{R}$$

denotes the realization operator.

5.8.2. Sampling Operator

Definition 5.40 (Sampling operator).

$$\mathcal{S}$$

denotes sampling from continuous to discrete representation.

5.8.3. Interpolation Operator

Definition 5.41 (Interpolation operator).

$$\mathcal{I}$$

denotes reconstruction from discrete data.

5.8.4. Mechanical Operator

Definition 5.42 (Mechanical operator).

$$\mathcal{M}$$

denotes mechanical or structural response of the railway system.

5.8.5. Adjustment Operator

Definition 5.43 (Adjustment operator).

$$\mathcal{T}$$

denotes a local construction or maintenance update operator.

5.9. CRS and Earth Geometry

5.9.1. Geodesic Curvature

Definition 5.44 (Geodesic curvature).

$$\kappa_g$$

denotes geodesic curvature on the Earth surface.

5.9.2. Normal Curvature

Definition 5.45 (Normal curvature).

$$\kappa_n$$

denotes curvature induced by Earth surface geometry.

5.9.3. Earth Surface Normal

Definition 5.46 (Earth surface normal).

$$\mathbf{n}_E$$

denotes the ellipsoid surface normal.

5.10. Vehicle-Space Terminology

5.10.1. Vehicle Space

Definition 5.47 (Vehicle space). Vehicle space denotes the physically occupied operational space of the railway vehicle.

5.10.2. Operational Geometry

Definition 5.48 (Operational geometry). Operational geometry denotes geometry relevant to:

- passenger comfort,
- platform interaction,
- clearance,
- vehicle behaviour,
- and dynamic vehicle-space interaction.

5.11. Canonical Terminology

Definition 5.49 (Preferred terminology). The following terminology is preferred:

alignment	intrinsic railway geometry system
curve	mathematical geometric object
geometry	geometric description
trajectory	motion-related path
pose	local railway state
frame	local spatial interpretation
state	runtime operational condition
realization	physical construction process
runtime	operational computational evaluation
world space	CRS-based spatial coordinates
track space	intrinsic alignment coordinates
track	physically realized railway structure

5.12. Avoided Ambiguities

Remark 5.50. The following ambiguities should be avoided:

- alignment $\neq$ merely curve,
- realization $\neq$ visualization,
- runtime $\neq$ software process only,
- pose $\neq$ frame,
- geometry $\neq$ realization,
- world coordinates $\neq$ track coordinates.

5.13. Connection to AIM

Summary 5.51. The AIM notation system separates:

- intrinsic geometry,
- realized geometry,
- runtime states,
- operational vehicle-space interpretation,
- and CRS-dependent world geometry.

This notation framework forms the canonical language of AIM.

6. State Model

6.1. Motivation

Remark 6.1. Railway alignment modelling involves interacting descriptions of geometry, orientation, cant, realization, runtime interpretation, and operational behaviour.

Remark 6.2. Classical alignment theory often treats these aspects separately:

- geometry as curves,
- cant as annotation,
- realization as construction detail,
- and optimization as external post-processing.

Remark 6.3. Within AIM, these components are interpreted as parts of a unified state-based system.

Remark 6.4. The purpose of the state model is:

- to define canonical railway states,
- to distinguish intrinsic and extrinsic representations,
- to separate target and realized states,
- and to establish a consistent basis for runtime services, projection, realization, and optimization.

6.2. State Concept

6.2.1. State Definition

Definition 6.5 (State). A state is the minimal set of variables required to describe the geometric and physical condition of a railway alignment at arc-length position s .

Remark 6.6. Within AIM, states are interpreted intrinsically along the alignment rather than globally in world coordinates.

Remark 6.7. The alignment is therefore interpreted as a spatial state trajectory parameterized by arc-length.

6.2.2. Canonical State Variable

Definition 6.8 (Canonical state). The canonical intrinsic railway state is:

$$x(s) = (\gamma(s), \theta(s), \phi(s)). \quad (6.1)$$

Remark 6.9. The state combines:

- intrinsic geometry,
- orientation,
- and cant evolution.

6.3. Pose2 State

6.3.1. Definition

Definition 6.10 (Pose2 state). The planar pose state is:

$$\text{pose2}(s) = (\gamma(s), \theta(s)). \quad (6.2)$$

Remark 6.11. The pose2 state describes:

- planar position,
- and planar orientation.

6.3.2. Pose2 Evolution

Remark 6.12. The pose2 evolution is governed by:

$$\theta'(s) = \kappa(s), \quad (6.3)$$

together with:

$$\gamma'(s) = (\cos \theta(s), \sin \theta(s)). \quad (6.4)$$

Remark 6.13. Curvature therefore acts as the primary geometric control variable.

6.4. Pose3 State

6.4.1. Definition

Definition 6.14 (Pose3 state). The extended railway pose state is:

$$\text{pose3}(s) = (\gamma(s), \theta(s), \phi(s)). \quad (6.5)$$

Remark 6.15. The pose3 state extends the planar state by including cant.

6. State Model

Remark 6.16. Cant is interpreted as a rotation around the local tangent direction.

6.4.2. Controls

Definition 6.17 (Pose3 controls). The canonical controls are:

$$u(s) = (\kappa(s), \omega(s)), \quad (6.6)$$

where:

$$\omega(s) = \phi'(s).$$

Remark 6.18. Curvature controls directional evolution, whereas cant-rate controls cross-level evolution.

6.4.3. Canonical Pose3 System

Remark 6.19. The canonical pose3 evolution system is:

$$\begin{aligned} \gamma'(s) &= (\cos \theta(s), \sin \theta(s)), \\ \theta'(s) &= \kappa(s), \\ \phi'(s) &= \omega(s). \end{aligned} \quad (6.7)$$

Remark 6.20. This formulation interprets railway alignment as a controlled dynamical state system.

6.5. Intrinsic and Extrinsic States

6.5.1. Intrinsic State

Definition 6.21 (Intrinsic state). An intrinsic state is defined relative to the alignment itself and uses arc-length s as primary parameter.

Remark 6.22. Examples include:

- curvature,
- cant,
- tangent direction,
- local railway frames.

6.5.2. Extrinsic State

Definition 6.23 (Extrinsic state). An extrinsic state is defined relative to a global coordinate reference system.

Remark 6.24. Examples include:

- WGS84 coordinates,
- projected CRS coordinates,
- BIM placement coordinates,
- sensor measurements.

6.5.3. Intrinsic–Extrinsic Coupling

Remark 6.25. The world–track transformation couples intrinsic and extrinsic states.

Remark 6.26. World-to-track projection therefore acts as a state conversion process between global and alignment-relative representations.

6.6. Target and Realized States

6.6.1. Target State

Definition 6.27 (Target state). The target state describes the analytically intended alignment:

$$\text{pose3}_{\text{target}}. \quad (6.8)$$

6.6.2. Realized State

Definition 6.28 (Realized state). The realized state describes the physically constructed alignment:

$$\text{pose3}_{\text{real}}. \quad (6.9)$$

6.6.3. Realization Relation

Remark 6.29. The realization operator connects both states:

$$\mathcal{R} : \text{pose3}_{\text{target}} \mapsto \text{pose3}_{\text{real}}. \quad (6.10)$$

Remark 6.30. The realized state includes:

- construction effects,
- machine behaviour,
- discretization,
- maintenance history,
- and measurement uncertainty.

6.7. Runtime States

6.7.1. Runtime Interpretation

Remark 6.31. Within AIM, states are not static stored objects but runtime-evaluated entities.

Remark 6.32. Runtime systems continuously evaluate:

- world-to-track projections,
- local frames,
- realized geometry,
- dynamic quantities,
- and optimization residuals.

6.7.2. Derived Runtime Quantities

Definition 6.33 (Derived runtime quantities). Derived runtime quantities include:

$$a_{\text{eff}}, \quad j, \quad \kappa_g, \quad d, \quad h. \quad (6.11)$$

Remark 6.34. These are not independent state variables, but quantities derived from the underlying pose and frame structure.

6.8. Sparse and Hybrid States

6.8.1. Sparse Representation

Definition 6.35 (Sparse state). A sparse state is a compact representation using only essential control parameters.

Remark 6.36. Examples include:

- fixed curvature segments,
- transition segments,
- cant transitions,
- anchor poses.

6.8.2. Hybrid Representation

Definition 6.37 (Hybrid state). A hybrid state combines:

- parametric structure,
- and local sampled corrections.

Remark 6.38. Hybrid states combine efficient modelling with local adaptation.

6.9. Optimization States

6.9.1. Optimization Variables

Remark 6.39. Optimization does not necessarily operate on full geometry.

Remark 6.40. Typical optimization variables include:

- curvature parameters,
- transition lengths,
- cant parameters,
- control point positions,
- and realization corrections.

6.9.2. Realization-Aware Optimization

Remark 6.41. Within AIM, optimization is evaluated on realized states:

$$J(\mathcal{R}(\text{pose3})). \tag{6.12}$$

Remark 6.42. Optimization therefore operates indirectly through the realization operator.

6.10. State Hierarchy

Remark 6.43. The AIM framework distinguishes:

- intrinsic states,
- extrinsic states,
- target states,
- realized states,
- runtime states,
- and optimization states.

6. State Model

Remark 6.44. These are not independent models, but interconnected layers of one unified railway state system.

6.11. Canonical Interpretation

Proposition 6.45. *Within AIM, railway alignment is interpreted as a controlled spatial state trajectory evolving along arc-length.*

Proposition 6.46. *Curvature and cant evolution form the primary control structure of this trajectory.*

Proposition 6.47. *Realization, projection, and optimization operate on states rather than on isolated geometric entities.*

6.12. Connection to AIM

Summary 6.48. The AIM state model provides a unified framework linking:

- geometry,
- cant,
- realization,
- runtime projection,
- dynamics,
- and optimization.

It establishes railway alignment as a coupled state-based system rather than merely a collection of curves and coordinates.

7. Curvature on the Ellipsoid

7.1. Motivation

Remark 7.1. Railway alignment modelling is traditionally performed in projected coordinate systems such as Gauss–Krüger or UTM.

Remark 7.2. Within these systems, curvature, arc-length, and derived dynamic quantities are evaluated in a planar approximation of the Earth.

Remark 7.3. However, such projections introduce geometric distortions and therefore do not exactly represent the physical geometry on which the railway exists.

Remark 7.4. This motivates a formulation of alignment geometry directly on the reference surface of the Earth, typically modelled as an ellipsoid.

7.2. Curve on a Surface

Definition 7.5 (Reference ellipsoid). Let $\mathcal{E} \subset \mathbb{R}^3$ denote a reference ellipsoid representing the Earth.

Definition 7.6 (Alignment on the ellipsoid). An alignment is defined as a curve

$$\gamma : [0, L] \rightarrow \mathcal{E}$$

parameterized by arc-length s .

Remark 7.7. The parameterization satisfies

$$\|\gamma'(s)\| = 1,$$

so that s represents physical distance along the curve.

7.3. Tangent and Normal Decomposition

Definition 7.8 (Tangent vector). The tangent vector is defined as

$$\mathbf{t}(s) = \gamma'(s).$$

Definition 7.9 (Surface normal). Let $\mathbf{n}_{\mathcal{E}}(s)$ denote the unit normal vector of the ellipsoid at $\gamma(s)$.

7. Curvature on the Ellipsoid

Remark 7.10. The second derivative $\gamma''(s)$ can be decomposed into:

- a component tangent to the surface,
- a component normal to the surface.

Definition 7.11 (Decomposition).

$$\gamma''(s) = \underbrace{\nabla_s \mathbf{t}(s)}_{\text{tangential}} + \underbrace{(\gamma''(s) \cdot \mathbf{n}_{\mathcal{E}}) \mathbf{n}_{\mathcal{E}}}_{\text{normal}}.$$

Remark 7.12. Here, $\nabla_s \mathbf{t}$ denotes the covariant derivative of the tangent vector along the curve on the surface.

7.4. Geodesic Curvature

Definition 7.13 (Geodesic curvature). The intrinsic curvature of the alignment on the ellipsoid is defined as

$$\kappa_g(s) = \|\nabla_s \mathbf{t}(s)\|.$$

Remark 7.14. Geodesic curvature measures the change of direction within the surface.

Remark 7.15. It is the surface-intrinsic counterpart of planar curvature and is the relevant curvature notion for alignment geometry constrained to the reference surface.

7.5. Normal Curvature

Definition 7.16 (Normal curvature). The normal curvature is defined as the normal component of the second derivative:

$$\kappa_n(s) = |\gamma''(s) \cdot \mathbf{n}_{\mathcal{E}}(s)|.$$

Remark 7.17. Normal curvature is induced by the curvature of the ellipsoid itself and does not describe the bending of the alignment within the surface.

7.6. Total Curvature

Proposition 7.18. *The total spatial curvature satisfies:*

$$\kappa^2(s) = \kappa_g^2(s) + \kappa_n^2(s).$$

Remark 7.19. Thus, the spatial curvature decomposes into an intrinsic surface contribution and an extrinsic surface-induced contribution.

7.7. Railway-Relevant Curvature

Proposition 7.20. *For railway alignment modelling on a reference surface, the railway-relevant curvature is the geodesic curvature:*

$$\kappa_{\text{rail}}(s) = \kappa_g(s).$$

Remark 7.21. Railway vehicles move along the surface, and dynamic quantities such as lateral acceleration depend on curvature within the surface rather than on the embedding curvature of the Earth.

7.8. Relation to Planar Models

Remark 7.22. In planar geometry, curvature is defined as

$$\kappa(s) = \theta'(s).$$

Remark 7.23. On a curved surface, the heading-angle derivative is replaced by the covariant change of the tangent direction:

$$\kappa_g(s) = \|\nabla_s \mathbf{t}(s)\|.$$

Remark 7.24. Thus, classical planar formulations are a special case of the intrinsic surface formulation.

7.9. Special Case: Geodesics

Definition 7.25 (Geodesic). A curve satisfies

$$\kappa_g(s) = 0$$

if and only if it is a geodesic on the surface.

Remark 7.26. Geodesics represent straightest paths on the ellipsoid in the intrinsic surface sense.

Remark 7.27. They are generally not straight in projected coordinate systems.

Remark 7.28. Railway alignment design is not equivalent to geodesic design. A railway alignment must satisfy engineering, dynamic, operational, and realization constraints.

7.10. Computational Formulation

Remark 7.29. In practical computations, geodesic curvature can be evaluated by projecting the second derivative onto the tangent plane of the surface:

$$\kappa_g = \|\gamma'' - (\gamma'' \cdot \mathbf{n}_\mathcal{E})\mathbf{n}_\mathcal{E}\|.$$

Remark 7.30. This formulation separates curvature within the surface from curvature caused by the embedding of the surface in three-dimensional space.

7.11. Implications for Engineering

Proposition 7.31. *Curvature evaluated in projected coordinate systems differs from intrinsic curvature on the ellipsoid.*

Remark 7.32. Consequently:

- arc-length is only approximately preserved,
- curvature is distorted by projection,
- dynamic quantities are evaluated on an approximate geometry.

Proposition 7.33. *Evaluating curvature intrinsically yields more physically consistent results.*

7.12. Vision

Remark 7.34. A consistent modelling framework may formulate alignment geometry directly on the ellipsoid and use projections only for visualization and data exchange.

Remark 7.35. This enables:

- physically meaningful curvature,
- accurate dynamic evaluation,
- improved robustness in optimization,
- and increased safety in alignment design.

Remark 7.36. This is a long-term research direction rather than a prerequisite for practical AIM implementations.

7.13. Connection to AIM

Summary 7.37. Curvature on the ellipsoid must be understood as an intrinsic quantity, measured within the surface rather than in ambient space.

For railway alignment on a reference surface, the relevant curvature is the geodesic curvature.

This distinction is essential for physically consistent modelling and provides a foundation for extending the AIM framework beyond projection-based engineering geometry.

8. Operators

8.1. Motivation

Remark 8.1. Railway alignment modelling is not merely based on geometric objects, but on transformations acting on states.

Remark 8.2. Within AIM, geometry, realization, projection, optimization, and runtime evaluation are formulated operator-theoretically.

Remark 8.3. Operators connect:

- intrinsic geometry,
- realized geometry,
- world-space representations,
- sampled representations,
- and optimization states.

Remark 8.4. This chapter introduces the canonical operators of AIM and their conceptual roles.

8.2. General Operator Interpretation

Definition 8.5 (Operator). An operator is a mapping acting on states, functions, or geometric representations.

Remark 8.6. Operators may:

- transform representations,
- evaluate quantities,
- modify states,
- or approximate physical processes.

Principle 8.7 (State-operator principle). Within AIM, operators act on states rather than on isolated geometric objects.

8.3. Geometric Operators

8.3.1. Differential Curvature Operator

Definition 8.8 (Curvature relation). The curvature operator is:

$$\theta'(s) = \kappa(s).$$

Remark 8.9. Curvature acts as the differential generator of directional evolution.

8.3.2. Pose Reconstruction Operator

Definition 8.10 (Pose reconstruction). The reconstruction operator is:

$$\gamma'(s) = (\cos \theta(s), \sin \theta(s)).$$

Remark 8.11. Together with curvature evolution, this defines intrinsic alignment reconstruction.

8.3.3. Cant Evolution Operator

Definition 8.12 (Cant evolution). Cant evolution is governed by:

$$\phi'(s) = \omega(s).$$

Remark 8.13. Cant-rate acts as the control quantity for cross-level evolution.

8.4. Projection Operators

8.4.1. Track-to-World Mapping

Definition 8.14 (Track-to-world operator). The forward projection operator is:

$$\mathcal{P}_{\text{tw}} : (s, d, h) \mapsto x.$$

Remark 8.15. This operator maps intrinsic track-space coordinates into world space.

8.4.2. World-to-Track Projection

Definition 8.16 (World-to-track operator). The inverse projection operator is:

$$\mathcal{P}_{\text{wt}} : x \mapsto (s, d, h).$$

Remark 8.17. The inverse projection is generally nonlinear and potentially ambiguous.

8.4.3. Projection Interpretation

Principle 8.18 (Projection principle). World-to-track transformation is an inverse runtime projection problem, not a static coordinate conversion.

8.5. Sampling and Reconstruction Operators

8.5.1. Sampling Operator

Definition 8.19 (Sampling operator). The sampling operator is:

$$\mathcal{S} : f(s) \mapsto \{f(s_i)\}.$$

Remark 8.20. Sampling converts continuous geometry into discrete representations.

8.5.2. Interpolation Operator

Definition 8.21 (Interpolation operator). The interpolation operator is:

$$\mathcal{I} : \{f(s_i)\} \mapsto \tilde{f}(s).$$

Remark 8.22. Interpolation reconstructs continuous approximations from sampled data.

8.5.3. Hybrid Representation Principle

Principle 8.23 (Hybrid representation principle). Efficient railway modelling combines:

- sampled representations for robustness,
- and parametric reconstruction for precision.

8.6. Realization Operators

8.6.1. Mechanical Operator

Definition 8.24 (Mechanical operator). The mechanical operator is:

$$\mathcal{M}.$$

Remark 8.25. The operator $\mathcal{M}$ represents:

- structural response,
- ballast behaviour,
- track elasticity,
- and mechanical smoothing.

8.6.2. Realization Operator

Definition 8.26 (Realization operator). The realization operator is:

$$\mathcal{R} : \text{pose3}_{\text{target}} \mapsto \text{pose3}_{\text{real}}.$$

Remark 8.27. Realization combines:

- sampling,
- interpolation,
- mechanical response,
- and construction processes.

8.6.3. Operator Composition

Definition 8.28 (Realization composition). The realization process may be interpreted as:

$$\mathcal{R} = \mathcal{M} \circ \mathcal{I} \circ \mathcal{S}.$$

Remark 8.29. This interpretation separates:

- discretization,
- reconstruction,
- and physical realization effects.

8.6.4. Adjustment Operator

Definition 8.30 (Adjustment operator). The local adjustment operator is:

$$\mathcal{T}.$$

Remark 8.31. The operator $\mathcal{T}$ represents localized construction or maintenance actions.

8.6.5. Iterative Realization

Definition 8.32 (Iterative realization). The realization process may be written as:

$$\gamma_{k+1} = \mathcal{T}(\gamma_k).$$

Remark 8.33. Thus, realization is interpreted as an iterative operator process.

8.7. Optimization Operators

8.7.1. Residual Operator

Definition 8.34 (Residual operator). The residual operator is:

$$r(x).$$

Remark 8.35. Residuals quantify deviations from desired properties.

8.7.2. Jacobian Operator

Definition 8.36 (Jacobian). The Jacobian operator is:

$$J(x) = \frac{\partial r}{\partial x}.$$

8.7.3. Gauss–Newton Operator

Definition 8.37 (Gauss–Newton approximation). The Gauss–Newton approximation is:

$$H \approx J^T J.$$

Remark 8.38. This approximation appears naturally in least-squares optimization.

8.7.4. Realization-Aware Optimization

Definition 8.39 (Realization-aware optimization). A realization-aware optimization problem takes the form:

$$\min J(\mathcal{R}(\text{pose3})).$$

Remark 8.40. Optimization therefore acts indirectly through realization.

8.8. Frame Operators

8.8.1. Frame Construction

Definition 8.41 (Frame construction operator). The frame operator constructs:

$$F(s) = (\mathbf{t}(s), \mathbf{n}(s), \mathbf{b}(s)).$$

Remark 8.42. Frames provide local spatial interpretation of the alignment state.

8.8.2. Cant Rotation

Definition 8.43 (Cant rotation operator). Cant acts as a rotation:

$$R_\phi(s)$$

around the tangent direction.

8.9. Runtime Operators

8.9.1. Runtime Evaluation

Remark 8.44. Many AIM operators are runtime operators:

- world-to-track projection,
- frame construction,
- realization evaluation,
- dynamic evaluation,
- and optimization residual computation.

8.9.2. Derived Runtime Quantities

Definition 8.45 (Derived runtime quantities). Runtime-evaluated quantities include:

$$a_{\text{eff}}, \quad \dot{j}, \quad \kappa_g.$$

Remark 8.46. These quantities are derived from underlying pose and frame states.

8.10. Operator Hierarchy

Remark 8.47. The AIM framework distinguishes:

- geometric operators,
- projection operators,
- realization operators,
- optimization operators,
- and runtime operators.

Remark 8.48. These operators act on different conceptual layers but form one coupled operator system.

8.11. Canonical Interpretation

Proposition 8.49. *Within AIM, railway alignment modelling is fundamentally operator-based.*

8. Operators

Proposition 8.50. *Operators transform and evaluate states across:*

- *intrinsic geometry,*
- *world-space embedding,*
- *realization,*
- *and operational runtime interpretation.*

Proposition 8.51. *Realization, projection, and optimization are therefore interpreted as operator layers acting on alignment states.*

8.12. Connection to AIM

Summary 8.52. The AIM operator framework unifies:

- geometry reconstruction,
- world–track projection,
- realization,
- runtime evaluation,
- and optimization.

Operators provide the dynamic transformation layer connecting all parts of the AIM framework.

9. Pose3 on the Ellipsoid

9.1. Motivation

Remark 9.1. The planar pose3 model describes railway state by position, direction, curvature, and cant in a projected engineering coordinate system.

Remark 9.2. For physically consistent modelling on the Earth, this state must be defined on the reference ellipsoid rather than in a planar projection.

9.2. Surface-Based Pose

Definition 9.3 (Surface pose). Let $\mathcal{E}$ be a reference ellipsoid and let

$$\gamma : [0, L] \rightarrow \mathcal{E}$$

be an arc-length parameterized alignment.

A surface pose is given by

$$\text{pose}_{\mathcal{E}}(s) = (\gamma(s), \mathbf{t}(s), \mathbf{n}_{\mathcal{E}}(s)),$$

where $\mathbf{t}(s) = \gamma'(s)$ is the tangent vector and $\mathbf{n}_{\mathcal{E}}(s)$ is the ellipsoid normal.

9.3. Railway Surface Frame

Definition 9.4 (Surface railway frame). The local railway frame is defined by

$$\mathbf{t}(s), \quad \mathbf{b}(s) = \mathbf{n}_{\mathcal{E}}(s) \times \mathbf{t}(s), \quad \mathbf{n}_{\mathcal{E}}(s).$$

Remark 9.5. Here, $\mathbf{t}$ is the longitudinal direction, $\mathbf{b}$ is the lateral direction within the surface, and $\mathbf{n}_{\mathcal{E}}$ is the local surface normal.

9.4. Cant as Rotation of the Cross-Section

Definition 9.6 (Cant on the ellipsoid). Cant is represented by a rotation of the cross-section around the tangent vector $\mathbf{t}(s)$:

$$R_{\mathbf{t}}(\phi(s)).$$

9. Pose3 on the Ellipsoid

Remark 9.7. This rotation acts on the lateral-normal plane spanned by

$$\mathbf{b}(s), \quad \mathbf{n}_{\mathcal{E}}(s).$$

Definition 9.8 (Canted lateral direction). The canted lateral direction is

$$\mathbf{b}_{\phi}(s) = R_{\mathbf{t}}(\phi(s)) \mathbf{b}(s).$$

Definition 9.9 (Canted normal direction). The canted normal direction is

$$\mathbf{n}_{\phi}(s) = R_{\mathbf{t}}(\phi(s)) \mathbf{n}_{\mathcal{E}}(s).$$

9.5. Gravity Direction

Remark 9.10. On the ellipsoid, gravity is not identical to the ellipsoid normal in full physical precision.

Definition 9.11 (Gravity vector). Let

$$\mathbf{g}(s)$$

denote the local gravity vector at $\gamma(s)$.

Remark 9.12. In a simplified model, one may approximate

$$\mathbf{g}(s) \parallel -\mathbf{n}_{\mathcal{E}}(s).$$

Remark 9.13. A refined model distinguishes between ellipsoid normal, geoid normal, and local gravity direction.

9.6. Curvature and Lateral Acceleration

Definition 9.14 (Railway curvature on the ellipsoid). The railway-relevant curvature is the geodesic curvature

$$\kappa_{\text{rail}}(s) = \kappa_g(s).$$

Remark 9.15. This curvature measures direction change within the reference surface and replaces the planar relation $\kappa = \theta'$.

Definition 9.16 (Surface lateral acceleration). For speed v , the curvature-induced lateral acceleration is

$$a_{\kappa}(s) = v^2 \kappa_g(s).$$

9.7. Effective Acceleration with Cant

Remark 9.17. The effective lateral acceleration is obtained by projecting both curvature-induced acceleration and gravity into the canted lateral direction.

Definition 9.18 (Effective lateral acceleration). A geometrically consistent formulation is

$$a_{\text{eff}}(s) = v^2 \kappa_g(s) - \mathbf{g}(s) \cdot \mathbf{b}_\phi(s).$$

Remark 9.19. In the planar approximation with

$$\mathbf{g} \parallel -\mathbf{n}_\mathcal{E},$$

this reduces to the familiar expression

$$a_{\text{eff}}(s) = v^2 \kappa(s) - g \sin \phi(s).$$

9.8. Pose3 State on the Ellipsoid

Definition 9.20 (Ellipsoidal pose3). The pose3 state on the ellipsoid is

$$\text{pose3}_\mathcal{E}(s) = (\gamma(s), \mathbf{t}(s), \mathbf{n}_\mathcal{E}(s), \kappa_g(s), \phi(s)).$$

Remark 9.21. Compared to planar pose3, the heading angle $\theta(s)$ is replaced by a tangent vector field on the surface.

9.9. Implication for AIM

Summary 9.22. Pose3 on the ellipsoid replaces planar heading-based geometry by a surface-based moving frame.

Cant becomes a rotation of the track cross-section around the tangent vector, while curvature is represented by geodesic curvature.

This provides a physically more consistent foundation for evaluating railway dynamics, especially where projection distortions become relevant.

10. Core Equations and Canonical Relations

10.1. Motivation

Remark 10.1. The AIM framework is based on a compact set of canonical equations that connect geometry, dynamics, realization, runtime interpretation, and optimization.

Remark 10.2. These equations do not represent isolated formulas, but a coupled state description for railway alignment modelling.

Remark 10.3. The purpose of this chapter is not to derive all relations in detail, but to define the canonical equations repeatedly used throughout AIM.

Remark 10.4. The equations collected here serve as:

- canonical reference relations,
- shared notation anchors,
- and structural building blocks of the AIM framework.

10.2. Canonical State Representation

10.2.1. Intrinsic State

Definition 10.5 (Canonical pose3 state). The canonical intrinsic railway state is:

$$x(s) = (\gamma(s), \theta(s), \phi(s)). \quad (10.1)$$

Remark 10.6. The state combines:

- intrinsic geometry,
- orientation,
- and cant evolution.

10.2.2. Control Variables

Definition 10.7 (Control variables). The canonical control variables are:

$$u(s) = (\kappa(s), \omega(s)), \quad (10.2)$$

10. Core Equations and Canonical Relations

where:

$$\omega(s) = \phi'(s).$$

Remark 10.8. Curvature and cant-rate act as the primary controls of railway state evolution.

10.3. Intrinsic Geometry

10.3.1. Arc-Length State Evolution

Definition 10.9 (State evolution relation). The intrinsic geometric state evolves according to:

$$\gamma'(s) = \mathbf{t}(s). \quad (10.3)$$

Remark 10.10. This relation defines the arc-length parameterization of the alignment.

10.3.2. Curvature Relation

Definition 10.11 (Curvature relation). Directional evolution is governed by:

$$\theta'(s) = \kappa(s). \quad (10.4)$$

Remark 10.12. Curvature is the differential quantity governing directional evolution along the alignment.

10.3.3. Planar Tangent Representation

Definition 10.13 (Planar tangent relation). In the planar case:

$$\mathbf{t}(s) = (\cos \theta(s), \sin \theta(s)). \quad (10.5)$$

Remark 10.14. This is the planar specialization of eq. (10.1).

10.3.4. Canonical Pose2 System

Definition 10.15 (Pose2 system). The canonical planar pose2 system is:

$$\begin{aligned} \gamma'(s) &= \mathbf{t}(s), \\ \theta'(s) &= \kappa(s). \end{aligned} \quad (10.6)$$

Remark 10.16. Together with eq. (10.3), this defines the classical curvature-based planar reconstruction system.

10.4. Cant Evolution

10.4.1. Cant State Equation

Definition 10.17 (Cant evolution). Cant evolution is governed by:

$$\phi'(s) = \omega(s), \quad (10.7)$$

where $\omega(s)$ denotes the cant-rate control.

Remark 10.18. Cant is treated as a state quantity rather than as a secondary annotation.

10.4.2. Canonical Pose3 System

Definition 10.19 (Pose3 system). The canonical pose3 system is:

$$\begin{aligned} \gamma'(s) &= \mathbf{t}(s), \\ \theta'(s) &= \kappa(s), \\ \phi'(s) &= \omega(s). \end{aligned} \quad (10.8)$$

Remark 10.20. The controls of the system are given by ??.

Remark 10.21. This formulation interprets railway alignment as a controlled dynamical state system.

10.4.3. Generalized State Form

Definition 10.22 (Generalized state equation). The pose evolution may be written abstractly as:

$$x'(s) = f(x(s), u(s)). \quad (10.9)$$

Remark 10.23. Within AIM, the canonical controls are:

$$u(s) = (\kappa(s), \omega(s)).$$

10.5. Dynamic Relations

10.5.1. Effective Lateral Acceleration

Definition 10.24 (Effective lateral acceleration). The effective lateral acceleration is:

$$a_{\text{eff}} = v^2 \kappa - g \sin \phi. \quad (10.10)$$

Remark 10.25. This relation expresses the coupling between curvature and cant.

10.5.2. Equilibrium Condition

Definition 10.26 (Equilibrium condition). A perfectly balanced alignment satisfies:

$$v^2 \kappa = g \sin \phi. \quad (10.11)$$

Remark 10.27. This relation defines ideal cant balance.

10.5.3. Jerk Relation

Definition 10.28 (Jerk relation). The jerk is:

$$j = v^3 \kappa' - g v \cos \phi \phi'. \quad (10.12)$$

Remark 10.29. The jerk depends jointly on curvature evolution and cant evolution.

10.6. Realization Relations

10.6.1. Realization Operator

Definition 10.30 (Realization operator). The realization operator maps target states to realized states:

$$\mathcal{R} : \text{pose3}_{\text{target}} \rightarrow \text{pose3}_{\text{real}}. \quad (10.13)$$

Remark 10.31. Realization includes:

- construction,
- machine behaviour,
- discretization,
- and structural response.

10.6.2. Curvature Realization Relation

Definition 10.32 (Curvature realization). At curvature level:

$$\mathcal{R}(\kappa_{\text{target}}) = \kappa_{\text{real}}. \quad (10.14)$$

10.6.3. Inverse Realization Problem

Definition 10.33 (Inverse realization). The inverse realization problem is:

$$\mathcal{R}(\kappa_{\text{target}}) \approx \kappa_{\text{desired}}. \quad (10.15)$$

Remark 10.34. Alignment design therefore becomes an inverse problem under realization constraints.

10.7. World–Track Transformation

10.7.1. Track-to-World Mapping

Definition 10.35 (Track-to-world mapping). The forward transformation is:

$$x_w(s, d) = \gamma(s) + d \mathbf{n}(s). \quad (10.16)$$

Remark 10.36. The vector $\mathbf{n}(s)$ denotes the local lateral direction.

10.7.2. World-to-Track Projection

Definition 10.37 (World-to-track projection). The inverse transformation is:

$$(s^*, d^*) = \arg \min_{s, d} \|x_w - (\gamma(s) + d \mathbf{n}(s))\|. \quad (10.17)$$

Remark 10.38. This is a nonlinear projection problem onto the alignment.

10.8. Ellipsoid Geometry

10.8.1. Curvature Decomposition

Definition 10.39 (Curvature decomposition). The total curvature satisfies:

$$\kappa^2 = \kappa_g^2 + \kappa_n^2. \quad (10.18)$$

Remark 10.40. The geodesic curvature κ_g is the physically relevant curvature for railway dynamics on the Earth surface.

10.8.2. Intrinsic Surface Curvature

Definition 10.41 (Geodesic curvature). The intrinsic surface curvature is:

$$\kappa_g = \|\nabla_s \mathbf{t}\|. \quad (10.19)$$

Remark 10.42. This replaces the purely planar curvature interpretation on curved reference surfaces.

10.9. Optimization Relations

10.9.1. General Optimization Problem

Definition 10.43 (Optimization problem). A general realization-aware optimization problem is:

$$\min_u J(\mathcal{R}(x(u))). \quad (10.20)$$

10. Core Equations and Canonical Relations

Remark 10.44. The objective is evaluated on realized states rather than purely analytical states.

10.9.2. Least-Squares Structure

Definition 10.45 (Least-squares structure). Many AIM optimization problems take the form:

$$\min_x \frac{1}{2} \|r(x)\|^2. \quad (10.21)$$

10.9.3. Gauss–Newton Approximation

Definition 10.46 (Gauss–Newton approximation). The Gauss–Newton approximation is:

$$H \approx J^T J. \quad (10.22)$$

Remark 10.47. This structure appears naturally in:

- curvature fitting,
- realization matching,
- trajectory optimization,
- and runtime adjustment problems.

10.10. Canonical AIM Interpretation

Proposition 10.48. *Within AIM, railway alignment is interpreted as a spatial state trajectory governed by curvature and cant evolution.*

Proposition 10.49. *Curvature, cant, realization, runtime interpretation, and optimization must therefore be treated as coupled components of one unified system.*

10.11. Connection to AIM

Summary 10.50. The equations collected in this chapter form the mathematical backbone of AIM.

They define:

- intrinsic state evolution,
- cant and dynamic relations,
- realization mappings,
- world–track transformations,

10. Core Equations and Canonical Relations

- ellipsoid-aware geometry,
- and optimization structures.

Together, these equations establish a unified mathematical framework connecting geometry, realization, runtime interpretation, and optimization.

11. System Layers

11.1. Motivation

Remark 11.1. Railway alignment modelling combines multiple fundamentally different domains.

Remark 11.2. Classical approaches often merge geometry, realization, coordinate systems, dynamics, and operational interpretation into a single implicit model.

Remark 11.3. Within AIM, these concerns are separated into interacting system layers.

Remark 11.4. This separation improves:

- conceptual clarity,
- numerical robustness,
- scalability,
- interoperability,
- and runtime flexibility.

11.2. Overview

Definition 11.5 (System layer). A system layer is a coherent subsystem operating on its own:

- state representation,
- operators,
- constraints,
- and interpretation rules.

Remark 11.6. The AIM framework consists primarily of:

- geometry layer,
- realization layer,
- runtime layer,

- vehicle layer,
- CRS layer,
- and optimization layer.

11.3. Geometry Layer

11.3.1. Role

Remark 11.7. The geometry layer defines the intrinsic mathematical structure of the alignment.

Remark 11.8. Its central quantities are:

- arc-length s ,
- curvature $\kappa(s)$,
- cant $\phi(s)$,
- and pose states.

11.3.2. Core Equations

Definition 11.9 (Intrinsic geometric relations). The geometry layer is governed by:

$$\theta'(s) = \kappa(s),$$

$$\gamma'(s) = \mathbf{t}(s).$$

11.3.3. Properties

Remark 11.10. The geometry layer is:

- intrinsic,
- continuous,
- analytical,
- and largely independent of realization.

Remark 11.11. It defines the target alignment rather than the physically realized track.

11.4. Realization Layer

11.4.1. Role

Remark 11.12. The realization layer models the transformation from analytical target geometry to physically realized track geometry.

11.4.2. Core Operator

Definition 11.13 (Realization operator).

$$\mathcal{R} : \text{pose3}_{\text{target}} \mapsto \text{pose3}_{\text{real}}.$$

11.4.3. Characteristics

Remark 11.14. The realization layer includes:

- construction constraints,
- machine behaviour,
- discretization,
- mechanical smoothing,
- and maintenance processes.

Remark 11.15. Realization acts approximately as a low-pass filter on geometry and curvature.

11.5. Runtime Layer

11.5.1. Role

Remark 11.16. The runtime layer connects intrinsic railway geometry with operational runtime systems.

11.5.2. Typical Services

Remark 11.17. Typical runtime services include:

- world-track projection,
- visualization,
- BIM placement,
- spatial queries,
- and sensor integration.

11.5.3. Runtime Nature

Remark 11.18. The runtime layer is:

- state-dependent,
- scale-dependent,
- and computationally adaptive.

Remark 11.19. Many operations are evaluated dynamically rather than stored explicitly.

11.6. Vehicle Layer

11.6.1. Role

Remark 11.20. The physically relevant object is not merely the alignment centerline, but the railway vehicle moving within space.

11.6.2. Operational Geometry

Remark 11.21. The vehicle layer includes:

- vehicle envelopes,
- bogie motion,
- wheelset orientation,
- center-of-gravity motion,
- and platform interaction.

11.6.3. Motivation

Remark 11.22. Classical railway alignment methods often approximate operational geometry through simplified clearance assumptions.

Remark 11.23. Within AIM, operational geometry may instead be modelled explicitly.

11.7. CRS Layer

11.7.1. Role

Remark 11.24. The CRS layer defines the relation between railway geometry and the Earth.

11.7.2. Coordinate Systems

Remark 11.25. Typical coordinate systems include:

- engineering CRS,
- projected CRS,
- geodetic coordinates,
- and ellipsoid-based representations.

11.7.3. Fundamental Observation

Proposition 11.26. *A railway alignment is generally not invariant under CRS transformations.*

Remark 11.27. In particular:

- straight lines,
- curvature,
- and transition behaviour

may change under projection.

11.8. Optimization Layer

11.8.1. Role

Remark 11.28. The optimization layer searches for railway states satisfying:

- geometric,
- dynamic,
- operational,
- realization,
- and regulatory constraints.

11.8.2. Realization-Aware Optimization

Remark 11.29. Within AIM, optimization is evaluated on realized states:

$$J(\mathcal{R}(\text{pose3})).$$

11.8.3. Hybrid Nature

Remark 11.30. Optimization combines:

- analytical models,
- sparse representations,
- discrete constraints,
- and runtime evaluation.

11.9. Layer Interactions

Remark 11.31. The system layers interact continuously.

Remark 11.32. For example:

- geometry feeds realization,
- realization affects runtime behaviour,
- runtime supports optimization,
- optimization modifies geometry,
- and CRS influences all spatial interpretation.

11.10. Canonical Transformation Chain

Definition 11.33 (Canonical transformation chain). A simplified AIM processing chain is:

$\text{CRS} \rightarrow \text{geometry} \rightarrow \text{realization} \rightarrow \text{runtime} \rightarrow \text{vehicle behaviour} \rightarrow \text{optimization}.$

Remark 11.34. This chain is generally iterative rather than strictly sequential.

11.11. Scale Dependence

Remark 11.35. Different layers dominate at different spatial and operational scales.

Remark 11.36. For example:

- CRS effects dominate at large scales,
- realization effects dominate at construction scales,
- vehicle-space effects dominate near platforms and switches.

11.12. Key Insight

Proposition 11.37. *Railway alignment modelling is not a single geometric problem, but a multi-layer interaction system.*

Remark 11.38. Many inconsistencies in classical railway modelling arise from implicit mixing of fundamentally different layers.

11.13. Connection to AIM

Summary 11.39. The AIM framework separates railway alignment modelling into interacting system layers.

This separation enables:

- rigorous geometry,
- realization-aware modelling,
- scalable runtime systems,
- operational vehicle interpretation,
- CRS consistency,
- and realization-aware optimization.

12. Ontology and Conceptual Structure

12.1. Motivation

Remark 12.1. Railway alignment modelling combines concepts originating from:

- geometry,
- mechanics,
- surveying,
- construction,
- vehicle dynamics,
- and operational railway engineering.

Remark 12.2. Many inconsistencies in classical systems arise because these concepts are mixed implicitly without a rigorous conceptual structure.

Remark 12.3. Within AIM, a distinction is made between:

- mathematical objects,
- physical objects,
- operational objects,
- and runtime representations.

12.2. Core Ontological Distinction

Definition 12.4 (Ontological layers). The AIM ontology distinguishes between:

1. abstract geometry,
2. realized physical geometry,
3. operational railway behaviour,
4. and computational runtime representation.

Remark 12.5. These layers are related, but not identical.

12.3. Alignment as Abstract Geometry

12.3.1. Alignment Concept

Definition 12.6 (Alignment). An alignment is an intrinsic geometric object primarily defined through:

$$\kappa(s), \quad \phi(s), \quad \text{pose}(s).$$

Remark 12.7. Within AIM, an alignment is not merely a spatial curve.

Remark 12.8. Instead, it represents a structured operational geometry along arc-length.

12.3.2. Intrinsic Nature

Remark 12.9. The alignment is:

- intrinsic,
- arc-length-based,
- and independent of global coordinates.

12.4. Target Geometry vs Realized Geometry

12.4.1. Target Geometry

Definition 12.10 (Target geometry). The target geometry is the intended analytical railway state defined by the design process.

12.4.2. Realized Geometry

Definition 12.11 (Realized geometry). The realized geometry is the physically existing railway geometry after construction and maintenance.

12.4.3. Fundamental Distinction

Proposition 12.12. *In general:*

$$\text{pose3}_{\text{real}} \neq \text{pose3}_{\text{target}}.$$

Remark 12.13. This distinction is fundamental within AIM.

12.5. Geometry vs Vehicle Space

12.5.1. Classical Simplification

Remark 12.14. Classical railway alignment systems often reduce the railway problem to a centerline geometry problem.

12.5.2. Operational Reality

Remark 12.15. The operationally relevant object, however, is the railway vehicle moving within space.

Remark 12.16. Important operational quantities include:

- center-of-gravity motion,
- vehicle envelope,
- bogie rotation,
- wheelset orientation,
- and platform interaction.

12.5.3. Consequence

Proposition 12.17. *The railway alignment is operationally meaningful only together with its vehicle-space interpretation.*

12.6. Pose, Frame, and State

12.6.1. Pose

Definition 12.18 (Pose). A pose describes a local geometric state attached to arc-length position s .

12.6.2. Frame

Definition 12.19 (Frame). A frame defines a local coordinate interpretation of the pose.

12.6.3. State

Definition 12.20 (State). A state combines:

- geometry,
- orientation,
- and operational interpretation.

12.6.4. Runtime Nature

Remark 12.21. Within AIM, states are runtime-evaluated rather than purely static data objects.

12.7. World Space vs Track Space

12.7.1. Track Space

Definition 12.22 (Track space). Track space is the intrinsic coordinate system attached to the alignment.

12.7.2. World Space

Definition 12.23 (World space). World space is a global spatial coordinate system defined through a CRS.

12.7.3. Transformation

Remark 12.24. The transformation:

$$\text{world} \leftrightarrow \text{track}$$

is nonlinear and context-dependent.

12.8. CRS and Earth Geometry

12.8.1. Projection Dependence

Remark 12.25. Railway geometry depends fundamentally on the underlying coordinate reference system.

12.8.2. Non-Invariance

Proposition 12.26. *Railway alignments are generally not invariant under CRS changes.*

12.8.3. Ellipsoid Reality

Remark 12.27. The physically realized railway exists on the Earth surface rather than within a projected Euclidean plane.

12.9. Runtime Ontology

12.9.1. Runtime Services

Remark 12.28. Many AIM objects exist operationally as runtime services rather than stored static entities.

12.9.2. Examples

Remark 12.29. Examples include:

- world-track projection,
- frame evaluation,
- local coordinate systems,
- and vehicle-space interpretation.

12.10. Realization Ontology

12.10.1. Construction Process

Remark 12.30. The physically realized railway emerges from:

- measurement,
- machine operation,
- maintenance,
- and mechanical response.

12.10.2. Iterative Nature

Remark 12.31. Realization is therefore not a static mapping, but an iterative physical process.

12.11. Optimization Ontology

12.11.1. Classical Optimization

Remark 12.32. Classical optimization often minimizes purely analytical target states.

12.11.2. AIM Optimization

Remark 12.33. Within AIM, optimization operates on:

- realized states,
- operational behaviour,
- and runtime-evaluated constraints.

12.11.3. Inverse Nature

Remark 12.34. Many railway optimization problems are inverse realization problems.

12.12. Sparse vs Runtime Reality

12.12.1. Sparse Representation

Definition 12.35 (Sparse representation). A sparse representation stores only the essential alignment-defining parameters.

12.12.2. Runtime Expansion

Remark 12.36. Operational runtime systems reconstruct:

- geometry,
- frames,
- vehicle-space behaviour,
- and projections

from sparse intrinsic data.

12.13. Fundamental AIM View

Proposition 12.37. *A railway alignment is not merely a geometric curve.*

Remark 12.38. It is a multi-layer operational system connecting:

- intrinsic geometry,
- realization,
- Earth geometry,
- vehicle-space behaviour,
- runtime services,
- and optimization.

12.14. Key Insight

Proposition 12.39. *Many limitations of classical railway systems arise from implicit collapsing of fundamentally different ontological layers into a single geometric abstraction.*

12.15. Connection to AIM

Summary 12.40. The AIM ontology separates:

- target geometry,
- realized geometry,
- operational vehicle behaviour,
- runtime representations,
- and Earth-related coordinate systems.

This separation forms the conceptual foundation for realization-aware, vehicle-aware, and Earth-aware railway alignment modelling.

Teil III.

Railway State Model

Remark 12.41. This part introduces the railway state model underlying the AIM framework. Instead of treating the track merely as a geometric curve, the state model captures position, orientation, curvature, and cant in a unified representation.

Remark 12.42. The purpose of this part is to define the variables and transformations required to connect intrinsic alignment modelling with dynamic railway interpretation.

Remark 12.43. This establishes the transition from purely geometric description towards a state-based view of railway alignment.

13. Pose2

13.1. Motivation

Remark 13.1. In railway alignment modelling, the geometric state of the track at a position along its length must be described in a minimal and consistent way.

Remark 13.2. While full spatial representation involves three-dimensional geometry, cant, and dynamic interpretation, many fundamental properties can already be captured in a planar setting.

Remark 13.3. The concept of *Pose2* provides such a minimal representation in two dimensions.

Remark 13.4. Within AIM, Pose2 is not merely a geometric helper structure. It forms the persistent reconstruction kernel of the alignment model.

13.2. Definition of Pose2

Definition 13.5 (Pose2). A planar pose at arc-length position s is defined as the tuple

$$\text{Pose2}(s) = (\gamma(s), \theta(s)),$$

where:

- $\gamma(s) = (x(s), y(s))$ is the planar position,
- $\theta(s)$ is the tangent direction.

Remark 13.6. Pose2 describes the local geometric state of the alignment in the plane.

13.3. Curvature-Driven Reconstruction

Principle 13.7 (Curvature-driven reconstruction). Within AIM, planar geometry is reconstructed from curvature by successive integration:

$$\kappa(s) \rightarrow \theta(s) \rightarrow \gamma(s).$$

13. Pose2

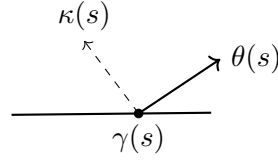


Abbildung 13.1.: Planar geometry emerges by successive integration of curvature. Within AIM, curvature is treated as the primary geometric quantity.

13.4. Kinematic Relations

Remark 13.8. The evolution of Pose2 is governed by curvature.

Proposition 13.9. *The tangent direction satisfies:*

$$\frac{d\theta}{ds} = \kappa(s).$$

Proposition 13.10. *The planar position satisfies:*

$$\frac{d\gamma}{ds} = \begin{pmatrix} \cos \theta(s) \\ \sin \theta(s) \end{pmatrix}.$$

Remark 13.11. Thus, geometry is not treated as primary. Instead, it emerges from integration of curvature.

13.5. Initial Conditions

Definition 13.12 (Initial pose). A Pose2 trajectory is uniquely determined by:

$$\gamma(0), \quad \theta(0),$$

together with the curvature law

$$\kappa(s).$$

Remark 13.13. The curvature law acts as the geometric control function of the planar state evolution.

13.6. Pose2 as Persistent Kernel

Proposition 13.14. *Within AIM, Pose2 forms the persistent geometric reconstruction kernel of the alignment model.*

Remark 13.15. Pose2 is sufficient for persistent planar reconstruction of the alignment geometry.

Remark 13.16. Higher-level geometric and dynamic quantities are derived from Pose2 rather than stored redundantly.

13.7. Connection to Sparse Alignment

Remark 13.17. Within the sparse alignment model, cGeom elements are interpreted as local curvature operators contributing to the reconstruction of the global Pose2 trajectory.

Remark 13.18. Fixed and transition elements therefore do not primarily store geometry itself, but rules governing geometric evolution.

Remark 13.19. The sparse alignment model can thus be interpreted as a compact integration program for reconstructing Pose2 trajectories.

13.8. Interpretation

Remark 13.20. Pose2 separates:

- intrinsic geometric behaviour (curvature),
- from derived geometric quantities (position and direction).

Remark 13.21. This separation is one of the central structural ideas of the AIM framework.

13.9. Relation to Pose3

Remark 13.22. Within AIM, Pose3 is not treated as a persistent extension of Pose2, but as a runtime-derived spatial evaluation state.

Remark 13.23. Pose3 is derived dynamically from:

- Pose2 reconstruction,
- cant evaluation,
- profile evaluation,
- and runtime coordinate services.

Remark 13.24. This avoids redundancy and synchronization instability between planar, cant, and profile representations.

13.10. Summary

Summary 13.25. Pose2 is the fundamental planar state representation of railway alignment.

Within AIM, Pose2 forms the persistent reconstruction kernel from which planar geometry is generated through curvature-driven integration.

Higher-dimensional runtime states such as Pose3 are derived from this kernel together with cant and profile information.

14. Pose3, Cant, and Spatial Railway State

14.1. Motivation

Remark 14.1. Classical railway alignment models are based on planar geometry. They describe the track centerline as a curve

$$\gamma : s \mapsto (x(s), y(s)).$$

Remark 14.2. However, real railway systems are inherently three-dimensional. In particular, the rotation of the track cross-section, commonly described through cant, plays a fundamental role for vehicle dynamics and operational behaviour [24, 6].

Remark 14.3. The pose3 formulation also opens the possibility of describing the motion of physically relevant vehicle reference points, such as the vehicle centre of gravity, instead of modelling only the track centerline.

Such ideas have occasionally appeared in advanced railway engineering concepts and patent literature [12].

Remark 14.4. This motivates the introduction of an extended runtime state description, referred to as *pose3*, which augments planar alignment geometry by vertical and rotational information.

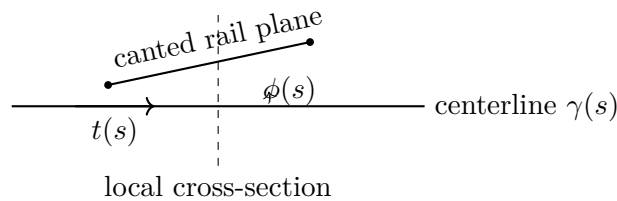


Abbildung 14.1.: Pose3 extends the planar centerline state by vertical and rotational information. In AIM, the roll angle $\phi(s)$ is derived from cant and gauge rather than stored as persistent truth.

14.2. Pose2 Revisited

Definition 14.5 (Pose2). The planar pose of a curve is defined as

$$\text{pose2}(s) = (\gamma(s), \theta(s)),$$

14. Pose3, Cant, and Spatial Railway State

where

$$\gamma(s) \in \mathbb{R}^2, \quad \theta(s) \in \mathbb{R}.$$

Remark 14.6. The evolution of pose2 is governed by

$$\theta'(s) = \kappa(s), \quad \gamma'(s) = (\cos \theta(s), \sin \theta(s)).$$

Remark 14.7. This is the planar reconstruction principle underlying the cGeom band of the sparse alignment model.

14.3. Cant as Superelevation Band

Definition 14.8 (Cant). Within AIM, cant is represented as a rail height difference

$$u(s),$$

not as a stored roll angle.

Remark 14.9. This follows railway engineering practice, where cant is commonly given as height difference between rails.

Definition 14.10 (Derived roll angle). Given a gauge value g_{auge} , the corresponding roll angle is derived by

$$\phi(s) = \arctan\left(\frac{u(s)}{g_{\text{auge}}}\right).$$

Remark 14.11. Thus, the roll angle depends on both cant and gauge. It is therefore a derived state quantity rather than persistent cant data.

Remark 14.12. The cant angle influences lateral acceleration, cant deficiency, and load distribution [24, 6].

14.3.1. Cant Reference Axis and Centerline Convention

Remark 14.13. In standard railway handling, cant is often specified by rotating the track plane around a relevant rail or by defining a height difference between rails.

Remark 14.14. The AIM state model uses the centerline point as the primary reference point of the alignment. Thus, cant must be translated into a centerline-based representation before it is coupled to profile, pose3, and optimization.

Definition 14.15 (Cant reference convention). A cant convention specifies:

- the reference axis of rotation,
- the sign convention of cant,
- the relation between rail height difference and roll angle,

14. Pose3, Cant, and Spatial Railway State

- and the induced vertical offset of the centerline.

Remark 14.16. This distinction is relevant in special cases such as track scissors, cant ramps, and overlapping transition zones, where the cant definition may influence the effective profile or gradient assigned to the centerline.

Remark 14.17. For implementation, AIM treats cant not merely as an isolated value, but as a band whose reference convention must be known when transforming between rail-based construction data and centerline-based modelling.

Definition 14.18 (Cant rate). The derivative

$$\phi'(s)$$

describes the rate of change of the derived roll angle along the track.

Remark 14.19. Cant rate is not merely a geometric quantity, but an engineering quantity constrained by operational and comfort-related requirements [6].

14.4. Persistent Model versus Runtime State

Principle 14.20 (Runtime pose3 principle). Pose3 is not persistent truth within AIM. It is a runtime-derived spatial state.

Remark 14.21. The persistent sparse model stores:

- cGeom as pose2-driven curvature geometry,
- cant as a longitudinal superelevation band,
- profile as a separate height band.

Remark 14.22. Pose3 is derived dynamically from these bands by runtime evaluation services.

Remark 14.23. This avoids redundant storage of three-dimensional points, roll angles, and derived spatial frames. It also avoids synchronization instability between geometry, cant, and profile.

Summary 14.24. The intelligence of the AIM state model does not reside in a stored pose3 object, but in the mathematical services that evaluate pose3 from sparse longitudinal bands.

14.5. Pose3 Definition

Definition 14.25 (Runtime Pose3). The spatial railway state is defined as

$$\text{pose3}(s) = (\Gamma(s), \mathbf{t}(s), \phi(s)),$$

where:

14. Pose3, Cant, and Spatial Railway State

- $\Gamma(s)$ is the reconstructed three-dimensional centerline point,
- $\mathbf{t}(s)$ is the spatial tangent direction,
- $\phi(s)$ is the derived roll angle.

Remark 14.26. In the projected engineering setting, $\Gamma(s)$ is obtained from planar pose2 and profile evaluation.

Remark 14.27. Pose3 captures both geometric and dynamically relevant aspects of the track, but remains a derived runtime state.

Remark 14.28. When a rail-based cant convention is used in source data, the centerline-consistent state must be resolved during import, normalization, or model construction.

14.6. State Evolution

Proposition 14.29 (Projected pose3 evolution). *In a projected engineering coordinate system, the runtime pose3 state is derived from:*

$$\begin{aligned}\gamma'(s) &= (\cos \theta(s), \sin \theta(s)), \\ \theta'(s) &= \kappa(s), \\ z(s) &= p(s), \\ \phi(s) &= \arctan\left(\frac{u(s)}{g_{\text{gauge}}}\right),\end{aligned}$$

where $p(s)$ denotes profile height and $u(s)$ denotes cant as rail height difference.

Remark 14.30. Thus, pose3 is not governed by one monolithic state equation. It is assembled from synchronized evaluations of cGeom, profile, and cant.

14.7. Reconstruction Principle

Proposition 14.31. *Given:*

$$\kappa(s), \quad p(s), \quad u(s),$$

together with initial pose2 data and gauge information, the runtime pose3 state is determined by evaluation.

Remark 14.32. This differs from storing pose3 directly. The sparse model stores the ingredients; pose3 is reconstructed when needed.

14.8. Relation to Physical Quantities

14.8.1. Effective Lateral Acceleration

Definition 14.33. In the planar engineering approximation, effective lateral acceleration is given by

$$a_{\text{eff}}(s) = v^2 \kappa(s) - g \sin \phi(s).$$

Remark 14.34. This quantity determines passenger comfort and safety and expresses the basic coupling between curvature and cant in operational railway design [24, 6].

14.8.2. Jerk

Definition 14.35. The jerk is given by

$$j(s) = v^3 \kappa'(s) - gv \cos \phi(s) \phi'(s).$$

Remark 14.36. The jerk measures the rate of change of acceleration and is a key criterion for comfort. This is one of the main reasons why curvature and cant development cannot be specified independently in high-quality alignment design [34, 27, 28].

14.9. Interpretation as Evaluation System

Remark 14.37. The pose3 model can be interpreted as an evaluation system:

persistent bands: cGeom, u , p ,
runtime state: pose3.

Remark 14.38. This formulation provides a natural bridge to simulation, visualization, validation, and optimization methods.

Remark 14.39. It is also consistent with the broader optimization-oriented view of alignment modelling proposed in railway alignment research [31, 30].

14.10. Cant and Curvature Coupling

Remark 14.40. Curvature and cant are not independent in engineering interpretation. They jointly determine dynamic behaviour.

Definition 14.41 (Equilibrium condition). A perfectly balanced design satisfies

$$v^2 \kappa(s) = g \sin \phi(s).$$

Remark 14.42. In practice, deviations from this condition are allowed but must remain within regulatory limits. This is the mathematical expression of the cant-balance logic used in railway design standards [6, 24].

Remark 14.43. The persistent model keeps cGeom and cant as separate synchronized bands. Their coupling is evaluated dynamically rather than stored redundantly.

14.11. Towards Schwerpunkt-Trassierung

Remark 14.44. The pose3 formulation enables a reinterpretation of alignment design. Instead of designing geometry alone, one designs a spatial state trajectory.

Remark 14.45. In particular, the concept of *Schwerpunkt-Trassierung* can be understood as the design of a trajectory that optimally balances curvature, cant, and dynamic response.

Remark 14.46. In the sparse AIM model, cant may remain coupled to the normalized curvature family, while curvature itself may receive additional correction terms for advanced modelling concepts.

RESEARCH: Formal definition of Schwerpunkt-based alignment design.

OPEN: Integration of vehicle dynamics into the pose3 framework.

14.12. Relation to Existing Standards

Remark 14.47. Existing railway design standards typically prescribe:

- limits on curvature,
- limits on cant,
- limits on cant gradient.

Remark 14.48. These rules can be interpreted as constraints on the runtime pose3 state and its derivatives [6].

Remark 14.49. From this perspective, pose3 does not contradict existing railway standards, but provides a mathematical structure in which such rules can be represented coherently.

Remark 14.50. The purpose of the pose3 formulation is not to replace such standards, but to make their geometric consequences explicit. In particular, rail-based cant rules must be converted consistently when the internal model uses the centerline as its reference geometry.

14.13. Connection to the AIM Framework

Summary 14.51. Pose3 extends planar alignment modelling to spatial railway state evaluation.

Within AIM, pose3 is not persistent truth. It is derived at runtime from cGeom, cant, profile, and coordinate services.

This provides:

- a unified runtime interpretation of geometry and dynamics,

14. Pose3, Cant, and Spatial Railway State

- a natural interface to validation and optimization methods,
- a foundation for advanced concepts such as Schwerpunkt-Trassierung.

15. Dynamics and Railway State Interpretation

15.1. Motivation

Remark 15.1. Geometric alignment alone is not sufficient to describe railway behaviour. The motion of a vehicle depends on both curvature and cant, as well as on operational parameters such as speed.

Remark 15.2. This motivates a dynamic interpretation of alignment in which geometric quantities are translated into physical effects. Such a perspective is consistent with both railway design standards and modern railway vehicle dynamics literature, where curvature, cant, and vehicle response are treated as strongly coupled aspects of the system [6, 4, 16, 49].

15.2. Kinematic Basis

Definition 15.3 (Velocity). Let $v > 0$ denote the velocity of the vehicle along the alignment.

Definition 15.4 (Curvature-induced acceleration). For a planar curve with curvature $\kappa(s)$, the lateral acceleration is

$$a_\kappa(s) = v^2 \kappa(s).$$

Remark 15.5. This is the classical centripetal acceleration associated with curved motion and forms the basic link between geometry and dynamics.

15.3. Cant and Effective Acceleration

Definition 15.6 (Cant angle). Let $\phi(s)$ denote the local roll angle induced by cant.

Definition 15.7 (Gravity component). The gravitational acceleration projected onto the lateral direction is

$$a_g(s) = g \sin \phi(s),$$

where g denotes gravitational acceleration.

Definition 15.8 (Effective lateral acceleration). The effective lateral acceleration acting on the vehicle is

$$a_{\text{eff}}(s) = v^2 \kappa(s) - g \sin \phi(s).$$

Remark 15.9. This expression captures the fundamental coupling between curvature and cant in railway engineering, where cant is introduced precisely to compensate part of the curvature-induced lateral acceleration [24, 6].

Remark 15.10. From a vehicle-dynamics perspective, this coupling is one of the central mechanisms governing running behaviour in curves [16, 49].

15.4. Equilibrium Condition

Definition 15.11 (Equilibrium cant). A state of equilibrium is achieved when

$$v^2 \kappa(s) = g \sin \phi(s).$$

Remark 15.12. In this case, the effective lateral acceleration vanishes:

$$a_{\text{eff}}(s) = 0.$$

Remark 15.13. This condition corresponds to the classical notion of balanced track, where centrifugal and gravitational effects cancel. The same general idea underlies standard cant and cant-deficiency formulations in railway design [6, 24].

15.5. Cant Deficiency and Excess

Definition 15.14 (Cant deficiency). Cant deficiency is defined as

$$\Delta(s) = v^2 \kappa(s) - g \sin \phi(s).$$

Remark 15.15. Positive values correspond to insufficient cant, negative values to excess cant.

Remark 15.16. Cant deficiency is a central design parameter in railway engineering, as it directly influences passenger comfort and safety [6, 4].

Remark 15.17. In more detailed vehicle-based interpretations, cant deficiency is also closely related to guiding forces, wheel unloading, and curving behaviour [16, 32].

15.6. Dynamic Smoothness

Definition 15.18 (Jerk). The rate of change of effective acceleration is given by

$$j(s) = \frac{d}{dt} a_{\text{eff}}(s).$$

Remark 15.19. Using $dt = \frac{1}{v} ds$, one obtains

$$j(s) = v \frac{d}{ds} a_{\text{eff}}(s).$$

Proposition 15.20. *The jerk is given by*

$$j(s) = v^3 \kappa'(s) - gv \cos \phi(s) \phi'(s).$$

Remark 15.21. This shows that both curvature variation and cant variation contribute to dynamic effects. Such higher-order quantities are directly relevant for comfort-oriented transition design [34, 27, 28].

Remark 15.22. In railway vehicle dynamics, smoothness criteria of this kind are important because abrupt changes in curvature or cant excite the vehicle-track system beyond what is visible from geometry alone [25, 16].

15.7. Time Parameterization

Remark 15.23. The alignment is parameterized by arc length s , while dynamic quantities evolve in time t .

Definition 15.24 (Arc-length/time relation). Assuming constant speed $v > 0$, we have

$$\frac{ds}{dt} = v.$$

Proposition 15.25. *For any function $f(s)$, the time derivative is given by*

$$\frac{df}{dt} = v \frac{df}{ds}.$$

Remark 15.26. This relation allows all spatial derivatives to be translated into temporal dynamics and provides the mathematical bridge between arc-length-based alignment modelling and time-based vehicle response.

15.8. Dynamic System Formulation

Definition 15.27 (State variables). The railway state is described by

$$x(s), y(s), \theta(s), \phi(s).$$

Proposition 15.28 (Kinematic system). *The system satisfies*

$$\begin{aligned} \frac{dx}{ds} &= \cos \theta(s), & \frac{dy}{ds} &= \sin \theta(s), \\ \frac{d\theta}{ds} &= \kappa(s). \end{aligned}$$

Remark 15.29. These equations correspond to the geometric reconstruction of the curve.

Definition 15.30 (Extended dynamic state). The extended state includes cant:

$$(\gamma(s), \theta(s), \phi(s)).$$

Remark 15.31. The functions $\kappa(s)$ and $\phi(s)$ act as control variables.

Remark 15.32. This formulation is compatible with reduced-order vehicle–track interpretations in which track geometry provides structured inputs to a dynamic system rather than merely static shape data [16, 43].

15.9. Time-Based Dynamics

Proposition 15.33. *In time parameterization, the heading angle satisfies*

$$\frac{d\theta}{dt} = v\kappa(s).$$

Proposition 15.34. *The lateral acceleration is given by*

$$a_\kappa(t) = v^2\kappa(s).$$

Remark 15.35. This formulation makes explicit that a purely geometric quantity, namely curvature, acquires direct physical meaning once speed is introduced.

15.10. Higher-Order Dynamics

Definition 15.36 (Jerk). The jerk is defined as

$$j(t) = \frac{d}{dt}a_{\text{eff}}(t).$$

Proposition 15.37. *The jerk is given by*

$$j(t) = v^3\kappa'(s) - gv \cos \phi(s) \phi'(s).$$

Beweis. Using

$$a_{\text{eff}} = v^2\kappa(s) - g \sin \phi(s),$$

we obtain

$$\frac{d}{dt}a_{\text{eff}} = v^2 \frac{d\kappa}{dt} - g \cos \phi(s) \frac{d\phi}{dt}.$$

With

$$\frac{d}{dt} = v \frac{d}{ds},$$

this yields

$$j(t) = v^3\kappa'(s) - gv \cos \phi(s) \phi'(s).$$

□

Remark 15.38. This explicit expression shows why transition design cannot be reduced to curvature alone if cant development is part of the system model.

15.11. Control-Theoretic Interpretation

Definition 15.39 (Control system). The railway alignment can be interpreted as a control system with

$$\text{state: } (\gamma, \theta, \phi), \quad \text{controls: } (\kappa, \phi').$$

Remark 15.40. The curvature $\kappa(s)$ controls the geometric evolution, while $\phi(s)$ controls the dynamic balancing of forces.

Remark 15.41. This formulation establishes a direct connection between alignment design and optimal control theory, and is consistent with the broader optimization-oriented interpretation of railway alignment proposed by Kufver [31, 30].

15.12. Coupled Dynamics

Remark 15.42. The effective acceleration depends on both curvature and cant:

$$a_{\text{eff}} = v^2 \kappa - g \sin \phi.$$

Remark 15.43. Thus, curvature and cant must be treated as a coupled system rather than independent design variables.

Proposition 15.44. *Perfect equilibrium requires*

$$v^2 \kappa(s) = g \sin \phi(s).$$

Remark 15.45. This coupling is also central from the vehicle–track interaction point of view, since the dynamic response depends on the joint evolution of track geometry, cant, and operating speed [25, 32].

15.13. Coupled State Interpretation

Remark 15.46. The expressions above show that railway behaviour depends on the coupled state variables

$$(\kappa(s), \phi(s)).$$

Remark 15.47. A purely geometric interpretation based on $\kappa(s)$ alone is therefore insufficient for describing the physical behaviour of the system. This is one of the core reasons why a state-based railway model is preferable to a purely planar alignment description [4, 6, 16].

15.14. Extension to Pose3

Remark 15.48. Within the pose3 framework, the railway state is described by

$$(x(s), y(s), z(s), \theta(s), \phi(s)).$$

Remark 15.49. This representation naturally integrates:

- geometry (position and direction),
- vertical profile,
- cant (through ϕ),
- and dynamic interpretation.

Remark 15.50. The extension from planar pose to spatial railway state is therefore not a cosmetic modification, but a structural requirement for coupling alignment description to engineering dynamics.

15.15. Relation to Optimization

Remark 15.51. The dynamic quantities introduced above provide natural objective terms for alignment optimization.

Remark 15.52. Typical optimization criteria include:

- minimizing $\int a_{\text{eff}}(s)^2 ds$,
- minimizing $\int j(s)^2 ds$,
- limiting peak values of cant deficiency.

Such objective terms arise naturally once alignment is interpreted as a coupled geometric–dynamic system rather than as static geometry alone [30, 16].

15.16. Engineering Interpretation

Remark 15.53. In practical railway design, acceptable ranges for cant, cant deficiency, and jerk are prescribed by standards and operational guidelines.

Remark 15.54. The mathematical formulation above provides a unified framework for expressing these rules in analytic form [6, 24, 48].

Remark 15.55. At a more detailed level, such limits are closely related to the running–safety and vehicle–track compatibility questions discussed in railway dynamics literature [48, 16, 49].

15.17. Connection to Schwerpunkt-Trassierung

Remark 15.56. The combined dependence on curvature and cant supports a Schwerpunkt-oriented view of alignment.

Remark 15.57. Rather than describing only a geometric path, alignment is interpreted as a controlled dynamic state along the track.

Remark 15.58. This opens the way toward formulations in which vehicle-relevant derived motions, rather than geometry alone, become the true design object.

15.18. Summary

Summary 15.59. Railway alignment must be interpreted dynamically rather than purely geometrically.

The coupled variables $\kappa(s)$ and $\phi(s)$ determine lateral acceleration, comfort, and operational feasibility.

This provides the foundation for extending alignment modelling toward state-based and optimization-based frameworks.

16. World–Track Coordinate Transformation

16.1. Motivation

Remark 16.1. Railway alignments are formulated intrinsically using arc-length-based coordinates.

Remark 16.2. Real-world data, however, is typically represented in external coordinate reference systems such as:

- WGS84,
- UTM,
- Gauss–Krüger,
- or local engineering CRS definitions.

Remark 16.3. A transformation between intrinsic alignment coordinates and external world coordinates is therefore required.

Remark 16.4. Within AIM, this transformation is interpreted as a runtime evaluation service connecting:

- intrinsic alignment states,
- measured data,
- visualization,
- realization,
- and operational geometry.

16.2. Track Coordinate System

Definition 16.5 (Track coordinate system). The intrinsic track coordinate system consists of:

$$(s, d, h),$$

where:

- s denotes arc-length,

16. World–Track Coordinate Transformation

- d denotes lateral offset,
- h denotes vertical offset.

Remark 16.6. In planar situations, only:

$$(s, d)$$

is required.

Remark 16.7. Track coordinates are intrinsic alignment coordinates and must not be confused with projected world coordinates.

16.2.1. Arc-Length and Stationing

Remark 16.8. Intrinsic arc-length is not identical to engineering stationing.

Remark 16.9. Operational kilometre systems may contain:

- station equations,
- kilometre jumps,
- overlapping station ranges,
- and topology-dependent references.

Remark 16.10. Thus, stationing systems are operational reference structures layered on top of intrinsic geometry.

16.3. Track-to-World Transformation

Definition 16.11 (Track-to-world mapping). The forward transformation is:

$$x(s, d) = \gamma(s) + d \mathbf{n}(s). \quad (16.1)$$

Remark 16.12. This corresponds to the canonical relation eq. (10.14).

16.3.1. Planar Tangent and Normal Directions

Remark 16.13. In the planar case:

$$\mathbf{t}(s) = (\cos \theta(s), \sin \theta(s)),$$

and:

$$\mathbf{n}(s) = (-\sin \theta(s), \cos \theta(s)).$$

16.3.2. Reference-Line Interpretation

Remark 16.14. The lateral coordinate d depends on the chosen reference interpretation.

Remark 16.15. Possible reference structures include:

- alignment centerline,
- track center,
- rail axis,
- or operational offset definitions.

OPEN: Formalize reference-line conventions for:

- multi-track systems,
- cant-dependent offsets,
- and vehicle-space interpretation.

16.4. World-to-Track Transformation

Definition 16.16 (World-to-track projection). The inverse transformation is:

$$(s^*, d^*) = \arg \min_{s, d} \|x - (\gamma(s) + d \mathbf{n}(s))\|. \quad (16.2)$$

Remark 16.17. This corresponds to the canonical relation eq. (10.15).

Remark 16.18. The term projection is used operationally and does not necessarily imply a unique orthogonal projection in the classical geometric sense.

16.5. Numerical Evaluation

Remark 16.19. The inverse transformation is generally nonlinear.

Remark 16.20. Typical numerical approaches include:

- nearest-point search,
- local Newton iteration,
- segment-wise projection,
- and hybrid lookup strategies.

16.6. Ambiguity and Context Dependence

Remark 16.21. World-to-track mapping is generally not globally unique.

Remark 16.22. Ambiguities may occur in:

- loops,
- switches and crossings,
- parallel tracks,
- closely spaced alignments,
- and station areas.

Remark 16.23. Therefore, runtime projection often requires contextual information such as:

- track identity,
- topology,
- expected station range,
- operational state,
- or interaction history.

16.7. Relation to Curvature

Remark 16.24. Projection stability depends directly on curvature behaviour.

Remark 16.25. High curvature increases projection sensitivity, whereas low curvature typically yields stable projection behaviour.

Remark 16.26. Errors in intrinsic coordinates propagate nonlinearly into world space.

16.8. Runtime Dependency

Remark 16.27. Within AIM, world–track transformation depends on runtime evaluation of the underlying alignment representation.

Remark 16.28. In particular:

- cGeom defines the intrinsic geometry,
- pose2 defines tangent evolution,
- pose3 extends spatial interpretation,
- and runtime operators evaluate resulting states.

Remark 16.29. Thus, world–track transformation is part of the runtime evaluation architecture rather than an isolated geometric utility.

16.9. LOD and Hybrid Evaluation

Remark 16.30. Exact world-to-track projection is computationally expensive.

Remark 16.31. Large-scale systems therefore require:

- LOD-dependent evaluation,
- approximate projection,
- and hybrid lookup strategies.

Proposition 16.32 (Selective projection principle). *Exact projection should only be evaluated where high precision is required.*

Remark 16.33. A hybrid strategy combines:

- precomputed samples for coarse lookup,
- and local parametric refinement for precision.

16.10. Connection to CRS and Realization

Remark 16.34. The complete transformation chain is:

$$\text{CRS} \rightarrow \text{world space} \rightarrow \text{track space}.$$

Remark 16.35. For high-precision applications, the alignment itself may be defined on a reference ellipsoid rather than in a projected plane.

Remark 16.36. Projection also depends on realized geometry:

$$\gamma_{\text{real}}(s) \neq \gamma_{\text{target}}(s).$$

Remark 16.37. Physical measurements should therefore reference realized rather than purely analytical geometry.

16.11. Key Insight

Proposition 16.38. *World–track transformation is an inverse runtime service rather than a static coordinate conversion.*

Proposition 16.39. *Within AIM, intrinsic alignment space is primary, whereas world coordinates are interpreted as external embeddings of alignment states.*

16.12. Connection to AIM

Summary 16.40. World–track transformation connects intrinsic railway geometry with external coordinate systems and measured reality.

Its nonlinear, ambiguous, and scale-dependent nature makes it a central runtime component of the AIM framework.

Efficient handling therefore requires hybrid and LOD-dependent runtime strategies operating directly on sparse alignment structures.

17. Pose3 Frames and Spatial Reference Systems

17.1. Motivation

Remark 17.1. The pose3 state describes railway geometry together with orientation, cant, and spatial interpretation.

Remark 17.2. The state variables alone do not fully determine how the alignment is interpreted in space. For this, local reference frames are required.

Remark 17.3. Within AIM, frames form the operational bridge between:

- intrinsic geometry,
- cant,
- dynamics,
- realization,
- world-track projection,
- and vehicle-space interpretation.

17.2. Pose3 and Frames

Remark 17.4. The canonical pose3 state is defined in ??.

Remark 17.5. A pose3 frame provides the local spatial interpretation of this state. It is derived from the alignment and evaluated along arc-length.

17.3. Tangent Direction

Definition 17.6 (Tangent vector). The tangent vector is:

$$\mathbf{t}(s) = \gamma'(s).$$

Remark 17.7. For arc-length parameterization,

$$\|\mathbf{t}(s)\| = 1.$$

Remark 17.8. The tangent direction defines the local forward direction of motion.

17.4. Lateral and Binormal Directions

Definition 17.9 (Lateral direction). The lateral direction is denoted by:

$$\mathbf{n}(s),$$

with

$$\mathbf{n}(s) \cdot \mathbf{t}(s) = 0.$$

Definition 17.10 (Binormal direction). The binormal direction is:

$$\mathbf{b}(s) = \mathbf{t}(s) \times \mathbf{n}(s).$$

Remark 17.11. The lateral and binormal directions define the local cross-section orientation of the railway frame.

17.5. Pose3 Frame

Definition 17.12 (Pose3 frame). The local pose3 frame is:

$$F(s) = (\mathbf{t}(s), \mathbf{n}(s), \mathbf{b}(s)).$$

Remark 17.13. The pose3 frame defines the local operational coordinate system of the railway alignment.

Remark 17.14. It is a runtime-evaluated structure rather than an independently stored geometric object.

17.6. Cant Rotation

Definition 17.15 (Cant rotation). Cant is interpreted as a rotation around the tangent direction:

$$R_\phi(s).$$

Definition 17.16 (Cant-rotated frame). The cant-rotated frame is:

$$F_\phi(s) = (\mathbf{t}(s), \mathbf{n}_\phi(s), \mathbf{b}_\phi(s)).$$

Remark 17.17. The cant-rotated frame determines:

- rail elevations,
- cross-level interpretation,

- vehicle orientation,
- effective gravity direction,
- and platform geometry.

17.7. Rail Plane

Remark 17.18. Gauge, rail elevation, and cross-level geometry are defined relative to the cant-rotated frame.

Remark 17.19. The rail plane is therefore not globally horizontal, but attached to the local pose3 frame and its cant rotation.

17.8. Frenet and Railway Frames

Remark 17.20. Classical Frenet frames are not fully adequate for railway alignment modelling.

Remark 17.21. In particular, Frenet frames become unstable in regions where

$$\kappa(s) \rightarrow 0.$$

Remark 17.22. Railway modelling therefore requires frame formulations that remain stable in straight and low-curvature sections.

Remark 17.23. In practice, railway frames are often transported continuously along the alignment rather than recomputed purely from differential geometry.

17.9. Surface and Ellipsoid Frames

Remark 17.24. When alignments are defined on the Earth surface, local frames must respect ellipsoid geometry.

Definition 17.25 (Surface-attached frame). A surface-attached frame consists of:

$$(\mathbf{t}, \mathbf{n}, \mathbf{n}_E),$$

where $\mathbf{n}_E$ denotes the ellipsoid surface normal.

Remark 17.26. In this setting:

- curvature splits into geodesic and normal curvature,
- gravity direction depends on Earth geometry,
- and world-to-track projection becomes surface-dependent.

17.10. Vehicle-Space Interpretation

Remark 17.27. The physically relevant object is not always the track centerline alone, but the motion of the railway vehicle within space.

Remark 17.28. Pose3 frames provide the spatial reference basis for:

- center-of-gravity motion,
- vehicle envelopes,
- bogie rotation,
- wheelset orientation,
- and platform clearance behaviour.

Remark 17.29. Thus, vehicle-space interpretation is derived from alignment states and their associated frames.

17.11. Connection to World–Track Transformation

Remark 17.30. World–track projection relies directly on the local frame.

Remark 17.31. In particular:

- tangent directions define longitudinal coordinates,
- lateral directions define offsets,
- and cant modifies the operational spatial interpretation.

Remark 17.32. This connects pose3 frames to the canonical world–track relation in eq. (10.15).

17.12. Frame Continuity

Remark 17.33. Frame continuity is essential for:

- smooth visualization,
- stable dynamics,
- cant interpretation,
- vehicle-space evaluation,
- and numerical robustness.

Remark 17.34. Discontinuous frame flips may lead to incorrect cant orientation, unstable vehicle interpretation, and invalid projection behaviour.

17.13. Key Insight

Proposition 17.35. *Pose3 frames are operational runtime structures rather than purely geometric differential objects.*

Remark 17.36. They connect intrinsic railway geometry with realization, dynamics, vehicle behaviour, and spatial interpretation.

17.14. Connection to AIM

Summary 17.37. Pose3 frames provide the spatial reference structure underlying railway alignment modelling.

Within AIM, they form the operational bridge between:

- curvature-based geometry,
- cant,
- realization,
- world-track projection,
- vehicle-space interpretation,
- and runtime system behaviour.

Teil IV.

Reality and Data

Remark 17.38. This part addresses the relation between the mathematical model and the real world. Railway alignments are not only theoretical objects, but are embedded in coordinate systems, measurement processes, construction workflows, and imperfect data.

Remark 17.39. A realistic alignment framework must therefore account for:

- coordinate reference systems,
- measurement uncertainty,
- special railway elements,
- and the distinction between designed and realized geometry.

Remark 17.40. The following chapters develop this reality layer as an essential part of alignment-based information modelling.

18. Measurement Data and Reality

18.1. Motivation

Remark 18.1. In practical applications, alignment geometry is rarely available in a clean, analytic form. Instead, it is derived from heterogeneous data sources with varying levels of precision, structure, and semantic completeness.

Remark 18.2. Therefore, the modelling process must explicitly distinguish between

- observed data,
 - interpreted structure,
 - and reconstructed geometry.
-

18.2. Sources of Data

Definition 18.3 (Data source). A data source is any origin of geometric or alignment-related information.

Remark 18.4. Typical sources include:

- survey measurements,
 - legacy engineering data,
 - container formats such as **LandXML**,
 - railway-specific formats,
 - point clouds and derived datasets.
-

18.3. Nature of Real-World Data

Remark 18.5. Real-world data exhibits several characteristic properties:

- **Incompleteness:** Not all required information is present.

- **Noise:** Measurements are subject to error.
- **Ambiguity:** Multiple interpretations may be possible.
- **Inconsistency:** Different parts of the data may contradict each other.

Remark 18.6. As a consequence, data cannot be directly equated with geometry.

18.4. Coordinate Systems

Remark 18.7. Geometric data is always embedded in a coordinate reference system (CRS).

Definition 18.8 (Coordinate reference system). A coordinate reference system defines how numerical coordinates relate to positions in physical space.

Remark 18.9. In infrastructure applications, multiple systems may coexist, including:

- global systems (e.g. WGS84),
- projected systems (e.g. UTM, Gauss–Krüger),
- domain-specific systems (e.g. railway reference systems).

OPEN: Formal integration of CRS handling into the alignment model.

RESEARCH: Transformation consistency across heterogeneous data sources.

18.5. Interpretation Layer

Definition 18.10 (Interpretation). Interpretation is the process of assigning structure and meaning to raw data.

Remark 18.11. Typical interpretation tasks include:

- identifying alignment segments,
- reconstructing curvature behaviour,
- associating auxiliary data such as profile or cant,
- resolving ambiguities in topology.

Remark 18.12. Interpretation is inherently non-unique and often requires domain knowledge.

OPEN: Formal separation between data and interpretation.

18.6. From Data to Sparse Model

Remark 18.13. The sparse alignment model provides a target structure for data interpretation.

Definition 18.14 (Reconstruction from data). Reconstruction is the process of deriving a sparse alignment model from a given data source.

Remark 18.15. This process typically involves:

- segmentation of data,
- classification of element types,
- estimation of curvature laws,
- assignment of initial conditions.

TODO: Formal pipeline from raw data to sparse alignment.

OPEN: Handling of incomplete or conflicting input data.

18.7. Container Formats

Remark 18.16. Many real-world datasets are provided in structured container formats.

Remark 18.17. Examples include:

- LandXML,
- railway-specific XML formats,
- spreadsheet-based engineering datasets.

Remark 18.18. Such formats often mix geometric, semantic, and metadata in ways that are not directly aligned with the sparse model.

RESEARCH: Mapping of container formats to sparse representation.

TODO: Definition of intermediate formats (e.g. landFAT).

18.8. Uncertainty and Error

Remark 18.19. All real-world data is subject to uncertainty.

Definition 18.20 (Measurement uncertainty). Measurement uncertainty describes the deviation between observed data and the true physical quantity.

Remark 18.21. Uncertainty propagates through:

- interpretation,
- reconstruction,
- numerical integration.

RESEARCH: Propagation of uncertainty through the reconstruction pipeline.

OPEN: Integration of uncertainty into the sparse model.

18.9. Topological vs Geometric Information

Remark 18.22. Data may contain both geometric and topological information.

Definition 18.23 (Geometric information). Geometric information describes positions, directions, and curvature.

Definition 18.24 (Topological information). Topological information describes connectivity and relationships between elements.

Remark 18.25. In many datasets, these two aspects are interwoven and not cleanly separated.

OPEN: Separation and integration of topology and geometry in the model.

18.10. Engineering Reality

Remark 18.26. Engineering data is often shaped by practical constraints rather than mathematical clarity.

Remark 18.27. Typical phenomena include:

- piecewise definitions without explicit continuity guarantees,
- implicit assumptions,
- domain-specific conventions,
- historical artefacts in legacy data.

RESEARCH: Systematic classification of real-world alignment data quality.

18.11. Role within the Overall Framework

Remark 18.28. The data layer connects the abstract modelling framework to real-world applications.

Remark 18.29. It forms the entry point for:

- import pipelines,
 - validation procedures,
 - reconstruction workflows.
-

18.12. Summary

Summary 18.30. Real-world alignment data is heterogeneous, noisy, and often ambiguous.

A clear separation between raw data, interpretation, and reconstructed geometry is essential.

The sparse alignment model provides a structured target for reconstruction, while numerical methods and domain knowledge are required to bridge the gap between data and geometry.

19. Coordinate Systems and Spatial Reference

19.1. Motivation

Remark 19.1. Railway alignment data is inherently spatial. Its interpretation depends fundamentally on the coordinate system in which it is expressed.

Remark 19.2. In practice, railway projects involve multiple coordinate reference systems (CRS), often used simultaneously.

Remark 19.3. Failure to handle coordinate systems consistently leads to:

- geometric inconsistencies,
- loss of accuracy,
- and incorrect integration of data sources.

CRS is not Metadata

In many software systems, the coordinate reference system (CRS) is treated as metadata.

It is stored, displayed, and occasionally converted — but rarely considered part of the model itself.

In railway alignment modelling, this view is fundamentally wrong.

The CRS determines:

- *how distances are measured,*
- *how curvature is interpreted,*
- *and how different datasets relate to each other.*

A shift of a few centimeters in coordinates may appear negligible.

A change in curvature is not.

If coordinate systems are inconsistent:

- *alignments no longer match,*

19. Coordinate Systems and Spatial Reference

- *curvature estimates become unstable,*
- *and optimization results lose physical meaning.*

The problem is not numerical.

It is structural.

A railway alignment does not exist independently of its spatial reference.

The coordinate system is part of the model.

19.2. Coordinate Reference Systems

Definition 19.4 (Coordinate reference system). A coordinate reference system (CRS) is a mathematical framework that defines how spatial coordinates relate to positions in the real world.

Remark 19.5. A CRS typically consists of:

- a reference surface (e.g. ellipsoid),
- a coordinate system (e.g. Cartesian or projected),
- and a transformation to real-world locations.

19.3. Global and Local Systems

Remark 19.6. Railway data commonly uses both global and local coordinate systems:

- global systems (e.g. WGS84),
- projected systems (e.g. UTM, Gauss–Krüger),
- engineering systems (e.g. track-based coordinates).

Remark 19.7. Each system serves a different purpose:

- global systems enable interoperability,
- projected systems provide metric consistency,
- local systems support engineering workflows.

19.4. Engineering Coordinate Systems

Remark 19.8. In railway engineering, alignments are often described in track-relative coordinate systems.

Definition 19.9 (Stationing). Let s denote the arc-length parameter along an alignment. This parameter is referred to as *stationing*.

Remark 19.10. Stationing defines a one-dimensional coordinate system along the track, independent of the global spatial reference.

Remark 19.11. Additional local coordinates may include:

- lateral offsets,
- vertical offsets,
- and local reference frames attached to the track.

19.5. Multiple CRS in Practice

Remark 19.12. Real-world projects frequently involve multiple CRS simultaneously.

Example 19.13. Typical combinations include:

- survey data in local engineering coordinates,
- GIS data in WGS84,
- design data in projected systems (UTM or GK).

Remark 19.14. These systems are often not perfectly consistent, due to:

- measurement errors,
- transformation approximations,
- and legacy definitions.

19.6. Transformations

Definition 19.15 (Coordinate transformation). A coordinate transformation is a mapping

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^n$$

that converts coordinates from one CRS to another.

Remark 19.16. Transformations may include:

- translations,
- rotations,
- scaling,

- and projection changes.

Remark 19.17. In practice, transformations are often defined by:

- reference points,
- parameter sets,
- or standardized transformation models.

19.7. Accuracy and Consistency

Remark 19.18. Coordinate transformations introduce uncertainties.

Remark 19.19. Even small inconsistencies can lead to:

- visible misalignment of datasets,
- incorrect curvature estimation,
- and unstable numerical reconstruction.

Remark 19.20. Therefore, a modelling framework must:

- track the CRS of all data,
- apply transformations explicitly,
- and avoid implicit assumptions.

19.8. CRS in Alignment Modelling

Remark 19.21. In a curvature-based alignment model, geometry is reconstructed from functions such as $\kappa(s)$.

Remark 19.22. This reconstruction requires a consistent spatial reference:

- initial position,
- initial orientation,
- and coordinate system definition.

Remark 19.23. Thus, CRS information is not auxiliary, but an essential component of the alignment model.

19.9. Challenges

Remark 19.24. The integration of CRS into alignment modelling raises several challenges:

- handling heterogeneous input data,
- maintaining numerical stability,
- supporting large coordinate values,
- and enabling local computations with high precision.

Remark 19.25. In particular, large global coordinates may lead to numerical issues in floating-point computations.

19.10. Local Coordinate Frames

Remark 19.26. To address numerical issues, local coordinate frames are often used.

Definition 19.27 (Local frame). A local frame is a coordinate system defined relative to a reference point, typically near the area of interest.

Remark 19.28. Using local frames:

- improves numerical stability,
- simplifies geometric computations,
- and allows efficient visualization.

19.11. Towards a Coordinate-Aware Model

Remark 19.29. A modern alignment modelling framework must be coordinate-aware.

Remark 19.30. This includes:

- explicit CRS representation,
- consistent transformation handling,
- integration with geometry reconstruction,
- and support for local and global coordinates.

19.12. Connection to the AIM Framework

Summary 19.31. Coordinate systems are a fundamental aspect of railway alignment data.

A consistent treatment of CRS is essential for integrating real-world data, ensuring numerical stability, and enabling reliable modelling.

Within the AIM framework, CRS is treated as an explicit component of the alignment state, rather than as external metadata.

19.13. Alignment Modelling Beyond Map Projections

19.13.1. Motivation

Remark 19.32. Conventional railway alignment design is typically performed in planar coordinate systems derived from map projections, such as Gauss–Krüger, UTM, or Mercator.

Remark 19.33. These projections introduce distortions in length, angle, and area, which are controlled but fundamentally unavoidable.

Remark 19.34. For large-scale infrastructure or high-precision applications, these distortions become conceptually and computationally relevant.

19.13.2. Intrinsic Surface Formulation

Remark 19.35. An alternative perspective is to formulate alignment geometry directly on the Earth’s reference surface, typically modeled as an ellipsoid.

Definition 19.36 (Surface-based alignment). An alignment is defined as a curve

$$\gamma : s \mapsto \mathcal{E},$$

where $\mathcal{E}$ denotes the reference ellipsoid.

Remark 19.37. In this formulation, arc-length is measured intrinsically on the surface, and curvature is defined with respect to the surface geometry.

19.13.3. Geometric Consequences

Remark 19.38. Curves on the ellipsoid are no longer planar:

- geodesics replace straight lines,
- curvature must be interpreted in a surface-intrinsic sense,
- normal vectors depend on the local surface geometry.

Remark 19.39. Thus, the classical decomposition into planar geometry and vertical profile becomes insufficient.

19.13.4. Relation to AIM

Remark 19.40. The AIM framework is inherently curvature-based and therefore not tied to a specific embedding space.

Remark 19.41. In particular, the representation

$$\kappa(s)$$

remains valid on curved surfaces, provided curvature is defined with respect to the surface metric.

19.13.5. Practical Considerations

Remark 19.42. Despite its conceptual appeal, a full ellipsoidal formulation introduces significant complexity:

- nonlinear surface geometry,
- increased computational cost,
- limited compatibility with existing engineering workflows.

Remark 19.43. For this reason, planar projections remain the dominant approach in practice.

19.13.6. Alignment as Curve on a Surface

Remark 19.44. The Earth is modeled as a reference surface $\mathcal{E}$, typically an ellipsoid.

Definition 19.45 (Surface alignment). An alignment can be interpreted as a curve

$$\gamma : s \mapsto \mathcal{E},$$

embedded in the surface.

Remark 19.46. In this setting, curvature decomposes into:

- intrinsic curvature (within the surface),
- normal curvature (with respect to embedding).

Remark 19.47. Straight lines in planar projections correspond to geodesics on the surface, which are not necessarily suitable as railway alignments.

Remark 19.48. Thus, alignment design is not equivalent to geodesic design.

19.13.7. Vision

Summary 19.49. AIM opens the possibility of alignment modelling directly on the Earth's reference surface.

While current engineering practice relies on planar projections, future systems may operate intrinsically on the ellipsoid, avoiding projection distortions and enabling globally consistent modelling.

This perspective represents a long-term research direction rather than an immediate engineering requirement.

Remark 19.50. In this sense, map projections can be understood as implementation choices rather than fundamental modelling constraints.

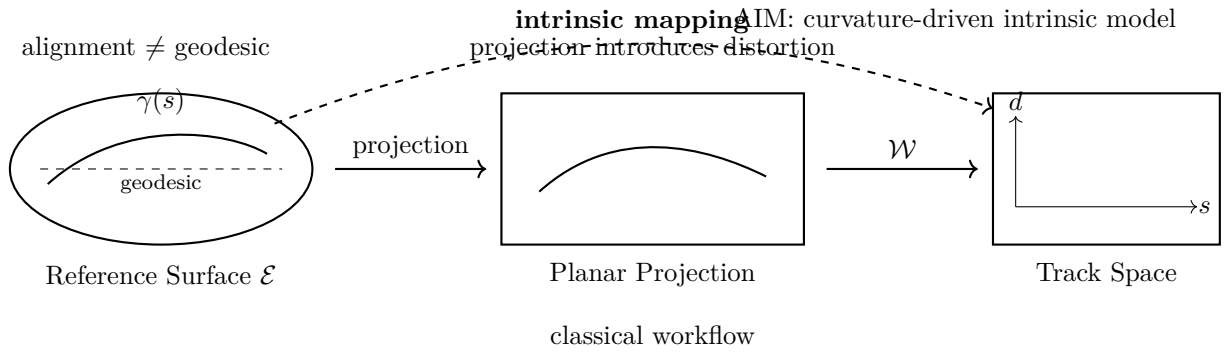


Abbildung 19.1.: Alignment modelling across geometric representations. On the reference surface, alignments are curves that generally differ from geodesics. Classical workflows project geometry to a plane before further processing, introducing distortions. The AIM perspective enables direct intrinsic modelling and mapping to track coordinates, independent of projection.

19.14. Physical Validity and Coordinate Systems

19.14.1. Current Practice

Remark 19.51. In current engineering workflows, railway alignment design and validation are typically performed in projected coordinate systems (e.g. Gauss–Krüger, UTM).

Remark 19.52. All geometric quantities, including curvature, distances, and derived dynamic measures, are evaluated within these planar representations.

Remark 19.53. However, such coordinate systems are approximations of the Earth’s geometry and introduce distortions:

- metric distortion (scale),
- angular distortion,
- non-uniform curvature representation.

19.14.2. Fundamental Limitation

Proposition 19.54. *Physical quantities derived in a projected coordinate system are not strictly consistent with the underlying geometry of the Earth.*

Remark 19.55. In particular:

- arc-length is only approximately preserved,
- curvature is distorted by projection,
- dynamic quantities (e.g. acceleration, jerk) are evaluated on an approximate geometry.

Remark 19.56. Thus, current alignment validation is performed in an *engineering approximation space*, not in the physical space in which the railway actually exists.

19.14.3. Ellipsoid-Based Perspective

Remark 19.57. A more physically consistent approach is to formulate alignment geometry directly on a reference ellipsoid.

Remark 19.58. In this setting:

- arc-length is defined intrinsically,
- curvature is defined with respect to the surface geometry,
- transformations to world coordinates are exact (up to numerical precision).

19.14.4. Differential-Geometric Formulation

Definition 19.59 (Intrinsic alignment on a surface). Let $\mathcal{E}$ denote a reference ellipsoid. An alignment is defined as a curve

$$\gamma : s \mapsto \mathcal{E},$$

with curvature defined intrinsically via differential geometry.

Remark 19.60. In this formulation:

- curvature is a geometric invariant,
- arc-length is physically meaningful,
- projection artifacts are eliminated.

19.14.5. Implication for Dynamics

Remark 19.61. Dynamic quantities such as lateral acceleration and jerk depend on curvature and arc-length derivatives.

Remark 19.62. If these are evaluated in a distorted coordinate system, the resulting values are only approximate.

Proposition 19.63. *Evaluating dynamics on an intrinsic (ellipsoid-based) geometry yields more physically accurate and reliable results.*

19.14.6. Safety Perspective

Remark 19.64. Railway alignment design directly affects safety-critical quantities:

- lateral acceleration,
- cant deficiency,
- jerk.

Remark 19.65. Errors introduced by coordinate distortions may be small individually, but can accumulate over long distances or in high-speed scenarios.

Proposition 19.66. *A geometrically consistent modelling framework improves the robustness and safety of alignment design.*

19.14.7. Vision

Remark 19.67. The long-term vision is to move from projection-based engineering geometry to intrinsic, surface-based modelling.

Remark 19.68. This implies:

- alignment design on the ellipsoid,
- differential-geometric formulation of curvature and dynamics,
- projection used only for visualization and data exchange.

Summary 19.69. Current railway alignment validation is performed in projected coordinate systems and is therefore only approximately physical.

A shift toward ellipsoid-based or differential-geometric modelling enables:

- physically consistent curvature,
- accurate dynamic evaluation,
- increased robustness,
- and ultimately safer alignment design.

This represents a fundamental extension of current engineering practice within the AIM framework.

Remark 19.70. Curvature evaluated in projected coordinate systems differs from intrinsic curvature on the ellipsoid.

See chapter 7.

20. Kilometre Line and Stationing

20.1. Motivation

Remark 20.1. Railway infrastructure is not only described geometrically, but also by stationing.

Remark 20.2. The kilometre line defines the longitudinal reference used for addressing positions along a railway route.

Remark 20.3. It must not be confused with the physical track centerline.

20.2. Definition

Definition 20.4 (Kilometre line). A kilometre line is a longitudinal reference alignment that assigns a station value to positions along a railway route.

Remark 20.5. The kilometre line may coincide with a track centerline, but this is not required.

20.3. Relation to Track Alignments

Remark 20.6. Several track alignments may refer to the same kilometre line.

Remark 20.7. This is common in double-track lines, junctions, station areas, and switching zones.

Remark 20.8. In single-track cases, the kilometre line and the track centerline may be identical.

20.4. Outer Stationing

Remark 20.9. When the physical track alignment differs from the kilometre line, positions along the track must be related back to the kilometre reference.

Definition 20.10 (Outer stationing). Outer stationing describes the mapping between a track alignment and an external kilometre reference.

20.5. Topological Role

Remark 20.11. The kilometre line acts as a topological and administrative reference rather than merely a geometric object.

Remark 20.12. It provides continuity across track changes, parallel tracks, and operationally relevant segments.

20.6. Implication for AIM

Summary 20.13. Within AIM, the kilometre line is treated as a reference structure separate from physical track alignments.

This distinction is essential for modelling multi-track situations, switches, stationing jumps, and real-world railway project data.

21. Kilometre Line, Stationing, and Spatial Reference

21.1. Motivation

Remark 21.1. Railway infrastructure is not only described by geometry, but also by stationing.

Remark 21.2. A point on a railway route is usually not addressed by global coordinates, but by a kilometre value along a reference line.

Remark 21.3. Thus, railway modelling must distinguish between:

- geometric location,
- spatial reference system,
- and operational addressing.

21.2. Three Reference Layers

Definition 21.4 (Spatial reference layer). The spatial reference layer defines where geometry lives. It may be based on:

- a planar projection,
- a local engineering coordinate system,
- or, in a long-term view, a reference ellipsoid.

Definition 21.5 (Geometric alignment layer). The geometric alignment layer describes the physical or designed track geometry, typically through curvature, pose, profile, and cant.

Definition 21.6 (Stationing layer). The stationing layer assigns address values to positions along a railway route.

Summary 21.7. Spatial reference answers: where is it? Geometry answers: what shape is it? Stationing answers: how is it addressed?

21.3. Kilometre Line

Definition 21.8 (Kilometre line). A kilometre line is a longitudinal reference structure used to assign station values to positions within a railway route or project.

Remark 21.9. The kilometre line may coincide with a physical track centerline, but this is not required.

Remark 21.10. It is therefore incorrect to treat the kilometre line simply as another track alignment.

21.4. Relation to Track Geometry

Remark 21.11. Several physical track alignments may refer to the same kilometre line.

Remark 21.12. This occurs in:

- double-track lines,
- station areas,
- junctions,
- switches and crossings,
- temporary or construction-stage alignments.

Remark 21.13. Conversely, in simple single-track situations, the kilometre line and track centerline may be identical.

21.5. Outer Stationing

Definition 21.14 (Outer stationing). Outer stationing describes the mapping between a physical track alignment and an external kilometre reference.

Remark 21.15. Outer stationing is required whenever the track geometry and the kilometre reference do not coincide.

Remark 21.16. In such cases, a point on the track has at least two relevant longitudinal coordinates:

- its intrinsic track coordinate s_{track} ,
- its assigned kilometre value s_{km} .

21.6. Relation to Ellipsoid and CRS

Remark 21.17. The kilometre line is independent of the coordinate reference system in which the geometry is represented.

Remark 21.18. A kilometre value is not a projected coordinate. It is an addressing value along a railway-specific reference structure.

Remark 21.19. Thus, even if geometry is formulated on a reference ellipsoid, in a Gauss–Krüger projection, or in a local engineering CRS, the kilometre line remains a separate semantic layer.

21.7. Reference Separation Principle

Proposition 21.20. *A railway project requires separation of at least three reference concepts:*

$$\textit{spatial reference} \neq \textit{geometric alignment} \neq \textit{stationing reference}.$$

Remark 21.21. Confusing these layers leads to systematic modelling errors, especially in multi-track and junction situations.

21.8. Implication for AIM

Remark 21.22. Within AIM, the kilometre line is treated as a reference object, not as the primary physical track geometry.

Remark 21.23. Track alignments may be assigned to kilometre lines by explicit stationing relations.

Remark 21.24. This allows AIM to model cases where:

- one kilometre line serves multiple tracks,
- a track has no independent kilometre line,
- stationing jumps occur,
- geometric and operational continuity differ.

21.9. Relation to the Seven-Line Model

Remark 21.25. The kilometre line plays a central role in the seven-line import and sorting model.

21. Kilometre Line, Stationing, and Spatial Reference

Remark 21.26. A simplified structure is:

$$\text{km-line} \rightarrow \begin{cases} \text{right track} \rightarrow (\text{profile, cant}), \\ \text{left track} \rightarrow (\text{profile, cant}), \\ \text{outer stationing.} \end{cases}$$

Remark 21.27. This expresses that the kilometre line provides the addressing context, while the tracks provide the physical geometry and engineering state.

21.10. Key Insight

Proposition 21.28. *The kilometre line is not primarily a geometric curve, but an addressing structure for railway infrastructure.*

Summary 21.29. The km-line, the geometric alignment, and the spatial reference system serve different roles.

The km-line addresses railway infrastructure.

The alignment describes physical geometry.

The CRS or ellipsoid defines the spatial embedding.

Keeping these layers separate is essential for robust railway modelling, especially in multi-track, station, and junction contexts.

22. Measurement and Imperfect Data

22.1. Motivation

Remark 22.1. Railway alignment modelling is not performed in an ideal mathematical environment, but based on measured and legacy data.

Remark 22.2. Such data is inherently imperfect. It contains noise, inconsistencies, and incomplete information.

Remark 22.3. A realistic modelling framework must therefore account explicitly for data imperfections.

22.2. Sources of Data

Remark 22.4. Typical sources of railway alignment data include:

- geodetic surveys,
- track measurement systems,
- legacy design files,
- and extracted data from infrastructure models.

Remark 22.5. These sources differ significantly in:

- accuracy,
- resolution,
- completeness,
- and coordinate reference systems.

22.3. Types of Imperfections

22.3.1. Measurement Noise

Definition 22.6 (Noise). Let $p_i \in \mathbb{R}^2$ denote measured points. Noise is the deviation between measured and true positions.

Remark 22.7. Noise arises from:

- instrument precision limits,
- environmental effects,
- and data processing steps.

22.3.2. Systematic Errors

Definition 22.8 (Systematic error). A systematic error is a consistent deviation affecting a dataset.

Remark 22.9. Examples include:

- incorrect CRS parameters,
- transformation offsets,
- calibration errors.

22.3.3. Incompleteness

Remark 22.10. Data may be incomplete:

- missing segments,
- missing metadata,
- or missing structural information.

22.3.4. Topological Ambiguity

Remark 22.11. In complex railway systems, data may not explicitly encode topology.

Remark 22.12. This leads to ambiguities such as:

- unclear connectivity of track segments,
- duplicated or overlapping elements,
- inconsistent ordering of points.

22.4. Representation of Measured Geometry

Definition 22.13 (Polyline representation). Measured alignments are often given as a sequence of points

$$P = (p_0, p_1, \dots, p_N), \quad p_i \in \mathbb{R}^2.$$

Remark 22.14. This representation is:

- simple,

- widely used,
- but not directly suitable for analysis or optimization.

Remark 22.15. In particular, curvature is not explicitly given, but must be inferred.

22.5. From Data to Model

Remark 22.16. A key task in alignment modelling is the transformation:

$$\text{raw data} \longrightarrow \text{structured model}.$$

Remark 22.17. This involves:

- filtering noise,
- reconstructing geometry,
- identifying structural elements,
- and assigning semantic meaning.

22.6. Curvature Estimation

Definition 22.18 (Discrete curvature). Given three consecutive points, curvature may be approximated by

$$\kappa_i \approx \frac{\theta_{i+1} - \theta_i}{\Delta s_i},$$

where θ_i denotes the local direction.

Remark 22.19. This approximation is sensitive to noise and sampling density.

Remark 22.20. Therefore, direct use of raw curvature estimates is often unstable.

22.7. Smoothing and Filtering

Remark 22.21. To obtain usable curvature information, smoothing is required.

Definition 22.22 (Smoothing operator). A smoothing operator transforms noisy data into a more regular form:

$$\kappa_i \mapsto \tilde{\kappa}_i.$$

Remark 22.23. Common techniques include:

- moving averages,
- Gaussian filtering,
- spline fitting,
- and regularization methods.

22.8. Model Reconstruction

Remark 22.24. The goal of reconstruction is to obtain a structured alignment model from imperfect data.

Remark 22.25. This includes:

- determining curvature functions,
- identifying transition regions,
- and constructing a consistent representation.

Remark 22.26. Reconstruction is not unique and depends on modelling assumptions.

22.9. Uncertainty

Definition 22.27 (Uncertainty). Uncertainty describes the lack of exact knowledge about the true alignment.

Remark 22.28. Uncertainty arises from:

- measurement errors,
- incomplete data,
- and modelling assumptions.

Remark 22.29. A robust modelling framework should:

- tolerate uncertainty,
- avoid overfitting,
- and provide stable results.

22.10. Impact on Optimization

Remark 22.30. Imperfect data directly affects optimization:

- objective functions may reflect noise,
- constraints may be violated by measurement errors,
- and solutions may become unstable.

Remark 22.31. Therefore, optimization must account for uncertainty, for example by:

- regularization,
- robust objective functions,
- or probabilistic models.

22.11. Engineering Perspective

Remark 22.32. In practice, engineers routinely work with imperfect data.

Remark 22.33. This includes:

- interpreting measurement results,
- reconciling inconsistent datasets,
- and making design decisions under uncertainty.

Remark 22.34. A modelling framework should support these tasks rather than assume ideal data.

22.12. Towards Data-Aware Modelling

Remark 22.35. A modern alignment model must be data-aware.

Remark 22.36. This includes:

- explicit handling of noise,
- robust reconstruction methods,
- integration with coordinate systems,
- and compatibility with real-world workflows.

22.13. Connection to the AIM Framework

Summary 22.37. Railway alignment data is inherently imperfect.

A realistic modelling framework must therefore incorporate measurement noise, incomplete data, and uncertainty.

Within the AIM framework, data is not assumed to be ideal, but is treated as an integral part of the modelling process.

23. Topology and Special Elements

23.1. Motivation

Remark 23.1. Railway systems are not isolated curves, but interconnected networks.

Remark 23.2. While mathematical models often focus on single alignments, real-world railway infrastructure consists of:

- branching tracks,
- merging lines,
- parallel alignments,
- and complex switching structures.

Remark 23.3. A complete modelling framework must therefore go beyond single curves and incorporate topology.

23.2. From Curves to Networks

Definition 23.4 (Alignment). An alignment is a parametrized curve

$$\gamma : [0, L] \rightarrow \mathbb{R}^2.$$

Definition 23.5 (Railway network). A railway network is a collection of alignments together with a set of connectivity relations.

Remark 23.6. The transition from a single alignment to a network introduces topological structure in addition to geometry.

23.3. Topological Relations

Remark 23.7. Alignments in a network may be related by:

- continuation (end-to-start connections),
- branching (one alignment splits into several),
- merging (multiple alignments join),

- parallel relations (tracks running side by side).

Definition 23.8 (Connection point). A connection point is a location where two or more alignments meet.

Remark 23.9. At connection points, geometric and dynamic consistency conditions must be satisfied.

23.4. Switches and Turnouts

Remark 23.10. Switches (turnouts) are fundamental elements of railway networks.

Definition 23.11 (Switch). A switch is a structure that allows a vehicle to move from one alignment to another.

Remark 23.12. Switches introduce:

- branching geometry,
- nontrivial curvature transitions,
- and constraints on feasible trajectories.

Remark 23.13. Unlike simple transitions, switches cannot be described by a single curvature function.

23.5. Geometric Complexity

Remark 23.14. Special elements often exhibit geometric properties that differ from standard alignment elements:

- discontinuities in curvature,
- piecewise-defined geometry,
- and locally non-smooth behaviour.

Remark 23.15. These properties must be handled explicitly in a modelling framework.

23.6. Topological Ambiguity in Data

Remark 23.16. Real-world data often does not explicitly encode topology.

Remark 23.17. Typical issues include:

- missing connectivity information,
- duplicated segments,
- and inconsistent ordering of elements.

Remark 23.18. Therefore, topology must often be inferred from geometric and contextual information.

23.7. Representation of Networks

Definition 23.19 (Graph representation). A railway network can be represented as a graph:

- nodes represent connection points,
- edges represent alignments.

Remark 23.20. This abstraction separates:

- topology (graph structure),
- and geometry (alignment shape).

23.8. Coupling of Topology and Geometry

Remark 23.21. Although topology and geometry can be separated conceptually, they are strongly coupled in practice.

Remark 23.22. For example:

- curvature constraints depend on connectivity,
- feasible transitions depend on network structure,
- and optimization must consider multiple interacting alignments.

23.9. Special Engineering Cases

Remark 23.23. Railway systems include a variety of special elements, such as:

- switches and crossings,
- sidings and yards,
- station layouts,
- and maintenance tracks.

Remark 23.24. These elements often violate assumptions of standard alignment models:

- unique parameterization by arc length,
- smooth curvature evolution,
- and single-track continuity.

23.10. Implications for Modelling

Remark 23.25. The presence of topology and special elements implies that:

- alignment models must support multiple connected curves,
- curvature-based descriptions must be extended,
- and reconstruction algorithms must handle branching structures.

Remark 23.26. In particular, optimization problems must be formulated on networks rather than individual curves.

23.11. Towards Topology-Aware Modelling

Remark 23.27. A topology-aware modelling framework must include:

- explicit representation of connectivity,
- integration of geometry and topology,
- support for special elements,
- and compatibility with real-world data.

23.12. Connection to the AIM Framework

Summary 23.28. Railway systems are inherently network-based and include complex special elements such as switches.

A complete modelling framework must therefore extend beyond single alignments and incorporate topology.

Within the AIM framework, topology is treated as an integral component of the alignment model, enabling consistent representation and optimization of railway networks.

24. Construction and Realization

24.1. Motivation

Remark 24.1. Theoretical railway alignments are defined as continuous geometric and curvature objects.

Remark 24.2. In practice, tracks are not constructed as continuous mathematical curves. They are realized through discrete construction and maintenance processes, local measurements, machine operations, correction limits, and structural response of the track.

Proposition 24.3. *The curve that is constructed is not identical to the curve that is designed.*

Remark 24.4. A realistic modelling framework must therefore include an explicit realization layer between analytical design and built geometry.

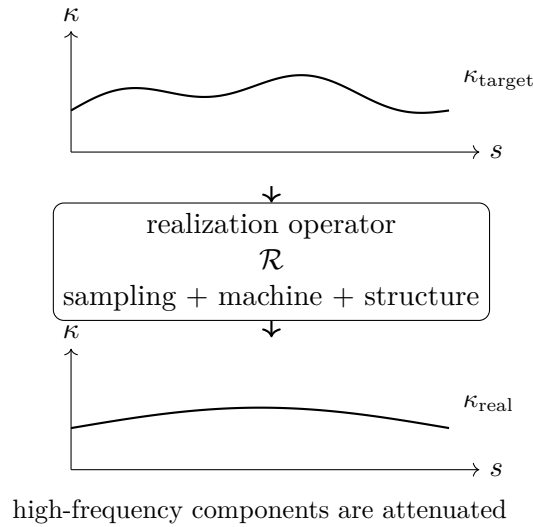


Abbildung 24.1.: Realization acts as a filtering process: target curvature may contain high-frequency components that are attenuated by sampling, machine action, and structural response.

24.2. Realized Alignment

Definition 24.5 (Realized alignment). The realized alignment is the physically constructed railway geometry, denoted by

$$\gamma_{\text{real}}(s).$$

24. Construction and Realization

Remark 24.6. It may be represented continuously, but it is usually generated from discrete construction, measurement, and maintenance data.

Definition 24.7 (Realized curvature). The realized curvature is

$$\kappa_{\text{real}}(s) = \frac{d\theta_{\text{real}}}{ds}.$$

Remark 24.8. The realized track should therefore not be judged only by positional agreement, but also by preservation of curvature evolution.

24.3. Realization Error

Definition 24.9 (Curvature realization error). A curvature-based realization error may be written as

$$\int_0^L (\kappa_{\text{real}} - \kappa_{\text{target}})^2 ds.$$

Definition 24.10 (Curvature derivative error). A higher-order realization error may be written as

$$\int_0^L (\kappa'_{\text{real}} - \kappa'_{\text{target}})^2 ds.$$

Remark 24.11. These measures reflect that different transition families may be close in position space while differing significantly in curvature evolution and dynamic interpretation.

24.4. Sampling and Control Points

Remark 24.12. Construction and maintenance operate through discrete control data.

Definition 24.13 (Control points). Let

$$0 = s_0 < s_1 < \cdots < s_N = L$$

be a strictly increasing sequence of arc-length positions. These positions are called control points.

Remark 24.14. The realized alignment may be reconstructed from these points using:

- piecewise linear interpolation,
- piecewise polynomial interpolation,
- spline-based reconstruction,
- or machine-specific reconstruction rules.

Proposition 24.15 (Adaptive sampling principle). *Sampling density should increase in regions of strong curvature variation:*

$$\rho(s) \sim f(|\kappa'(s)|, |\kappa''(s)|),$$

for a suitable increasing function f .

Remark 24.16. Nearly constant-curvature regions require fewer control points, whereas transition zones require denser control.

24.5. Adjustment as Iterative Process

Remark 24.17. In railway construction and maintenance, track geometry is adjusted through a sequence of localized operations, for example by tamping machines.

Definition 24.18 (Adjustment operator). A local construction or maintenance step is represented by an adjustment operator

$$\mathcal{T}.$$

Definition 24.19 (Iterative realization). Starting from an initial alignment γ_0 , the iterative process is

$$\gamma_{k+1} = \mathcal{T}_{\gamma_{\text{target}}}(\gamma_k).$$

Remark 24.20. The operator depends on local measurement data, machine parameters, correction limits, and physical properties of the track.

Definition 24.21 (Realized alignment as limit). If the sequence converges, the realized alignment is

$$\gamma_{\text{real}} = \lim_{k \rightarrow \infty} \gamma_k.$$

Proposition 24.22 (Fixed point interpretation). *If the iterative realization converges, then*

$$\gamma_{\text{real}} = \mathcal{T}_{\gamma_{\text{target}}}(\gamma_{\text{real}}).$$

Remark 24.23. In practice, convergence cannot be assumed a priori, since it depends on measurement feedback, machine limits, and local track conditions.

24.6. Locality and Correction Limits

Definition 24.24 (Local correction). A guided adjustment may be written as

$$\mathcal{T}_{\gamma_{\text{target}}}(\gamma) = \gamma + \Delta(\gamma, \gamma_{\text{target}}),$$

where Δ is a local correction operator.

Remark 24.25. Typically, corrections are bounded:

$$\|\Delta(\gamma, \gamma_{\text{target}})\| \leq \Delta_{\text{max}}.$$

24. Construction and Realization

Proposition 24.26 (Locality). *The adjustment operator is local in the sense that*

$$(\mathcal{T}_{\gamma_{\text{target}}}\gamma)(s)$$

depends only on γ and γ_{target} in a neighbourhood of s .

Remark 24.27. This reflects the finite working range of construction and maintenance equipment.

24.7. Kernel-Based Realization Model

Definition 24.28 (Kernel-based adjustment). A linearized model of the adjustment process is

$$(\mathcal{T}_{\gamma_{\text{target}}}\gamma)(s) = \gamma(s) + \int_I K(s, \sigma)(\gamma_{\text{target}}(\sigma) - \gamma(\sigma)) \, d\sigma,$$

where $K(s, \sigma)$ is a localized influence kernel.

Remark 24.29. The kernel represents the spatial influence of local construction or maintenance action.

Proposition 24.30 (Localization). *There exists a characteristic length $\ell > 0$ such that*

$$K(s, \sigma) \approx 0 \quad \text{for } |s - \sigma| > \ell.$$

Definition 24.31 (Discrete adjustment model). For control points s_i , the kernel model becomes

$$\gamma_i^{(k+1)} = \gamma_i^{(k)} + \sum_j K_{ij}(\gamma_j^{\text{target}} - \gamma_j^{(k)}).$$

Remark 24.32. This form is compatible with sparse numerical implementation.

24.8. Filter Interpretation

Proposition 24.33. *The realization process acts approximately as a low-pass filter on curvature.*

Remark 24.34. High-frequency curvature components are attenuated during construction and maintenance, while low-frequency structure is preserved.

Remark 24.35. This explains why transition curves with similar low-order behaviour may become nearly indistinguishable in position space after realization, while still differing in curvature evolution and dynamic interpretation.

24.9. Relation to Realization Operator

Remark 24.36. In the AIM operator framework, realization is represented by

$$\mathcal{R} : \text{pose3}_{\text{target}} \rightarrow \text{pose3}_{\text{real}}.$$

24. Construction and Realization

Remark 24.37. The construction process described in this chapter provides one interpretation of how such an operator may arise physically.

Remark 24.38. The simplified operator composition

$$\mathcal{R} \approx \mathcal{M} \circ \mathcal{I} \circ \mathcal{S}$$

separates sampling, reconstruction, and mechanical response.

24.10. Design Implication

Remark 24.39. The designed transition curve is not simply the curve that is built. It is a target that produces a realized behaviour after construction and maintenance.

Remark 24.40. Thus, alignment design implicitly includes assumptions about construction, machine behaviour, and structural response.

Proposition 24.41. *The relevant physical design space is the image of the realization operator, not the full space of analytical curvature laws.*

Remark 24.42. This connects construction and realization to realization-aware optimization.

24.11. Connection to AIM

Summary 24.43. Construction and realization describe the transformation from analytical target alignment to physically realized railway geometry.

Within AIM, this transformation is treated as an explicit operator layer.

The realization layer explains why physical railway geometry must be understood through construction processes, measurement feedback, local correction limits, and mechanical response rather than through analytical geometry alone.

25. Realized Pose3 and Dynamic Realization

25.1. Motivation

Remark 25.1. Classical alignment theory typically assumes that the designed geometry is identical to the realized geometry.

Remark 25.2. In reality, however, railway tracks are produced through iterative and physically constrained realization processes.

Remark 25.3. As a consequence, not only the geometric position of the track differs from the theoretical target alignment, but also its orientation, cant, and dynamic behaviour.

Remark 25.4. Thus, realization affects the full spatial railway state rather than geometry alone.

Remark 25.5. This motivates the introduction of a realized pose3 state:

$$\text{pose3}_{\text{real}}(s).$$

25.2. Target and Realized States

Definition 25.6 (Target pose3). The target railway state is defined as

$$\text{pose3}_{\text{target}}(s) = (\gamma_{\text{target}}(s), \theta_{\text{target}}(s), \phi_{\text{target}}(s)).$$

Definition 25.7 (Realized pose3). The realized railway state is defined as

$$\text{pose3}_{\text{real}}(s) = (\gamma_{\text{real}}(s), \theta_{\text{real}}(s), \phi_{\text{real}}(s)).$$

Remark 25.8. The realized state is the physically constructed and measurable state of the railway track.

Remark 25.9. In practice,

$$\text{pose3}_{\text{real}} \neq \text{pose3}_{\text{target}}.$$

25.3. Sources of Realization Deviations

Remark 25.10. Deviations between target and realized state arise from:

- measurement uncertainty,

- machine limitations,
- discrete control points,
- structural elasticity,
- ballast behaviour,
- rail bending,
- settlement processes,
- and operational degradation.

Remark 25.11. Thus, realization is not a purely geometric transformation, but a mechanical and operational process.

25.4. Realized Curvature and Cant

Definition 25.12 (Realized curvature). The realized curvature is

$$\kappa_{\text{real}}(s) = \frac{d\theta_{\text{real}}}{ds}.$$

Definition 25.13 (Realized cant). The realized cant angle is

$$\phi_{\text{real}}(s).$$

Remark 25.14. Both quantities differ from their target values due to realization effects.

Remark 25.15. In particular,

$$\kappa_{\text{real}} \neq \kappa_{\text{target}}, \quad \phi_{\text{real}} \neq \phi_{\text{target}}.$$

25.5. Cant Realization

Remark 25.16. Cant realization is particularly sensitive because cant is not only a geometric quantity but also a dynamically relevant operational state.

Remark 25.17. Cant realization is affected by:

- tamping accuracy,
- rail fastening systems,
- ballast stiffness,
- torsion limits,
- local support conditions,

- and measurement procedures.

Remark 25.18. In practice, cant ramps are realized only approximately and may exhibit local irregularities.

Remark 25.19. This is especially relevant in:

- transition zones,
- switches and crossings,
- station areas,
- and overlapping transition regions.

25.6. Coupling of Curvature and Cant

Remark 25.20. Curvature and cant are dynamically coupled quantities.

Remark 25.21. Therefore, realization errors in curvature and cant cannot be treated independently.

Remark 25.22. The physically relevant quantity is the realized dynamic state:

$$(\kappa_{\text{real}}, \phi_{\text{real}}).$$

Definition 25.23 (Realized equilibrium condition). A realized track is dynamically balanced if

$$v^2 \kappa_{\text{real}}(s) = g \sin \phi_{\text{real}}(s).$$

Remark 25.24. In practice, deviations from equilibrium are unavoidable and constrained only within admissible limits.

25.7. Dynamic Realization Error

Definition 25.25 (Realized effective lateral acceleration). The realized effective lateral acceleration is

$$a_{\text{eff,real}}(s) = v^2 \kappa_{\text{real}}(s) - g \sin \phi_{\text{real}}(s).$$

Remark 25.26. This quantity is operationally more relevant than geometric deviation alone.

Definition 25.27 (Realized jerk). The realized jerk is

$$j_{\text{real}}(s) = v^3 \kappa'_{\text{real}}(s) - g v \cos \phi_{\text{real}}(s) \phi'_{\text{real}}(s).$$

Remark 25.28. Passenger comfort depends directly on the realized jerk rather than on the analytical target functions.

25.8. Realization as State Transformation

Definition 25.29 (Pose3 realization operator). The realization process may be written as

$$\mathcal{R}_{\text{pose3}} : \text{pose3}_{\text{target}} \mapsto \text{pose3}_{\text{real}}.$$

Remark 25.30. This operator includes:

- geometric realization,
- cant realization,
- mechanical filtering,
- and operational constraints.

Remark 25.31. Thus, realization acts on the full railway state rather than only on position space.

25.9. Filtering Interpretation

Proposition 25.32. *The realization operator acts as a low-pass filter on:*

- *curvature,*
- *cant,*
- *and higher-order derivatives.*

Remark 25.33. High-frequency components are attenuated through:

- machine limitations,
- structural smoothing,
- ballast mechanics,
- and measurement discretization.

Remark 25.34. Therefore, highly detailed analytical target states may collapse into similar realized states.

25.10. Operational Interpretation

Remark 25.35. Railway operation interacts with the realized state rather than with the theoretical target geometry.

Remark 25.36. Consequently, the relevant engineering question is not:

“Was the theoretical geometry built exactly?”

but rather:

“Does the realized state produce the intended dynamic behaviour?”

25.11. Connection to Optimization

Remark 25.37. Optimization should therefore operate on realized states:

$$J = J(\text{pose3}_{\text{real}}) = J(\mathcal{R}_{\text{pose3}}(\text{pose3}_{\text{target}})).$$

Remark 25.38. This extends classical alignment optimization toward realization-aware optimization.

25.12. Connection to AIM

Summary 25.39. Real railway tracks are not analytical curves but realized physical states.

The realization process acts on the complete pose3 state, including geometry, curvature, and cant.

Thus, railway alignment design must be understood as the design of a realizable dynamic state rather than merely a geometric target curve.

Within the AIM framework, realized pose3 forms the bridge between geometry, mechanics, construction, and operation.

Teil V.

Alignment Modeling

Remark 25.40. This part develops the internal representation of railway alignments within the AIM framework. The emphasis lies on structured, curvature-based models that are compact, interpretable, and suitable for computation.

Remark 25.41. The goal is not merely to store geometry, but to represent alignments in a way that supports reconstruction, realization, and optimization.

Remark 25.42. The following chapters therefore introduce sparse representations, transition functions, classification principles, and higher-level modelling concepts such as Schwerpunkt-Trassierung.

26. Sparse Alignment Model

26.1. Motivation

Remark 26.1. A continuous curvature law is mathematically natural, but it is not by itself an operational representation for editing, exchange, validation, or reconstruction from heterogeneous engineering data. For this reason, a sparse model is introduced as a structured, piecewise-defined representation of an alignment.

Remark 26.2. The sparse model is not intended as a denial of continuous geometry. On the contrary, it serves as an intermediate layer between continuous curvature theory and practical modelling tasks such as import, computation, optimization, and interoperability.

Remark 26.3. Dense geometric representations such as point clouds or sampled polylines preserve positional information, but largely destroy the structural semantics of the alignment itself.

26.2. General Idea

Definition 26.4 (Sparse alignment model). A sparse alignment model is a structured longitudinal representation consisting of synchronized alignment bands together with sufficient reconstruction information.

Remark 26.5. The adjective *sparse* refers here to the fact that the alignment is not stored as a dense point cloud, but as a compact sequence of semantically meaningful elements.

Remark 26.6. Within AIM, the sparse model is not interpreted as a static 3D geometry, but as a persistent structural kernel from which geometry and dynamics are derived.

26.3. Element Types

Definition 26.7 (Alignment element). An alignment element is a segment of an alignment with a prescribed local curvature behaviour on a parameter interval.

Definition 26.8 (Fixed element). A fixed element is an alignment element with constant curvature

$$\kappa(s) = \kappa_0.$$

Typical special cases are straight segments with $\kappa_0 = 0$ and circular arcs with $\kappa_0 \neq 0$.

26. Sparse Alignment Model

Definition 26.9 (Transition element). A transition element is an alignment element with non-constant curvature, i.e.

$$\kappa(s) = f(s)$$

for a suitable curvature law f .

Remark 26.10. Within AIM, transition elements are primarily interpreted through their curvature evolution rather than through explicit geometric shape.

Remark 26.11. At this level of abstraction, no commitment is yet made to a specific family of transition laws. This includes classical clothoids as well as polynomial, lookup-based, or hybrid constructions.

TODO: Formal classification of transition families.

26.4. Minimal Parameterization

Definition 26.12 (Arc length of an element). Each element e_i is associated with an arc length

$$L_i > 0.$$

Definition 26.13 (Local curvature law). Each element e_i is associated with a curvature law

$$\kappa_i : [0, L_i] \rightarrow \mathbb{R}.$$

Remark 26.14. For fixed elements, κ_i is constant. For transition elements, κ_i varies along the local parameter.

Definition 26.15 (Sparse alignment). A sparse alignment is an ordered tuple

$$A = (e_1, e_2, \dots, e_n)$$

together with sufficient initial data for geometric reconstruction.

26.5. Initial Data and Pose

Definition 26.16 (Planar start pose). A planar start pose is a tuple

$$P_0 = (x_0, y_0, \theta_0)$$

with position $(x_0, y_0) \in \mathbb{R}^2$ and initial direction $\theta_0 \in \mathbb{R}$.

Proposition 26.17. *The planar alignment is reconstructed by successive integration of the curvature law:*

$$\kappa(s) \rightarrow \theta(s) \rightarrow \gamma(s).$$

Remark 26.18. Together with the ordered element sequence and the local curvature laws, the start pose determines the alignment geometry by successive integration.

Definition 26.19 (Reconstructable sparse alignment). A sparse alignment is called reconstructable if its element sequence and its initial data suffice to determine a unique planar alignment up to the intended modelling conventions.

26.6. Sparse Longitudinal Band Structure

Principle 26.20 (Sparse longitudinal band principle). Railway alignments are represented as synchronized longitudinal bands rather than as monolithic geometric objects.

Remark 26.21. Within the AIM framework, the persistent alignment model is intentionally kept sparse and decomposed into synchronized longitudinal bands.

26.6.1. cGeom

Definition 26.22 (Core geometric band). The central geometric band is denoted by

$$\text{cGeom.}$$

It represents the curvature-driven planar alignment core and consists of:

- fixed elements,
- transition elements,
- pose2-based reconstruction anchors.

Remark 26.23. The cGeom band forms the primary integration core of the alignment.

26.6.2. Cant Band

Definition 26.24 (Cant band). Cant is represented as an independent longitudinal band correlated with fixed alignment elements.

Remark 26.25. The cant band:

- does not define an independent stationing system,
- stores rail height differences rather than roll angles,
- follows the same normalized curvature family as the associated cGeom element,
- may contain local station offsets at element boundaries.

Remark 26.26. Since cant evolution follows the same normalized curvature family as the associated cGeom element, no independent geometric evolution model is required for the cant band itself.

26. Sparse Alignment Model

Remark 26.27. The corresponding roll angle is derived via:

$$\phi(s) = \arctan\left(\frac{u(s)}{g}\right),$$

where:

- $u(s)$: rail height difference,
- g : gauge.

26.6.3. Profile Band

Definition 26.28 (Profile band). Vertical geometry is represented independently through a profile band.

Remark 26.29. Thus, cGeom, cant, and profile are treated as synchronized but distinct bands sharing a common arc-length domain.

26.6.4. Runtime State

Proposition 26.30. *Pose3 is not treated as persistent truth within the sparse model.*

Remark 26.31. Persisting pose3 directly would introduce redundancy and synchronization instabilities between geometry, cant, and profile representations.

Remark 26.32. Instead, pose3 is derived dynamically from:

- cGeom,
- cant,
- profile,
- and runtime evaluation services.

Remark 26.33. Consequently, the primary intelligence of the AIM framework resides not in the storage format itself, but in the mathematical operators and evaluation services acting on the sparse model.

26.7. Concatenation

Definition 26.34 (Element concatenation). Two consecutive elements are concatenated if the end state of the first element provides the start state of the second element.

Definition 26.35 (Positional continuity). A sparse alignment is positionally continuous if consecutive elements meet in a common endpoint.

Definition 26.36 (Directional continuity). A sparse alignment is directionally continuous if consecutive elements share the same tangent direction at their common boundary.

26. Sparse Alignment Model

Remark 26.37. In most engineering contexts, positional and directional continuity are minimal requirements. Higher continuity requirements may depend on the chosen transition families and on the intended application.

TODO: Formalize continuity classes such as C^0 , C^1 , and curvature-related continuity conditions.

26.8. Normal Form and Alternation

Remark 26.38. In practical sparse representations, a frequent pattern is the alternation between fixed elements and transition elements. This corresponds closely to engineering intuition and to many traditional alignment descriptions.

OPEN: Should strict alternation between fixed and transition elements be formulated as a mathematical property, as a preferred normal form, or merely as an implementation convention?

Remark 26.39. It is plausible to regard alternation as a normal form rather than a fundamental axiom. Multiple neighbouring fixed elements may be merged, and certain degenerate transition cases may collapse into fixed elements.

26.9. Normalized Representation of Elements

Definition 26.40 (Normalized local parameter). A normalized local parameter is a variable

$$u \in [0, 1]$$

used to represent an element independently of its physical length.

Definition 26.41 (Normalized curvature law). A normalized curvature law is a law defined on the unit interval, typically of the form

$$\hat{\kappa} : [0, 1] \rightarrow \mathbb{R}.$$

Remark 26.42. Normalization separates shape from scale. It allows the same transition family to be reused for different lengths and different curvature magnitudes.

Remark 26.43. Normalized representations form the basis for reusable transition registries, family classification, and implementation-independent comparison of alignment elements.

26.10. Structural Role of the Sparse Model

Remark 26.44. The sparse model occupies an intermediate conceptual level:

- below the purely abstract curvature-theoretic foundation,

26. Sparse Alignment Model

- but above dense numerical sampling, imported point lists, or format-specific container structures.

Remark 26.45. The sparse model is intentionally not a dense geometric storage format, nor a direct copy of external exchange containers.

Remark 26.46. This intermediate status makes the sparse model suitable as

- an internal modelling layer,
- a reconstruction target from imported data,
- a source for numerical evaluation,
- and a bridge toward standards and BIM-related representations.

26.11. Relation to Imported and Standardized Data

Remark 26.47. Existing engineering data rarely arrives directly in sparse form. Instead, it is typically given through container formats, survey data, legacy descriptions, or derived coordinate sequences.

Remark 26.48. Such data may contain:

- inconsistent stationing,
- incomplete cant information,
- broken coordinate reference systems,
- discontinuities,
- partial geometry fragments,
- or non-synchronized alignment components.

Remark 26.49. The sparse model therefore should not be identified with any one exchange standard. It is more appropriate to view it as a compact, internal representation that can be derived from, and mapped back to, multiple external formats.

RESEARCH: Dock sparse model to railML alignment structures. **RESEARCH:** Dock sparse model to LandXML alignment representations. **OPEN:** Define a minimal common abstraction layer for standards mapping. **OPEN:** Clarify the relation between topology-oriented standards and purely geometric container models.

26.12. Relation to BIM and BIMalignment

Remark 26.50. A sparse alignment model may support BIM-oriented workflows, but it should not merely imitate BIM container logic. Its primary responsibility is to represent alignment structure in a mathematically and operationally coherent way.

RESEARCH: Formulate a BIM-compatible alignment core without losing geometric precision. **OPEN:** Position the sparse model between generic BIM, infrastructure BIM, and explicit BIMalignment concepts. **TODO:** State explicitly where the manuscript aims to push BIMalignment rather than only satisfy BIM conformity.

26.13. Summary

Summary 26.51. The sparse alignment model is a compact, structured representation of an alignment by means of synchronized longitudinal bands together with reconstruction data and concatenation rules.

Within AIM, the sparse alignment model is not interpreted as a static 3D geometry, but as a persistent structural kernel from which geometry, runtime state, and dynamic interpretation are derived.

It serves as a bridge between continuous curvature theory and practical engineering operations such as import, validation, editing, numerical evaluation, optimization, and interoperability.

27. Transition Functions

27.1. Motivation

Remark 27.1. Transitions are the decisive element type for curvature-driven alignment modelling. While fixed elements describe geometric states of constant curvature, transitions describe the controlled change between such states.

Remark 27.2. In railway engineering, transitions are essential for ensuring smooth vehicle motion, limiting dynamic forces, and satisfying comfort and safety requirements, as discussed in both engineering-oriented studies and review literature [26, 4].

Remark 27.3. Within AIM, transitions are not primarily interpreted through explicit geometric shape, but through the evolution of curvature.

27.2. Definition by Curvature Evolution

Principle 27.4 (Curvature evolution principle). Within AIM, transitions are classified primarily by their curvature evolution rather than by explicit geometric shape.

Definition 27.5 (Transition). A transition is an alignment element whose curvature varies over its parameter interval.

Definition 27.6 (Transition curvature law). Let $L > 0$. A transition curvature law is a function

$$\kappa : [0, L] \rightarrow \mathbb{R}$$

with prescribed boundary values

$$\kappa(0) = \kappa_0, \quad \kappa(L) = \kappa_1.$$

Remark 27.7. This formulation directly follows the curvature-based description of planar curves [8].

27.3. Normalized Representation

Definition 27.8 (Normalized transition law). A normalized transition is defined by

$$\hat{\kappa} : [0, 1] \rightarrow \mathbb{R}$$

27. Transition Functions

with

$$\hat{\kappa}(0) = 0, \quad \hat{\kappa}(1) = 1.$$

Remark 27.9. Normalization separates the geometric shape of a transition from its scale and allows systematic comparison of transition families.

Remark 27.10. Normalization separates intrinsic transition behaviour from physical scale and therefore enables reusable transition registries and family-based modelling.

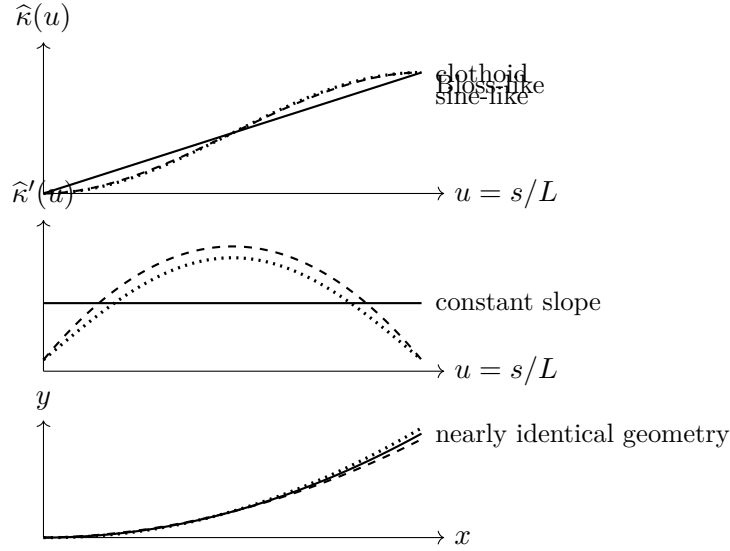


Abbildung 27.1.: Transition families may produce very similar geometric paths while differing strongly in curvature evolution and curvature derivatives. This is why transition design must be evaluated in curvature and dynamic terms, not only in position space.

Remark 27.11. Transitions that appear nearly identical in position space may differ substantially in curvature derivatives and therefore in dynamic behaviour.

27.4. Scaling

Proposition 27.12. Let $\hat{\kappa}$ be a normalized transition law. Then a physical transition is given by

$$\kappa(s) = \kappa_0 + (\kappa_1 - \kappa_0) \hat{\kappa}\left(\frac{s}{L}\right).$$

Remark 27.13. This separates shape from scale and provides the basis for reusable transition families and parameterized design.

27.5. Regularity and Smoothness

Definition 27.14 (Regularity class). A transition is of class C^k if its curvature function is k -times continuously differentiable.

Remark 27.15. In engineering practice, higher-order properties such as curvature derivatives are directly related to ride comfort and dynamic loading, as shown in both theoretical and engineering studies [34, 27].

OPEN: Formal role of κ' and κ'' in engineering rules.

27.6. Transition Families

Definition 27.16 (Transition family). A transition family is a class of curvature laws sharing a common functional structure.

27.6.1. Classical Families

Remark 27.17. The classical transition curve in railway engineering is the clothoid, characterized by linear curvature evolution:

$$\kappa(s) \propto s.$$

Remark 27.18. Historically, this line of development can be traced through early work by Glover [9], Horsburgh [15], and Higgins [13].

27.6.2. Polynomial and Alternative Families

Remark 27.19. Polynomial transition curves provide additional flexibility in shaping curvature evolution and its derivatives.

Remark 27.20. Such approaches are associated with modern polynomial smoothing research [51] as well as control-oriented constructions proposed by Klauder [19, 22].

TODO: Formal definition of polynomial transition families.

27.6.3. Geometric Families

Remark 27.21. Some transition curves are introduced through geometric or implicit relations rather than explicit curvature laws.

Example 27.22. Lemniscate-type or sinusoid-related constructions belong to this class after suitable reparameterization into curvature space [37].

Remark 27.23. These curves require transformation into $\kappa(s)$ representation before they can be integrated into the present framework.

27.6.4. Engineering and Practice-Oriented Families

Remark 27.24. Engineering practice often introduces transition curves through design rules and operational considerations rather than through purely theoretical derivation.

Remark 27.25. The Bloss transition is a notable example of a practical engineering family, originally described by Bloss [3] and further discussed in modern summaries [7].

27.6.5. Modern and Hybrid Families

Remark 27.26. Modern engineering practice considers a variety of transition families with improved dynamic behaviour, smoother endpoint properties, or higher-order control.

Remark 27.27. Examples include smoothed-curvature transitions proposed by Koc [27, 29, 28] and Wiener-Bogen-related approaches [50].

RESEARCH: Systematic classification of modern transition families.

27.7. Decomposition Approaches

Remark 27.28. A transition need not always be treated as a single indivisible object. It may be useful to represent it as a composition of sub-elements.

Remark 27.29. Within AIM, complex transition behaviour may be constructed from composable normalized suboperators such as half-wave and clothoid components.

TODO: Formalize halfWave – clothoid – halfWave decomposition.

OPEN: Is decomposition fundamental or implementation-specific?

27.8. Operator View

Principle 27.30 (Transition operator principle). A transition element acts as a local curvature evolution operator transforming one geometric state into another.

Remark 27.31. Transitions can therefore be interpreted as state evolution operators rather than merely as predefined geometric curve shapes.

RESEARCH: Formal operator algebra of transitions.

27.9. Implementation Perspective

Remark 27.32. In computational systems, transition curves may be represented using:

- analytic expressions,
- polynomial approximations,
- lookup tables,
- compiled hybrid representations.

Remark 27.33. Transition families may be represented through registries containing normalized curvature laws together with derivative and integration operators.

TODO: Registry-based transition system.

OPEN: Trade-off between analytic precision and computational efficiency.

27.10. Relation to Railway Engineering

Remark 27.34. In railway applications, transitions are tightly coupled to operational parameters such as speed and cant, and must satisfy strict engineering constraints [30, 24].

Remark 27.35. The review literature confirms that transition curves must be understood not only geometrically, but also with respect to comfort, maintenance, and dynamic response [4].

27.11. Summary

Summary 27.36. Transitions define the controlled evolution of curvature between geometric states.

Within AIM, they are interpreted as normalized curvature evolution operators rather than merely as predefined geometric curve types.

Their normalized representation enables reusable transition families, operator-based modelling, and runtime reconstruction of alignment geometry and dynamics.

28. Transition Families and Classification

28.1. Motivation

Remark 28.1. Transition curves form a central component of alignment design, yet their treatment in the literature remains fragmented across geometric, engineering, and dynamic perspectives.

Remark 28.2. A unified classification is therefore required in order to compare, evaluate, and optimize transition concepts systematically.

28.2. Curvature-Based Representation

Proposition 28.3. *Any transition curve can be represented by a curvature function*

$$\kappa : [0, L] \rightarrow \mathbb{R}.$$

Remark 28.4. This representation is independent of the original definition of the curve and provides a common mathematical framework.

28.2.1. Normalization

Definition 28.5. A normalized transition is defined on

$$s \in [0, 1]$$

with

$$\kappa(0) = 0, \quad \kappa(1) = 1.$$

Remark 28.6. Normalization removes scale and allows direct comparison of transition families.

28.3. Classification by Functional Form

28.3.1. Linear Curvature: Clothoid

Definition 28.7.

$$\kappa(s) = as.$$

Remark 28.8. The clothoid is characterized by constant curvature derivative

$$\kappa'(s) = \text{const.}$$

Remark 28.9. This simplicity explains its widespread adoption in railway engineering.

28.3.2. Polynomial Curvature Families

Definition 28.10.

$$\kappa(s) = \sum_{i=0}^n a_i s^i.$$

Remark 28.11. Polynomial transitions allow explicit control of higher-order derivatives of curvature.

Remark 28.12. Such flexibility enables optimization with respect to dynamic criteria.

Remark 28.13. Klauder-type transitions belong to this class, incorporating dynamic considerations directly into the curvature design.

28.3.3. Geometric Families

Remark 28.14. Some transition curves are defined implicitly or through geometric relations rather than explicit curvature functions.

Remark 28.15. These curves must be transformed into curvature representation to be integrated into the present framework.

28.3.4. Engineering-Based Families

Remark 28.16. Engineering approaches often define transition curves through design rules or empirical considerations.

Remark 28.17. The Bloss transition represents a notable example of such a practical engineering compromise.

28.3.5. Optimized and Hybrid Families

Remark 28.18. Modern approaches introduce transition curves designed explicitly to optimize dynamic performance.

Remark 28.19. These include smoothed curvature transitions and jerk-optimized polynomial curves.

—

28.4. Classification by Higher-Order Behaviour

28.4.1. Curvature Derivatives

Definition 28.20.

$$\kappa'(s), \quad \kappa''(s)$$

denote the first and second derivatives of curvature.

—

Remark 28.21. These quantities determine dynamic behaviour, including acceleration variation and smoothness of force development.

—

28.4.2. Smoothness Classes

Definition 28.22. A transition is of class C^k if κ is k -times continuously differentiable.

—

Remark 28.23. Higher smoothness typically improves dynamic performance and reduces wear.

—

28.5. Comparison of Families

Family	κ	κ'	κ''
Clothoid	linear	constant	0
Polynomial	flexible	flexible	flexible
Engineering	implicit	implicit	implicit
Optimized	designed	controlled	controlled

—

28.6. Interpretation as Control Functions

Proposition 28.24. *Transition curves can be interpreted as control functions acting on system dynamics through curvature evolution.*

Remark 28.25. This interpretation links geometric design directly to dynamic performance.

28.7. Connection to Railway Engineering

Remark 28.26. Railway transition design is governed by dynamic constraints such as acceleration, jerk, and wear.

Remark 28.27. These constraints translate directly into requirements on

$$\kappa(s), \quad \kappa'(s), \quad \kappa''(s).$$

28.8. Main Insight

Theorem 28.28. *Transition curves are not merely geometric connectors, but control functions governing system behaviour.*

28.9. Conclusion

Summary 28.29. Transition curves form a unified class of curvature functions.

Their classification by functional form and higher-order behaviour provides a systematic framework for comparison, optimization, and integration into engineering systems.

This perspective establishes a direct link between geometry and system dynamics.

29. Schwerpunkt-Trassierung

29.1. Motivation

Remark 29.1. Classical railway alignment design separates:

- geometric design (alignment),
- cant design,
- dynamic evaluation.

Remark 29.2. This separation leads to suboptimal solutions, as geometry and dynamics are intrinsically coupled. In railway practice, this coupling appears through the interaction of curvature, cant, speed, and vehicle response [24, 6, 4].

Remark 29.3. The concept of *Schwerpunkt-Trassierung* aims to unify these aspects by defining a single design principle based on the motion of a representative point.

Remark 29.4. The idea that railway alignment should be related to the motion of the vehicle centre of gravity rather than merely to the track centerline has also appeared in engineering practice and patent literature [12].

Remark 29.5. This supports the broader AIM interpretation that alignment should be understood as a dynamically relevant spatial state trajectory rather than as a purely geometric curve.

29.2. Conceptual Idea

Definition 29.6 (Reference trajectory). Let

$$\gamma_c : I \rightarrow \mathbb{R}^2$$

denote a reference trajectory representing the motion of a characteristic point of the vehicle system.

Remark 29.7. This point may be interpreted as:

- the center of mass,
- a dynamically relevant reference point,
- or an abstract design trajectory.

Remark 29.8. The conceptual shift consists in evaluating track design through the motion of such a reference point rather than through geometry alone.

29.3. Relation to Track Geometry

Remark 29.9. The physical track centerline $\gamma(s)$ and cant angle $\phi(s)$ induce a spatial configuration of the vehicle.

Definition 29.10 (Induced reference trajectory). Given a pose3 state

$$(\gamma(s), \theta(s), \phi(s)),$$

the induced reference trajectory is

$$\gamma_c(s) = \gamma(s) + d n(s, \phi(s)),$$

where:

- d is a characteristic offset,
- $n(s, \phi)$ is a direction depending on orientation and cant.

Remark 29.11. The exact form of $n(s, \phi)$ depends on the chosen mechanical model.

RESEARCH: Precise definition of offset direction based on vehicle model.

OPEN: Consistent integration of local track frames and global CRS transformations in the Schwerpunkt model.

29.4. Schwerpunkt Principle

Definition 29.12 (Schwerpunkt principle). A railway alignment is said to satisfy the Schwerpunkt principle if the induced reference trajectory $\gamma_c(s)$ possesses prescribed smoothness and dynamical properties.

Remark 29.13. In this formulation, the primary design object is not the track itself, but the induced trajectory of the reference point.

Remark 29.14. This makes the design task closer to vehicle-oriented engineering than to purely geometric construction.

29.5. Variational Formulation

Definition 29.15 (Schwerpunkt functional). Let

$$J_c[\gamma_c] = \int_0^L \|\gamma_c''(s)\|^2 ds.$$

29. Schwerpunkt-Trassierung

Remark 29.16. This functional penalizes curvature of the reference trajectory and therefore promotes smooth motion.

Definition 29.17 (Schwerpunkt optimization problem). Find

$$(\kappa, \phi)$$

such that the induced trajectory γ_c minimizes

$$J_c[\gamma_c]$$

subject to the pose3 dynamics.

Remark 29.18. This formulation extends the variational interpretation of transition design from curvature laws alone to vehicle-relevant derived motion.

29.6. Coupling to Pose3

Remark 29.19. The trajectory γ_c depends on:

$$\gamma(s), \quad \theta(s), \quad \phi(s).$$

Remark 29.20. Thus, the Schwerpunkt problem becomes a constrained optimization problem in the variables:

$$\kappa(s), \quad \phi(s).$$

Remark 29.21. This places Schwerpunkt-Trassierung naturally within the pose3-based state model introduced earlier and within the broader optimization perspective of railway alignment [31, 30].

29.7. Interpretation

Remark 29.22. The Schwerpunkt formulation shifts the perspective:

- from track geometry
- to vehicle motion.

Remark 29.23. This leads to a unified framework in which:

- geometry,
- cant,
- and dynamics

are optimized simultaneously.

Remark 29.24. Such a viewpoint is consistent with the general observation that railway transition and alignment quality cannot be judged from geometry alone [4, 6].

29.8. Relation to Classical Design

Proposition 29.25. *Classical transition curves correspond to special cases of the Schwerpunkt formulation where the reference trajectory coincides with the track centerline.*

Remark 29.26. In this case, the problem reduces to curvature-based optimization in the plane.

Remark 29.27. Thus, classical alignment design is not contradicted, but embedded as a special case of a more general formulation.

29.9. Advantages

- unified modelling of geometry and dynamics,
- natural integration of cant,
- compatibility with optimization frameworks,
- extensibility to vehicle-specific models.

Remark 29.28. These advantages derive from the fact that the model treats alignment as a state-governed physical system rather than as a static geometric line.

29.10. Open Problems

OPEN: Precise mechanical interpretation of the reference trajectory.

OPEN: Integration with full vehicle dynamics models.

OPEN: Extension to network-level alignment design.

OPEN: Explicit treatment of CRS, local track frames, and coordinate transformations in large-scale alignment systems.

29.11. Connection to the AIM Framework

Summary 29.29. Schwerpunkt-Trassierung provides a conceptual and mathematical extension of classical alignment design.

Instead of prescribing geometric elements, it defines the design problem in terms of a reference trajectory derived from the railway state.

This aligns with the AIM philosophy of treating alignment as a functional object governed by curvature and its derived quantities.

29.12. Vehicle Model

29.12.1. Rigid Body Approximation

Remark 29.30. We model the railway vehicle as a rigid body with a characteristic center of mass.

Definition 29.31 (Center of mass). Let

$$C(s) \in \mathbb{R}^3$$

denote the spatial position of the vehicle center of mass.

Remark 29.32. For planar track geometry, we consider the projection

$$\gamma_c(s) \in \mathbb{R}^2$$

of the center of mass trajectory.

Remark 29.33. This is intentionally a simplified model. Its purpose is not to replace full multibody simulation, but to introduce a mathematically tractable vehicle-related state layer.

29.12.2. Track Coordinate Frame

Definition 29.34 (Frenet frame). Let

$$t(s) = (\cos \theta(s), \sin \theta(s))$$

be the tangent direction and

$$n(s) = (-\sin \theta(s), \cos \theta(s))$$

the normal direction.

Remark 29.35. The pair (t, n) defines a local coordinate system attached to the track.

Remark 29.36. This local frame is sufficient for the present formulation, but future extensions should consistently relate it to a global project CRS.

29.12.3. Cant Rotation

Definition 29.37 (Rotated normal). Let the cant angle $\phi(s)$ define a rotation of the cross-section.

Remark 29.38. In a simplified model, the lateral offset direction remains aligned with $n(s)$, while vertical effects are captured implicitly.

Remark 29.39. This simplification is acceptable at the present conceptual stage, but a full treatment should explicitly include the three-dimensional rotated cross-section and its transformation into the global frame.

29.12.4. Center of Mass Offset

Definition 29.40 (Offset model). Let $d > 0$ denote a characteristic lateral offset. The center of mass trajectory is approximated by

$$\gamma_c(s) = \gamma(s) + d n(s).$$

Remark 29.41. More refined models may include dependence on $\phi(s)$, e.g.

$$\gamma_c(s) = \gamma(s) + d \cos \phi(s) n(s).$$

RESEARCH: Inclusion of full 3D vehicle geometry.

29.13. Dynamics of the Center of Mass

29.13.1. Velocity and Acceleration

Proposition 29.42. *The velocity of the center of mass is*

$$\gamma'_c(s) = t(s) + d n'(s).$$

Remark 29.43. Using

$$n'(s) = -\kappa(s) t(s),$$

we obtain

$$\gamma'_c(s) = (1 - d\kappa(s)) t(s).$$

Proposition 29.44. *The acceleration of the center of mass is proportional to*

$$\gamma''_c(s) = -d\kappa'(s) t(s) + (1 - d\kappa(s))\kappa(s) n(s).$$

Remark 29.45. Thus, both curvature and its derivative influence the motion of the center of mass.

Remark 29.46. This is one of the key conceptual benefits of the formulation: it makes the dynamic relevance of higher-order geometric quantities explicit.

29.13.2. Effective Forces

Definition 29.47 (Lateral acceleration). The lateral acceleration of the center of mass is

$$a_c(s) = v^2(1 - d\kappa(s))\kappa(s).$$

29. Schwerpunkt-Trassierung

Remark 29.48. Including cant yields the effective acceleration

$$a_{\text{eff}}(s) = v^2 \kappa(s) - g \sin \phi(s).$$

Remark 29.49. The latter term reflects the same curvature–cant coupling that appears in railway design standards and in dynamic alignment interpretation [24, 6].

29.14. Coupled Design Problem

Definition 29.50 (Vehicle-based objective). Define

$$J_{\text{veh}}[\kappa, \phi] = \int_0^L \|\gamma_c''(s)\|^2 ds.$$

Remark 29.51. This functional directly measures the smoothness of the motion of the center of mass.

Definition 29.52 (Dynamic objective). Define

$$J_{\text{dyn}}[\kappa, \phi] = \int_0^L \left(v^2 \kappa(s) - g \sin \phi(s) \right)^2 ds.$$

Definition 29.53 (Combined vehicle functional). A combined objective is

$$J[\kappa, \phi] = \alpha J_{\text{veh}} + \beta J_{\text{dyn}} + \gamma \int_0^L (\kappa'(s))^2 ds + \delta \int_0^L (\phi'(s))^2 ds.$$

Remark 29.54. This combines geometric smoothing, dynamic balancing, and higher-order regularization within a single optimization framework.

29.15. Schwerpunkt-Trassierung (Formal Definition)

Definition 29.55 (Schwerpunkt alignment). A Schwerpunkt alignment is a pair

$$(\kappa, \phi)$$

that minimizes the functional

$$J[\kappa, \phi]$$

subject to:

- pose3 dynamics,
- boundary conditions,
- engineering constraints.

Remark 29.56. This turns alignment design into the optimization of a vehicle-related state trajectory rather than the mere composition of geometric elements.

29.16. Interpretation

Remark 29.57. In this formulation:

- the track is no longer the primary design object,
- the motion of the vehicle becomes central.

Remark 29.58. Classical alignment design is recovered as a special case where

$$d = 0 \quad \text{and} \quad \phi(s) = 0.$$

29.17. Discussion

Remark 29.59. The presented model is intentionally simplified. However, it already captures:

- coupling between curvature and cant,
- influence of curvature derivatives,
- relation between geometry and dynamics.

Remark 29.60. Its main contribution is therefore conceptual and structural: it defines a mathematically coherent bridge between alignment geometry and vehicle-oriented design thinking.

RESEARCH: Extension to full multibody vehicle models.

RESEARCH: Calibration using measured track data.

OPEN: Integration into practical design workflows.

29.18. Connection to the AIM Framework

Summary 29.61. The combination of pose3 and a vehicle-based model provides a rigorous foundation for Schwerpunkt-Trassierung.

It transforms alignment design into a problem of optimizing the motion of a representative physical system.

This supports the central AIM idea that alignment is not merely a geometric object, but a functional entity governed by curvature and its physical implications.

Teil VI.

System and Pipeline

30. System and Pipeline

Remark 30.1. This part connects the individual components of the AIM framework into a single system view. It describes alignment modelling as a pipeline of transformations linking coordinate systems, intrinsic track coordinates, modelling, realization, and optimization.

Remark 30.2. The purpose of this part is to show that AIM is not merely a collection of methods, but an operator-based framework in which data, geometry, construction reality, and computation interact.

Remark 30.3. The following chapters introduce the AIM pipeline, its interpretation, its failure modes, and the design principles that follow from it.

30.1. AIM Processing Pipeline

30.1.1. Motivation

Remark 30.4. Railway alignment modelling involves multiple domains:

- global coordinate systems (CRS),
- geometric alignment representation,
- physical realization,
- measurement data,
- optimization.

Remark 30.5. These domains are often treated separately, leading to fragmented models and inconsistent results.

Remark 30.6. Within the AIM framework, these components are unified into a single processing pipeline.

30.1.2. Operator Chain

Definition 30.7 (AIM pipeline). The complete modelling and optimization process is described as a composition of operators:

$$\mathcal{P} = \mathcal{W}^{-1} \circ \mathcal{R} \circ \mathcal{M} \circ \mathcal{W} \circ \mathcal{C}$$

Remark 30.8. The operators are:

- $\mathcal{C}$: CRS transformation,
- $\mathcal{W}$: world-to-track transformation,
- $\mathcal{M}$: alignment model (parametric or hybrid),
- $\mathcal{R}$: realization operator,
- $\mathcal{W}^{-1}$: track-to-world transformation.

—

30.1.3. Detailed Structure

Remark 30.9. The pipeline maps raw data to realized geometry:

CRS data $\xrightarrow{\mathcal{C}}$ world coordinates $\xrightarrow{\mathcal{W}} (s, d) \xrightarrow{\mathcal{M}} \kappa(s) \xrightarrow{\mathcal{R}} \gamma_{\text{real}}(s) \xrightarrow{\mathcal{W}^{-1}}$ world geometry.

—

30.1.4. Hybrid Model Integration

Remark 30.10. The alignment model operator is hybrid:

$$\mathcal{M} : (\mathbf{T}, \delta\kappa) \mapsto \kappa(s).$$

Remark 30.11. Thus, the pipeline operates on both:

- structured parameters,
- local corrections.

—

30.1.5. Inverse Problem

Definition 30.12 (Pipeline inversion). Given observed data x_i , the inverse problem is:

$$\min_{\mathbf{x}} \sum_i \|\mathcal{P}(\mathbf{x})(s_i) - x_i\|^2.$$

Remark 30.13. This corresponds to fitting a model such that the realized geometry matches observations.

—

30.1.6. Optimization Embedding

Remark 30.14. The pipeline is embedded into an optimization problem:

$$\min_{\mathbf{x}} J(\mathcal{P}(\mathbf{x})).$$

Remark 30.15. The objective is evaluated on the realized geometry, not on the parametric model.

30.1.7. Jacobian Structure

Remark 30.16. The derivative of the pipeline follows from the chain rule:

$$\frac{\partial \mathcal{P}}{\partial \mathbf{x}} = D\mathcal{W}^{-1} \cdot D\mathcal{R} \cdot D\mathcal{M} \cdot D\mathcal{W}.$$

Remark 30.17. Each component has specific properties:

- $D\mathcal{M}$: sparse and structured,
 - $D\mathcal{R}$: smoothing operator,
 - $D\mathcal{W}$: nonlinear projection,
 - $D\mathcal{C}$: linear transformation.
-

30.1.8. Computational Implications

Remark 30.18. The pipeline exhibits:

- sparsity (model level),
- nonlinearity (projection level),
- smoothing (realization level),
- scaling effects (CRS level).

Remark 30.19. Efficient implementation requires:

- hybrid representations,
 - block-structured solvers,
 - LOD-dependent evaluation.
-

30.1.9. Level-of-Detail Interpretation

Remark 30.20. Different parts of the pipeline dominate at different scales:

- large scale: CRS and world geometry,
 - medium scale: parametric alignment,
 - small scale: corrections and realization.
-

30.1.10. Key Insight

Proposition 30.21. *Alignment modelling is not a single mapping, but a composition of geometric, physical, and computational operators.*

30.1.11. Connection to AIM

Summary 30.22. The AIM framework is fundamentally defined by its operator pipeline.

It integrates coordinate systems, geometry, physics, and optimization into a single coherent model.

This unified perspective enables:

- consistent data integration,
- physically meaningful optimization,
- scalable computation.

Thus, AIM transforms alignment modelling from a collection of isolated methods into a structured computational system.

30.2. AIM Pipeline Overview

30.3. AIM Master Architecture

30.4. Pipeline Story: From Measurement to Alignment

30.4.1. A Single Point

Remark 30.23. Consider a single measured point x in the real world.

Remark 30.24. It may originate from:

- a surveying campaign,
- a track recording vehicle,

- or a design dataset.

Remark 30.25. At this stage, the point exists only in a global coordinate system.

—

30.4.2. Step 1: CRS Transformation

Remark 30.26. The point is first interpreted within a coordinate reference system:

$$x_{\text{crs}} \xrightarrow{\mathcal{C}} x_{\text{world}}.$$

Remark 30.27. This step ensures that all data is expressed in a consistent spatial frame.

—

30.4.3. Step 2: Projection onto the Track

Remark 30.28. The point is projected onto the alignment:

$$x_{\text{world}} \xrightarrow{\mathcal{W}} (s, d).$$

Remark 30.29. This is a nonlinear operation:

- the longitudinal position s is determined,
- the lateral deviation d is computed.

Remark 30.30. At this moment, the point becomes part of the intrinsic alignment coordinate system.

—

30.4.4. Step 3: Interpretation through the Model

Remark 30.31. The alignment model provides a curvature description:

$$(s, d) \xrightarrow{\mathcal{M}} \kappa(s).$$

Remark 30.32. Here, two components interact:

- the parametric structure (design intent),
- local corrections (data-driven adjustments).

Remark 30.33. The point is no longer treated as isolated data, but as part of a structured geometric system.

—

30.4.5. Step 4: Physical Realization

Remark 30.34. The theoretical model is transformed by physical processes:

$$\kappa(s) \xrightarrow{\mathcal{R}} \gamma_{\text{real}}(s).$$

Remark 30.35. This step reflects:

- construction constraints,
- mechanical behaviour,
- maintenance operations.

Remark 30.36. The resulting geometry is not identical to the theoretical model.

30.4.6. Step 5: Back to World Coordinates

Remark 30.37. The realized alignment is mapped back into world space:

$$\gamma_{\text{real}}(s) \xrightarrow{\mathcal{W}^{-1}} x_{\text{real}}.$$

Remark 30.38. This produces the geometry that is visible and measurable in reality.

30.4.7. Step 6: Comparison and Feedback

Remark 30.39. The original measurement and the realized position are compared:

$$x_{\text{real}} \approx x_{\text{world}}.$$

Remark 30.40. The deviation drives optimization:

- parameters are adjusted,
 - corrections are updated,
 - the pipeline is evaluated again.
-

30.4.8. Interpretation

Remark 30.41. The point has passed through multiple representations:

- global coordinates,
- intrinsic alignment coordinates,

- parametric model,
- realized geometry.

Remark 30.42. Each transformation introduces structure, constraints, and interpretation.

30.4.9. Key Insight

Proposition 30.43. *A measured point is not directly part of the alignment.*

It becomes part of the alignment only after passing through the world-to-track transformation and the alignment model.

30.4.10. Connection to AIM

Summary 30.44. The AIM framework transforms raw spatial data into structured alignment information through a sequence of operators.

This process integrates:

- coordinate systems,
- geometric modelling,
- physical realization,
- optimization.

Thus, alignment modelling is not a static representation, but a dynamic process linking measurement, theory, and construction.

30.5. Failure Modes and Limitations of the AIM Pipeline

30.5.1. Motivation

Remark 30.45. Any modelling and optimization framework must account not only for its capabilities, but also for its limitations.

Remark 30.46. Within the AIM pipeline, failure modes arise from the interaction of:

- coordinate systems,
 - geometric transformations,
 - physical realization,
 - numerical optimization.
-

30.5.2. CRS-Related Errors

Remark 30.47. Errors in coordinate reference systems include:

- incorrect CRS identification,
- inconsistent transformations,
- mixed coordinate systems within datasets.

Proposition 30.48. *CRS inconsistencies lead to systematic spatial distortions that cannot be corrected by local alignment optimization.*

30.5.3. Projection Errors (World-to-Track)

Remark 30.49. The world-to-track transformation may fail due to:

- ambiguous nearest points,
- high curvature regions,
- sparse or noisy alignment representations.

Proposition 30.50. *Projection errors introduce nonlocal distortions in the intrinsic coordinate s , affecting all downstream computations.*

30.5.4. Model Mis-Specification

Remark 30.51. The parametric model may fail to represent the true alignment due to:

- insufficient transition families,
- incorrect parameterization,
- overly restrictive structure.

Remark 30.52. Hybrid correction terms mitigate but do not eliminate this issue.

30.5.5. Overfitting vs. Underfitting

Remark 30.53. Hybrid models introduce a trade-off:

- underfitting: parametric model too rigid,
- overfitting: correction terms capture noise.

Proposition 30.54. *Regularization is required to balance model fidelity and stability.*

30.5.6. Realization Effects

Remark 30.55. The realization operator introduces:

- smoothing of curvature,
- loss of high-frequency information,
- dependence on construction processes.

Proposition 30.56. *Certain features of the target alignment are not physically realizable and will be systematically suppressed.*

30.5.7. Inverse Problem Instability

Remark 30.57. The inverse realization problem is ill-posed:

- multiple target alignments yield similar realized geometry,
- small changes in data may cause large parameter variations.

Proposition 30.58. *Regularization and prior knowledge are essential for stable solutions.*

30.5.8. Numerical Optimization Issues

Remark 30.59. Optimization may fail due to:

- poor initial guesses,
- nonconvex objective functions,
- ill-conditioned Jacobians.

Remark 30.60. Despite structured sparsity, convergence is not guaranteed.

30.5.9. Block Structure Degeneracy

Remark 30.61. The block structure may become degenerate if:

- coupling terms dominate,
- insufficient data separates variables,
- parameters become redundant.

Proposition 30.62. *Loss of block separability reduces computational efficiency and interpretability.*

30.5.10. Measurement Noise and Bias

Remark 30.63. Measurement data may contain:

- random noise,
- systematic bias,
- missing or inconsistent segments.

Proposition 30.64. *Noise propagates through the pipeline and may be amplified by inverse operations.*

30.5.11. Scale-Dependent Failures

Remark 30.65. Different failure modes dominate at different scales:

- large scale: CRS and projection errors,
 - medium scale: model mis-specification,
 - small scale: realization and measurement noise.
-

30.5.12. Non-Uniqueness of Solutions

Remark 30.66. Different alignment models may produce nearly identical realized geometry.

Proposition 30.67. *The mapping*

$$\kappa_{\text{target}} \mapsto \gamma_{\text{real}}$$

is not injective.

30.5.13. Practical Engineering Limits

Remark 30.68. Certain effects are outside the scope of the model:

- material-dependent behaviour,
- long-term degradation,
- construction variability.

Remark 30.69. These factors must be treated as external constraints.

30.5.14. Key Insight

Proposition 30.70. *The AIM pipeline is inherently approximate and limited by both physical and computational constraints.*

30.5.15. Connection to AIM

Summary 30.71. Failure modes in the AIM pipeline arise from the interaction of geometry, physics, data, and computation.

Understanding these limitations is essential for:

- robust model design,
- stable optimization,
- reliable engineering application.

Thus, failure analysis is not an auxiliary component, but an integral part of the AIM framework.

30.6. Design Principles of Alignment-Based Information Modelling

30.6.1. Motivation

Remark 30.72. The AIM framework integrates geometry, coordinate systems, realization, and optimization into a unified model.

Remark 30.73. From this integration, a set of fundamental design principles emerges.

30.6.2. Principle 1: Curvature-Centric Modelling

Proposition 30.74. *Railway alignments should be modelled primarily through curvature functions $\kappa(s)$, not through point-based geometry.*

Remark 30.75. Curvature provides:

- intrinsic representation,
 - direct link to dynamics,
 - natural integration with optimization.
-

30.6.3. Principle 2: Separation of Representation and Realization

Proposition 30.76. *The theoretical **alignment** model must be separated from its physical realization.*

Remark 30.77. The realization operator introduces:

- smoothing,
- constraints,
- deviations from the ideal model.

Remark 30.78. Ignoring this separation leads to systematic modelling errors.

30.6.4. Principle 3: Hybrid Modelling

Proposition 30.79. *Alignment models should combine parametric structure with local correction terms.*

Remark 30.80. This allows:

- interpretable design intent,
 - robust handling of imperfect data,
 - flexible refinement.
-

30.6.5. Principle 4: Operator-Based Architecture

Proposition 30.81. *Alignment modelling should be formulated as a composition of operators:*

$$\mathcal{P} = \mathcal{W}^{-1} \circ \mathcal{R} \circ \mathcal{M} \circ \mathcal{W} \circ \mathcal{C}.$$

Remark 30.82. Each operator represents a distinct domain:

- coordinate systems,
 - geometric modelling,
 - physical processes,
 - data transformation.
-

30.6.6. Principle 5: Design in Realized Space

Proposition 30.83. *Optimization objectives must be evaluated on realized geometry, not on theoretical models.*

Remark 30.84. The relevant mapping is:

$$\kappa_{\text{target}} \mapsto \gamma_{\text{real}}.$$

Remark 30.85. Design must account for the transformation induced by realization.

30.6.7. Principle 6: Exploitation of Structure

Proposition 30.86. *The intrinsic block and sparsity structure of the problem must be exploited for efficient computation.*

Remark 30.87. This includes:

- locality in arc-length,
 - block decomposition (curvature vs. cant),
 - sparse Jacobians.
-

30.6.8. Principle 7: Scale Awareness

Proposition 30.88. *Different components of the pipeline must be evaluated at appropriate scales.*

Remark 30.89. • CRS dominates at large scale,

- parametric modelling at medium scale,
 - corrections and realization at small scale.
-

30.6.9. Principle 8: Measurement Integration

Proposition 30.90. *Measured data must be integrated through the world-to-track transformation, not directly in world space.*

Remark 30.91. This ensures consistency with the intrinsic alignment representation.

30.6.10. Principle 9: Regularization as Necessity

Proposition 30.92. *Regularization is an essential component of alignment modelling and optimization.*

Remark 30.93. It stabilizes:

- inverse problems,
 - hybrid corrections,
 - numerical optimization.
-

30.6.11. Principle 10: Acceptance of Non-Uniqueness

Proposition 30.94. *Alignment solutions are not unique and must be evaluated based on engineering criteria.*

Remark 30.95. Different models may produce nearly identical realized geometry.

Remark 30.96. Thus, solution quality must be defined through:

- dynamics,
 - constraints,
 - robustness.
-

30.6.12. Summary

Summary 30.97. The AIM framework is governed by a set of design principles that emerge from the interaction of geometry, physics, data, and computation.

These principles define:

- how alignments should be represented,
- how they should be transformed,
- how they should be optimized,
- and how their limitations must be handled.

Together, they provide a coherent foundation for alignment-based information modelling as a unified engineering and computational discipline.

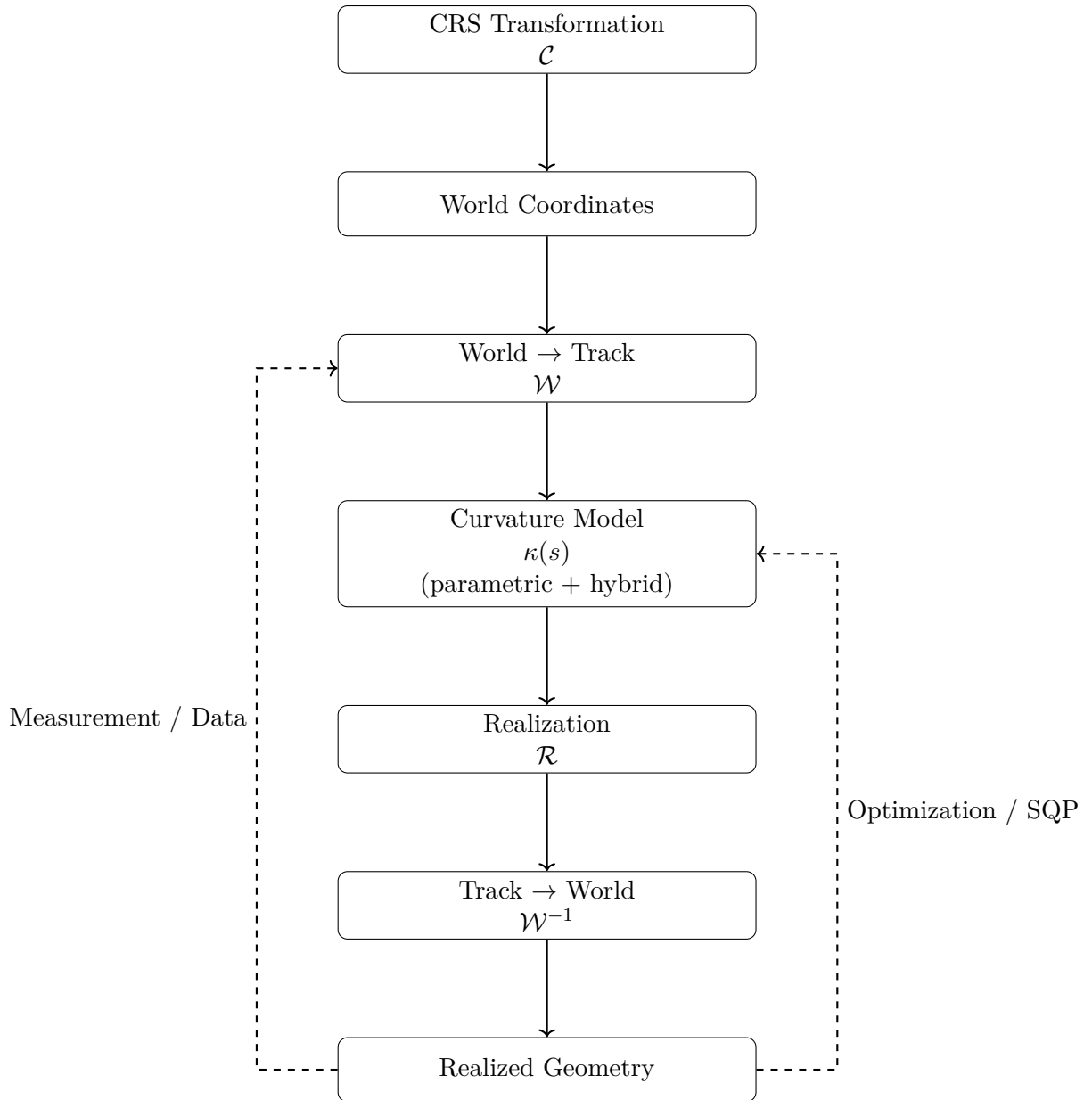


Abbildung 30.1.: AIM processing pipeline as composition of operators linking coordinate systems, alignment modelling, realization, and optimization.

30. System and Pipeline

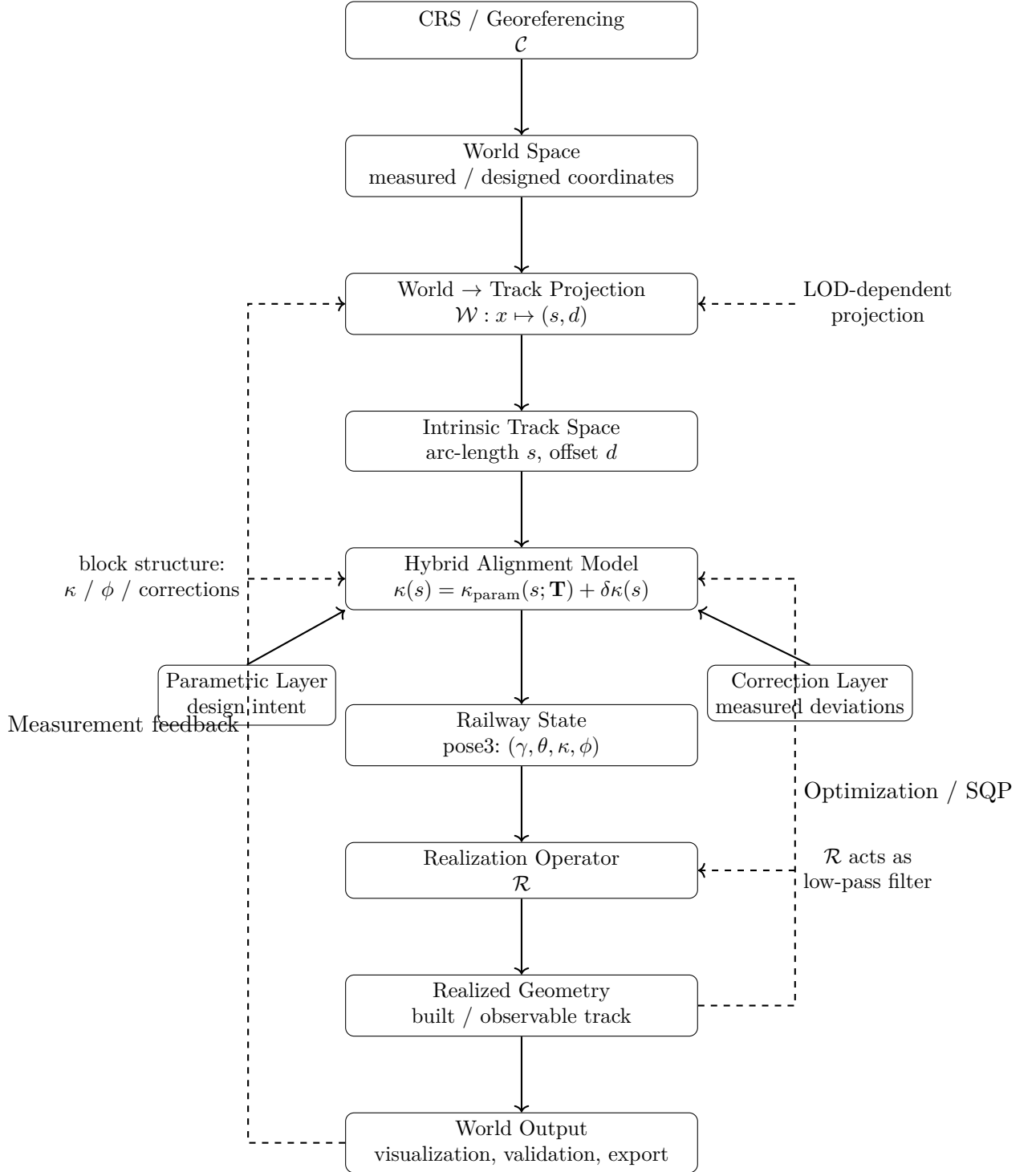


Abbildung 30.2.: Level-2 AIM master architecture. The framework combines CRS handling, nonlinear world-to-track projection, hybrid curvature modelling, pose3 state reconstruction, realization, and optimization feedback. The central modelling object is not the geometric curve itself, but the curvature-driven and realization-aware operator system producing it.

Teil VII.

Optimization

Remark 30.98. This part formulates railway alignment design and reconstruction as optimization problems. Within the AIM framework, optimization is not an external add-on, but a natural consequence of curvature-based modelling.

Remark 30.99. The main focus lies on:

- continuous and discrete formulations,
- structured residual models,
- analytical Jacobians,
- sparse and block-structured solution methods.

Remark 30.100. The following chapters show how alignment modelling leads directly to an efficient and interpretable optimization framework.

31. Numerical Reconstruction

31.1. Motivation

Remark 31.1. In the curvature-based framework, geometric objects are not given explicitly as coordinate sets. Instead, they are reconstructed from curvature information by integration.

Remark 31.2. As a consequence, numerical methods are not merely implementation details, but an essential part of the modelling process.

31.2. Reconstruction Principle

Definition 31.3 (Reconstruction from curvature). Let $\kappa(s)$ be a curvature function and let

$$(x_0, y_0, \theta_0)$$

be a start pose. The corresponding curve is obtained by solving

$$\theta'(s) = \kappa(s), \quad \gamma'(s) = (\cos \theta(s), \sin \theta(s)).$$

Remark 31.4. Even if κ is given analytically, closed-form solutions for γ are generally not available.

31.3. Discretization

Definition 31.5 (Discretization). A discretization of the parameter interval $[0, L]$ is a finite sequence

$$0 = s_0 < s_1 < \cdots < s_n = L.$$

Definition 31.6 (Discrete reconstruction). A discrete reconstruction computes approximate values

$$\gamma(s_i), \quad \theta(s_i)$$

at the discretization points.

Remark 31.7. The choice of discretization strongly influences both computational cost and geometric accuracy.

TODO: Formal criteria for discretization strategies (uniform vs adaptive).

31.4. Integration Methods

Remark 31.8. The reconstruction problem can be formulated as an initial value problem for a system of ordinary differential equations.

31.4.1. Basic Scheme

Definition 31.9 (Explicit step). Given (γ_i, θ_i) at s_i , an explicit step computes

$$\theta_{i+1} = \theta_i + \kappa(s_i) \Delta s,$$

$$\gamma_{i+1} = \gamma_i + (\cos \theta_i, \sin \theta_i) \Delta s.$$

Remark 31.10. This corresponds to a first-order method and is generally insufficient for high-precision reconstruction.

31.4.2. Higher-Order Methods

TODO: Runge–Kutta methods for coupled systems.

TODO: Romberg integration for curvature integration.

TODO: Adaptive step size control.

OPEN: Separation of curvature integration and geometric integration.

31.5. Error Analysis

Definition 31.11 (Local error). The local error of a step is the deviation introduced at a single integration step.

Definition 31.12 (Global error). The global error is the accumulated deviation over the entire interval.

Remark 31.13. Errors arise from:

- discretization,
- numerical integration,
- approximation of curvature laws.

RESEARCH: Quantitative bounds for error propagation in curvature-driven reconstruction.

31.6. Sensitivity

Remark 31.14. The reconstruction process is sensitive to perturbations in curvature. Small deviations in κ may lead to significant geometric drift.

RESEARCH: Sensitivity analysis with respect to curvature perturbations.

OPEN: Relation between curvature noise and positional uncertainty.

31.7. Sampling and Representation

Definition 31.15 (Sampling). Sampling refers to the extraction of a finite set of points from a continuous curve representation.

Remark 31.16. Sampling is required for:

- visualization,
- collision detection,
- export to point-based formats.

TODO: Error-controlled sampling strategies.

OPEN: Difference between sampling for visualization and sampling for computation.

31.8. Reconstruction in the Sparse Model

Remark 31.17. In the sparse alignment model, reconstruction is performed element-wise. Each element contributes a local curvature law, which is integrated sequentially.

Definition 31.18 (Sequential reconstruction). Let $(e_1, \dots, e_n)$ be a sparse alignment. The reconstruction proceeds by integrating each element in sequence, using the end state of the previous element as initial state for the next.

OPEN: Handling of numerical drift across element boundaries.

31.9. Numerical Stability

Remark 31.19. Numerical stability is a central concern, especially for long alignments or highly curved geometries.

RESEARCH: Stability criteria for curvature-based integration.

TODO: Techniques for drift correction and re-normalization.

31.10. Connection to Optimization

Remark 31.20. Numerical reconstruction is closely linked to optimization problems. In fitting and design tasks, reconstruction is performed repeatedly within iterative algorithms.

TODO: SQP formulation for alignment fitting.

RESEARCH: Connection to optimal control problems.

31.11. Summary

Summary 31.21. Numerical reconstruction is the process of deriving geometric representations from curvature data by integration.

It introduces discretization, approximation, and error propagation as intrinsic components of the modelling framework.

Therefore, numerical methods are not merely technical tools, but an essential layer connecting mathematical definitions with practical geometry.

31.12. Numerical Reconstruction from Curvature

31.12.1. Problem Statement

Remark 31.22. Given a curvature function

$$\kappa : [0, L] \rightarrow \mathbb{R},$$

and initial conditions

$$\gamma(0), \quad \theta(0),$$

the geometric alignment is obtained by solving:

$$\gamma'(s) = (\cos \theta(s), \sin \theta(s)), \quad \theta'(s) = \kappa(s).$$

Remark 31.23. This defines a nonlinear initial value problem.

31.12.2. Discrete Approximation

Definition 31.24 (Discretization). Let

$$0 = s_0 < s_1 < \cdots < s_N = L$$

be a partition with step sizes Δs_i .

31. Numerical Reconstruction

Remark 31.25. We approximate:

$$\kappa_i \approx \kappa(s_i).$$

—

31.12.3. Forward Integration Scheme

Definition 31.26 (Explicit integration). Given (γ_i, θ_i) , define:

$$\theta_{i+1} = \theta_i + \kappa_i \Delta s_i,$$

$$\gamma_{i+1} = \gamma_i + \Delta s_i (\cos \theta_i, \sin \theta_i).$$

Remark 31.27. This corresponds to an explicit Euler scheme.

—

31.12.4. Midpoint Scheme

Definition 31.28 (Midpoint method). Define:

$$\theta_{i+\frac{1}{2}} = \theta_i + \frac{1}{2} \kappa_i \Delta s_i,$$

$$\gamma_{i+1} = \gamma_i + \Delta s_i (\cos \theta_{i+\frac{1}{2}}, \sin \theta_{i+\frac{1}{2}}),$$

$$\theta_{i+1} = \theta_i + \kappa_i \Delta s_i.$$

Remark 31.29. This improves accuracy and stability compared to the explicit scheme.

—

31.12.5. Extension to Pose3

Remark 31.30. Including cant, we obtain:

$$\phi_{i+1} = \phi_i + \omega_i \Delta s_i.$$

Remark 31.31. Thus, the full discrete system is:

$$(\gamma_i, \theta_i, \phi_i) \longrightarrow (\gamma_{i+1}, \theta_{i+1}, \phi_{i+1}).$$

—

31.12.6. Numerical Stability

Remark 31.32. The reconstruction is sensitive to:

- step size Δs ,

- noise in κ_i ,
- and large coordinate values.

Remark 31.33. To ensure stability:

- sufficiently small step sizes must be used,
- curvature should be smoothed,
- and local coordinate frames are recommended.

—

31.12.7. Floating Origin Technique

Remark 31.34. For large coordinates, numerical precision degrades.

Definition 31.35 (Floating origin). A local coordinate system is introduced such that:

$$\gamma_i^{\text{local}} = \gamma_i - \gamma_{\text{ref}}.$$

Remark 31.36. This reduces numerical errors in computation and visualization.

—

31.12.8. Consistency with CRS

Remark 31.37. Reconstruction depends on:

- initial position,
- initial orientation,
- and coordinate system definition.

Remark 31.38. Thus, numerical reconstruction must be performed within a consistent CRS.

—

31.12.9. Interpretation

Summary 31.39. Numerical reconstruction provides the link between curvature-based models and geometric representation.

It transforms abstract functions into concrete spatial geometry, while preserving the structure required for analysis and optimization.

32. Optimization of Alignments

32.1. Motivation

Remark 32.1. In practical applications, alignments are not only reconstructed from data, but also designed, modified, and improved.

Remark 32.2. This leads naturally to optimization problems in which alignment parameters are adjusted to satisfy geometric, physical, and regulatory constraints, as already emphasized in railway alignment literature [30, 24].

32.2. General Optimization Problem

Definition 32.3 (Alignment optimization problem). An alignment optimization problem consists of:

- a parameterized alignment model,
- an objective function,
- a set of constraints.

Definition 32.4 (Objective function). An objective function is a mapping

$$J : \mathcal{A} \rightarrow \mathbb{R}$$

that assigns a cost or quality measure to an alignment $\mathcal{A}$.

Remark 32.5. Typical objectives include:

- minimizing curvature variation,
- minimizing deviation from measurement data,
- minimizing construction cost,
- maximizing comfort.

Such objective classes are consistent with both general numerical optimization literature [36] and railway-specific optimization approaches [30].

32.3. Parameterization

Remark 32.6. The choice of parameters is central to the formulation of the optimization problem.

Definition 32.7 (Parameter vector). An alignment is described by a parameter vector

$$\mathbf{x} \in \mathbb{R}^n$$

encoding quantities such as:

- element lengths,
- curvature values,
- transition parameters,
- geometric constraints.

Remark 32.8. The sparse alignment model provides a natural parameterization by element-wise variables.

TODO: Explicit parameter sets for fixed and transition elements.

32.4. Constraints

Definition 32.9 (Constraint). A constraint is a condition of the form

$$g_i(\mathbf{x}) \leq 0, \quad h_j(\mathbf{x}) = 0.$$

Remark 32.10. Constraints arise from:

- geometric continuity,
- engineering design rules,
- physical limitations,
- regulatory requirements.

32.4.1. Geometric Constraints

- continuity of position and direction,
- continuity of curvature.

32.4.2. Engineering Constraints

- minimum radius,
- maximum cant,
- limits on curvature derivatives.

Remark 32.11. Such conditions reflect standard railway design practice and are closely related to design rules discussed in engineering literature and standards [24, 6].

OPEN: Formal representation of engineering rules in constraint form.

32.5. Relation to Numerical Reconstruction

Remark 32.12. Evaluation of the objective function requires reconstruction of the alignment geometry from the parameter vector.

Remark 32.13. Thus, optimization is intrinsically linked to numerical reconstruction, as discussed in the previous chapter. From a numerical perspective, this places alignment optimization within the broader context of differential equations and numerical approximation [11, 41].

32.6. Sequential Quadratic Programming

Remark 32.14. Sequential Quadratic Programming (SQP) is a powerful method for solving nonlinear constrained optimization problems [36].

Definition 32.15 (SQP iteration). An SQP method approximates the original problem by a sequence of quadratic subproblems.

Remark 32.16. Each iteration solves a quadratic approximation of the objective function subject to linearized constraints.

TODO: Explicit formulation of SQP for alignment parameters.

RESEARCH: Adaptation of SQP to sparse alignment structures.

32.7. Alignment Optimization as Control Problem

Remark 32.17. The curvature function $\kappa(s)$ may be interpreted as a control variable in a continuous system.

Definition 32.18 (Control formulation). The alignment problem can be formulated as an optimal control problem with state variables (γ, θ) and control κ .

Remark 32.19. This viewpoint is compatible with the curvature-centred perspective on railway alignment advocated by Kufver [31, 30].

RESEARCH: Connection between alignment optimization and optimal control theory.

32.8. Network-Level Optimization

Remark 32.20. Real-world railway systems involve not only single alignments, but networks with branching and merging structures.

RESEARCH: Extension of optimization from single alignments to networks.

OPEN: Handling of switches, crossings, and parallel tracks.

32.9. Robustness and Uncertainty

Remark 32.21. Optimization must account for uncertainty in input data.

RESEARCH: Robust optimization under uncertain geometry.

OPEN: Trade-off between optimality and robustness.

32.10. Implementation Perspective

Remark 32.22. Practical optimization systems must balance:

- numerical stability,
- computational efficiency,
- model expressiveness.

Remark 32.23. This balance is central both in generic numerical optimization [36] and in railway-specific optimization settings [30].

TODO: Integration of optimization into the modelling workflow.

32.11. Historical Context

Remark 32.24. Classical systems such as AXTRAN have demonstrated the feasibility of optimization-based alignment design.

Remark 32.25. Modern approaches aim to extend these ideas using more flexible models and improved numerical methods, in line with the development from classical railway design methods toward explicit optimization formulations [30].

TODO: Reconstruction and generalization of AXTRAN-like methods.

32.12. Formal Optimization in Curvature Space

32.12.1. Problem Setting

Remark 32.26. Let $L > 0$ be fixed and let

$$\kappa : [0, L] \rightarrow \mathbb{R}$$

32. Optimization of Alignments

denote the curvature function of a transition.

Remark 32.27. We assume the boundary conditions

$$\kappa(0) = 0, \quad \kappa(L) = \kappa_1,$$

and optionally

$$\kappa'(0) = 0, \quad \kappa'(L) = 0.$$

32.12.2. Admissible Set

Definition 32.28 (Admissible transition set).

$$\mathcal{A} = \left\{ \kappa \in H^1(0, L) \mid \kappa(0) = 0, \kappa(L) = \kappa_1 \right\}.$$

32.12.3. Variational Problem

Definition 32.29 (Quadratic curvature-gradient functional).

$$J[\kappa] = \int_0^L (\kappa'(s))^2 ds.$$

Definition 32.30 (Optimization problem). Find $\kappa^* \in \mathcal{A}$ such that

$$J[\kappa^*] = \min_{\kappa \in \mathcal{A}} J[\kappa].$$

32.12.4. Euler–Lagrange Equation

Proposition 32.31. *If κ^* is a minimizer, then*

$$\kappa^{*''}(s) = 0.$$

32.12.5. Interpretation

Proposition 32.32. *The minimizer is*

$$\kappa^*(s) = \frac{\kappa_1}{L} s.$$

Remark 32.33. Thus, the clothoid appears as the solution of a variational problem. This connects classical railway transition design with an optimization view in curvature space and provides a formal counterpart to the historical role of the clothoid in railway engineering [9, 13].

32.12.6. Higher-Order Model

Definition 32.34.

$$J_2[\kappa] = \int_0^L (\kappa''(s))^2 ds.$$

32. Optimization of Alignments

Proposition 32.35. *Minimizers satisfy*

$$\kappa^{(4)}(s) = 0.$$

Remark 32.36. This explains the occurrence of cubic transition laws and links them to modern polynomial transition design [51].

32.12.7. Generalized View

Remark 32.37. General functionals of the form

$$J[\kappa] = \int_0^L F(\kappa, \kappa', \kappa''; v, c) \, ds$$

allow incorporation of dynamic and engineering criteria.

32.12.8. Connection to the Main Thesis

Summary 32.38. Transition design can be formulated as an optimization problem in curvature space.

Classical transition curves arise as solutions of such problems, supporting the interpretation of transitions as curvature control functions.

32.13. Coupled Optimization of Curvature and Cant

32.13.1. Motivation

Remark 32.39. In railway applications, curvature and cant cannot be optimized independently. Both jointly determine the effective lateral acceleration and its rate of change.

Remark 32.40. This motivates a coupled optimization problem in which both

$$\kappa(s) \quad \text{and} \quad \phi(s)$$

are treated as design variables.

Remark 32.41. Such coupling is fully consistent with railway alignment design standards, where curvature, speed, and cant are not independent design quantities [6, 24].

32.13.2. Effective Acceleration Functional

Definition 32.42 (Effective lateral acceleration). The effective lateral acceleration is

$$a_{\text{eff}}(s) = v^2 \kappa(s) - g \sin \phi(s).$$

Remark 32.43. A natural optimization objective is to reduce the magnitude of unbalanced lateral acceleration.

32. Optimization of Alignments

Definition 32.44 (Acceleration-based functional). Define

$$J_a[\kappa, \phi] = \int_0^L a_{\text{eff}}(s)^2 ds = \int_0^L (v^2 \kappa(s) - g \sin \phi(s))^2 ds.$$

32.13.3. Jerk-Based Functional

Definition 32.45 (Jerk). The jerk is given by

$$j(s) = v^3 \kappa'(s) - gv \cos \phi(s) \phi'(s).$$

Definition 32.46 (Jerk functional). Define

$$J_j[\kappa, \phi] = \int_0^L j(s)^2 ds.$$

That is,

$$J_j[\kappa, \phi] = \int_0^L (v^3 \kappa'(s) - gv \cos \phi(s) \phi'(s))^2 ds.$$

Remark 32.47. This functional penalizes rapid variation of effective acceleration and therefore provides a natural dynamic smoothness criterion. Such criteria are closely related to engineering concerns about comfort and dynamic compatibility [6, 4].

32.13.4. Combined Objective

Definition 32.48 (Coupled objective functional). A general coupled objective may be written as

$$J[\kappa, \phi] = \alpha J_a[\kappa, \phi] + \beta J_j[\kappa, \phi] + \gamma J_\kappa[\kappa] + \delta J_\phi[\phi],$$

where:

$$J_\kappa[\kappa] = \int_0^L (\kappa'(s))^2 ds, \quad J_\phi[\phi] = \int_0^L (\phi'(s))^2 ds.$$

Remark 32.49. The weights

$$\alpha, \beta, \gamma, \delta \geq 0$$

determine the relative importance of equilibrium, comfort, curvature smoothness, and cant smoothness.

32.13.5. Admissible Set

Definition 32.50 (Admissible railway state set). Let

$$\mathcal{B} = \left\{ (\kappa, \phi) \mid \kappa \in H^1(0, L), \phi \in H^1(0, L), \kappa(0) = 0, \kappa(L) = \kappa_1 \right\}.$$

Remark 32.51. Additional boundary conditions may be imposed, for example:

$$\phi(0) = 0, \quad \phi(L) = \phi_1,$$

32. Optimization of Alignments

or derivative constraints such as

$$\kappa'(0) = \kappa'(L) = 0, \quad \phi'(0) = \phi'(L) = 0.$$

32.13.6. Coupled Optimization Problem

Definition 32.52 (Coupled railway optimization problem). Find

$$(\kappa^*, \phi^*) \in \mathcal{B}$$

such that

$$J[\kappa^*, \phi^*] = \min_{(\kappa, \phi) \in \mathcal{B}} J[\kappa, \phi].$$

32.13.7. Interpretation

Remark 32.53. This problem extends classical transition design in two directions:

- from pure geometry to coupled geometry–cant design,
- from curve selection to functional optimization.

Remark 32.54. In this framework, the transition curve is no longer chosen from a catalogue, but emerges as part of an optimal coupled state.

32.13.8. Special Cases

Proposition 32.55. *If $\phi(s) \equiv 0$, the coupled problem reduces to a pure curvature optimization problem.*

Proposition 32.56. *If $a_{\text{eff}}(s) = 0$ for all s , then*

$$v^2 \kappa(s) = g \sin \phi(s),$$

and the design is perfectly balanced.

Remark 32.57. These special cases show that classical equilibrium design appears as a subcase of the coupled optimization framework.

32.13.9. Relation to Optimal Control

Remark 32.58. The coupled optimization problem may be interpreted as an optimal control problem with controls

$$\kappa(s), \quad \phi'(s),$$

state variables

$$x(s), y(s), \theta(s), \phi(s),$$

and cost functional $J[\kappa, \phi]$.

32. Optimization of Alignments

Remark 32.59. This provides a direct bridge from railway alignment design to modern control-theoretic and variational methods.

32.13.10. Connection to the Main Thesis

Summary 32.60. The coupled optimization problem shows that railway alignment design is not a purely geometric task.

Instead, it is the optimization of a coupled state system in which curvature and cant jointly determine dynamic behaviour.

This supports the central thesis of the manuscript that transition curves and cant evolution should be understood as control functions in a curvature-based railway state model.

32.14. Summary

Summary 32.61. Alignment optimization treats geometry as a variable rather than a fixed input.

It combines parameterization, numerical reconstruction, and constrained optimization methods to produce alignments that satisfy both geometric and engineering requirements.

32.15. Discrete Formulation of Alignment Optimization

32.15.1. Discretization of the Domain

Definition 32.62 (Discretization). Let

$$0 = s_0 < s_1 < \cdots < s_N = L$$

be a partition of the interval $[0, L]$.

Remark 32.63. We denote

$$\Delta s_i = s_{i+1} - s_i.$$

32.15.2. Discrete Variables

Definition 32.64 (Discrete curvature). Let

$$\kappa_i \approx \kappa(s_i)$$

denote the curvature values at the nodes.

Definition 32.65 (Discrete cant). Let

$$\phi_i \approx \phi(s_i)$$

32. Optimization of Alignments

denote the cant values.

Definition 32.66 (Parameter vector). The optimization variable is

$$\mathbf{x} = (\kappa_0, \dots, \kappa_N, \phi_0, \dots, \phi_N) \in \mathbb{R}^{2(N+1)}.$$

32.15.3. Discrete Pose Reconstruction

Remark 32.67. The pose3 state is reconstructed by numerical integration.

Definition 32.68 (Discrete integration). Given initial conditions

$$\gamma_0, \theta_0, \phi_0,$$

we compute iteratively:

$$\theta_{i+1} = \theta_i + \kappa_i \Delta s_i,$$

$$\gamma_{i+1} = \gamma_i + \Delta s_i (\cos \theta_i, \sin \theta_i),$$

$$\phi_{i+1} = \phi_i + \omega_i \Delta s_i.$$

Remark 32.69. Here, ω_i may be expressed via differences of ϕ_i . This discrete reconstruction corresponds to standard numerical integration ideas [11, 41].

32.15.4. Finite Differences

Definition 32.70 (First derivative). Approximate derivatives by

$$\kappa'(s_i) \approx \frac{\kappa_{i+1} - \kappa_i}{\Delta s_i}, \quad \phi'(s_i) \approx \frac{\phi_{i+1} - \phi_i}{\Delta s_i}.$$

Definition 32.71 (Second derivative).

$$\kappa''(s_i) \approx \frac{\kappa_{i+1} - 2\kappa_i + \kappa_{i-1}}{(\Delta s_i)^2}.$$

32.15.5. Discrete Objective Functions

Curvature Smoothness

$$J_\kappa(\mathbf{x}) = \sum_{i=0}^{N-1} \left(\frac{\kappa_{i+1} - \kappa_i}{\Delta s_i} \right)^2 \Delta s_i.$$

Cant Smoothness

$$J_\phi(\mathbf{x}) = \sum_{i=0}^{N-1} \left(\frac{\phi_{i+1} - \phi_i}{\Delta s_i} \right)^2 \Delta s_i.$$

Dynamic Objective

$$J_{\text{dyn}}(\mathbf{x}) = \sum_{i=0}^N (v^2 \kappa_i - g \sin \phi_i)^2 \Delta s_i.$$

Schwerpunkt Objective

Remark 32.72. Using finite differences, we approximate

$$\|\gamma_c''(s_i)\|^2$$

by second-order differences of γ_c .

$$J_{\text{veh}}(\mathbf{x}) = \sum_{i=1}^{N-1} \|\gamma_{c,i+1} - 2\gamma_{c,i} + \gamma_{c,i-1}\|^2.$$

32.15.6. Combined Discrete Problem

Definition 32.73 (Discrete objective).

$$J(\mathbf{x}) = \alpha J_{\text{veh}} + \beta J_{\text{dyn}} + \gamma J_{\kappa} + \delta J_{\phi}.$$

32.15.7. Constraints

Definition 32.74 (Equality constraints).

$$\kappa_0 = 0, \quad \kappa_N = \kappa_1.$$

Definition 32.75 (Inequality constraints).

$$|\kappa_i| \leq \kappa_{\max}, \quad |\phi_i| \leq \phi_{\max}.$$

Remark 32.76. Further constraints may include:

- bounds on derivatives,
- continuity conditions,
- alignment anchoring points.

32.15.8. Final Optimization Problem

Definition 32.77 (Discrete optimization problem). Find

$$\mathbf{x}^* \in \mathbb{R}^{2(N+1)}$$

such that

$$J(\mathbf{x}^*) = \min_{\mathbf{x}} J(\mathbf{x})$$

subject to the given constraints.

32.15.9. Structure of the Problem

Remark 32.78. The problem has the structure of a nonlinear least-squares problem.

Remark 32.79. It is therefore well suited for:

- Sequential Quadratic Programming,
- Gauss–Newton methods,
- sparse optimization techniques.

32.15.10. Connection to Implementation

Remark 32.80. The discrete formulation directly corresponds to a computational model:

- κ_i correspond to element parameters,
- ϕ_i correspond to cant samples,
- reconstruction matches numerical evaluation in software.

Summary 32.81. The continuous alignment optimization problem can be reduced to a finite dimensional nonlinear optimization problem.

This provides the direct link between theoretical modelling and practical implementation.

32.16. Sequential Quadratic Programming (SQP)

32.16.1. Problem Structure

Remark 32.82. We consider the discrete optimization problem

$$\min_{\mathbf{x}} J(\mathbf{x}) \quad \text{subject to} \quad g(\mathbf{x}) = 0, \quad h(\mathbf{x}) \leq 0.$$

Definition 32.83 (Parameter vector).

$$\mathbf{x} = (\kappa_0, \dots, \kappa_N, \phi_0, \dots, \phi_N).$$

32.16.2. Residual Formulation

Definition 32.84 (Residual vector).

$$J(\mathbf{x}) = \frac{1}{2} \|\mathbf{r}(\mathbf{x})\|^2.$$

Remark 32.85. Typical residual components:

$$r_i^{(\kappa)} = \sqrt{\gamma} \frac{\kappa_{i+1} - \kappa_i}{\Delta s}, \quad r_i^{(\phi)} = \sqrt{\delta} \frac{\phi_{i+1} - \phi_i}{\Delta s},$$

$$r_i^{(\text{dyn})} = \sqrt{\beta} (v^2 \kappa_i - g \sin \phi_i).$$

32.16.3. Gauss–Newton Step

Definition 32.86 (SQP step). At iteration k , compute $\mathbf{p}_k$ from:

$$(J_r^T J_r) \mathbf{p}_k = -J_r^T \mathbf{r}.$$

Remark 32.87. The Hessian is approximated by:

$$H \approx J_r^T J_r.$$

32.16.4. Sparse Structure

Remark 32.88. The system matrix has band structure:

- tridiagonal coupling for smoothness terms,
- diagonal contributions from dynamic terms.

32.16.5. Constraints Handling

Remark 32.89. Constraints are handled via:

- projection,
- penalty terms,
- Lagrange multipliers.

32.16.6. Iteration Scheme

Algorithm 1 SQP iteration

```

initialize  $\mathbf{x}_0$ 
for  $k = 0, 1, \dots$  do
  compute  $\mathbf{r}(\mathbf{x}_k)$ 
  compute  $J_r(\mathbf{x}_k)$ 
  solve  $(J_r^T J_r) \mathbf{p}_k = -J_r^T \mathbf{r}$ 
  choose step size  $\alpha_k$ 
  update  $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k$ 
end for
```

32.16.7. Interpretation

Summary 32.90. SQP provides a practical and efficient method for solving the discrete alignment optimization problem.

Its efficiency follows from the sparse and local structure of the underlying model.

32.17. Construction of Initial Guesses

32.17.1. Motivation

Remark 32.91. The performance of SQP methods strongly depends on the choice of the initial parameter vector.

Remark 32.92. In railway applications, initial data is often available in the form of:

- measurement polylines,
- existing alignments,
- partially structured design data.

32.17.2. Input Representation

Definition 32.93 (Input polyline). Let

$$P = (p_0, \dots, p_M), \quad p_i \in \mathbb{R}^2$$

denote a measured or imported polyline.

32.17.3. Arc-Length Parameterization

Definition 32.94. Define cumulative arc-length

$$s_0 = 0, \quad s_{i+1} = s_i + \|p_{i+1} - p_i\|.$$

Remark 32.95. This defines a parameterization of the input data.

32.17.4. Initial Curvature Estimation

Definition 32.96 (Discrete curvature). For three consecutive points, define:

$$\kappa_i \approx \frac{\theta_{i+1} - \theta_i}{\Delta s_i},$$

where

$$\theta_i = \arg(p_{i+1} - p_i).$$

Remark 32.97. This provides a first estimate of curvature from geometric data.

32.17.5. Smoothing

Remark 32.98. Raw curvature estimates are typically noisy.

Definition 32.99 (Smoothing operator). Apply a smoothing filter:

$$\kappa_i^{(0)} = \sum_j w_{ij} \kappa_j.$$

32. Optimization of Alignments

Remark 32.100. Possible choices:

- moving average,
- Gaussian filter,
- low-pass filtering.

32.17.6. Initial Cant Estimation

Remark 32.101. Cant may be initialized based on equilibrium:

$$\phi_i^{(0)} = \arcsin \left(\frac{v^2 \kappa_i^{(0)}}{g} \right).$$

Remark 32.102. Alternatively, cant may be:

- set to zero,
- taken from input data,
- approximated by design rules.

32.17.7. Projection to Discrete Grid

Definition 32.103. Interpolate the values $\kappa_i^{(0)}, \phi_i^{(0)}$ onto the optimization grid:

$$s_0, \dots, s_N.$$

32.17.8. Heuristic Segmentation

Remark 32.104. The input alignment may be segmented into:

- straight sections,
- circular arcs,
- transition regions.

Definition 32.105 (Segment detection). A segment is classified based on curvature:

$$\kappa_i \approx 0 \quad \Rightarrow \text{straight},$$

$$\kappa_i \approx \text{const} \quad \Rightarrow \text{arc}.$$

Remark 32.106. This structure can be used to improve the initial guess and is compatible with railway alignment reconstruction approaches such as those discussed by Kufver [31, 30].

32.17.9. Final Initial Guess

Definition 32.107 (Initial parameter vector). The initial guess is

$$\mathbf{x}_0 = (\kappa_0^{(0)}, \dots, \kappa_N^{(0)}, \phi_0^{(0)}, \dots, \phi_N^{(0)}).$$

32.17.10. Interpretation

Remark 32.108. The initial guess combines:

- measured geometry,
- smoothing,
- physical approximation.

Summary 32.109. A robust initial guess transforms raw geometric input into a structured parameter vector suitable for optimization.

This step is essential for bridging real-world data and numerical methods.

32.18. Analytical Structure of the Optimization Problem

32.18.1. Motivation

Remark 32.110. In classical numerical optimization, Jacobians are often computed by finite differences or automatic differentiation.

Remark 32.111. Within the AIM framework, this is not necessary. The alignment model is inherently differentiable.

32.18.2. Fundamental Observation

Proposition 32.112. *All state variables are obtained via arc-length integration:*

$$\begin{aligned}\theta(s) &= \theta_0 + \int_0^s \kappa(\sigma) \, d\sigma, \\ \gamma(s) &= \gamma_0 + \int_0^s (\cos \theta, \sin \theta) \, d\sigma.\end{aligned}$$

32.18.3. Parameter Types

Definition 32.113. Primary parameter classes:

- curvature parameters dK ,
- length parameters dL ,
- transition parameters dT .

32.18.4. Analytical Derivatives

Proposition 32.114.

$$\frac{\partial \theta(s)}{\partial p} = \int_0^s \frac{\partial \kappa}{\partial p} d\sigma.$$

Proposition 32.115.

$$\frac{\partial \gamma(s)}{\partial p} = \int_0^s \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix} \frac{\partial \theta}{\partial p} d\sigma.$$

Remark 32.116. All derivatives reduce to arc-length integrals.

32.18.5. Key Insight

Proposition 32.117. *The optimization problem admits exact Jacobians without numerical approximation.*

Remark 32.118. This eliminates:

- finite differences,
- numerical instability,
- high computational cost.

32.18.6. Block Structure

Remark 32.119. The parameter vector decomposes as:

$$(\kappa, \phi).$$

Remark 32.120. This induces a block structure:

$$H = \begin{pmatrix} H_{\kappa\kappa} & H_{\kappa\phi} \\ H_{\phi\kappa} & H_{\phi\phi} \end{pmatrix}.$$

32.18.7. Interpretation

Summary 32.121. The structure of the optimization problem is not imposed numerically, but follows directly from the curvature-based modelling approach.

This yields:

- exact derivatives,
- sparse matrices,
- natural block structure.

This is a central advantage of the AIM framework.

32.19. Parametric Representation instead of Sampling

32.19.1. Motivation

Remark 32.122. Classical discretization represents curvature and cant by samples:

$$\kappa_i \approx \kappa(s_i), \quad \phi_i \approx \phi(s_i).$$

Remark 32.123. This leads to high-dimensional optimization problems with weak structure and limited interpretability.

Remark 32.124. Within the AIM framework, curvature and cant are instead represented by parameterized functions.

—

32.19.2. Parametric Curvature Model

Definition 32.125 (Parametric curvature). A transition is described by:

$$\kappa(s) = \kappa_0 + (\kappa_1 - \kappa_0) \hat{\kappa}\left(\frac{s}{L}, \mathbf{T}\right),$$

where:

- L is the element length,
- $\mathbf{T}$ is a vector of transition parameters,
- $\hat{\kappa}$ is a normalized prototype function.

—

32.19.3. Parametric Cant Model

Definition 32.126 (Parametric cant). Cant is represented analogously:

$$\phi(s) = \phi_0 + (\phi_1 - \phi_0) \hat{\phi}\left(\frac{s}{L}, \mathbf{T}_\phi\right).$$

—

32.19.4. Parameter Vector

Definition 32.127 (Reduced parameter vector). The optimization variable becomes:

$$\mathbf{x} = (L, \kappa_0, \kappa_1, \mathbf{T}, \phi_0, \phi_1, \mathbf{T}_\phi).$$

Remark 32.128. The dimension is independent of discretization density.

—

32.19.5. Dimensional Reduction

Remark 32.129. A sampled representation uses $O(N)$ variables.

Remark 32.130. The parametric representation uses:

$$O(1) \text{ parameters per element.}$$

Remark 32.131. This yields a drastic reduction in problem size.

—

32.19.6. Derivative Structure

Proposition 32.132. *Derivatives reduce to:*

$$\frac{\partial \kappa}{\partial p} = (\kappa_1 - \kappa_0) \frac{\partial \hat{\kappa}}{\partial p}.$$

Remark 32.133. Thus, all derivatives are computed analytically from prototype functions.

—

32.19.7. Integration into Reconstruction

Remark 32.134. The parametric representation is evaluated through:

$$\theta(s) = \int \kappa(s) ds, \quad \gamma(s) = \int (\cos \theta, \sin \theta) ds.$$

Remark 32.135. This yields a smooth and consistent geometry without discretization artifacts.

—

32.19.8. Comparison to Sample-Based Methods

	Sample-based	Parametric
dimension	high	low
smoothness	enforced numerically	intrinsic
interpretability	low	high
constraints	many local	few global

—

32.19.9. Interpretation

Remark 32.136. The parametric approach shifts the problem from:

- fitting values

to:

- designing functions.

—

32.19.10. Connection to Transition Families

Remark 32.137. Different transition curves correspond to different prototype functions $\hat{\kappa}$.

Remark 32.138. Thus, the optimization problem includes selection and tuning of transition families.

—

32.19.11. Computational Implications

- smaller optimization problems,
- better conditioning,
- faster convergence,
- direct integration with analytical Jacobians.

—

32.19.12. Connection to Classical Systems

Remark 32.139. This formulation is conceptually close to classical systems such as AXTRAN, which operate on element parameters rather than sampled data.

—

32.19.13. Connection to AIM

Summary 32.140. The parametric representation replaces discrete sampling by structured functions.

It enables a compact, interpretable, and analytically differentiable model, which forms the basis for efficient and robust alignment optimization.

This approach is central to the AIM philosophy.

32.20. Hybrid Parametric–Discrete Model

32.20.1. Motivation

Remark 32.141. Purely parametric models provide low-dimensional structure but may lack flexibility for fitting real-world data.

32. Optimization of Alignments

Remark 32.142. Purely sampled models provide flexibility but lead to large and ill-conditioned optimization problems.

Remark 32.143. A hybrid approach combines both advantages.

—

32.20.2. Hybrid Representation

Definition 32.144 (Hybrid curvature model).

$$\kappa(s) = \kappa_{\text{param}}(s; \mathbf{T}) + \delta\kappa(s),$$

where:

- κ_{param} is a parametric base model,
- $\delta\kappa(s)$ is a correction term.

Definition 32.145 (Hybrid cant model).

$$\phi(s) = \phi_{\text{param}}(s; \mathbf{T}_\phi) + \delta\phi(s).$$

—

32.20.3. Discretization of Correction

Remark 32.146. The correction term is represented discretely:

$$\delta\kappa_i = \delta\kappa(s_i), \quad \delta\phi_i = \delta\phi(s_i).$$

—

32.20.4. Extended Parameter Vector

Definition 32.147.

$$\mathbf{x} = (\mathbf{T}, \mathbf{T}_\phi, \delta\kappa, \delta\phi).$$

—

32.20.5. Regularization of Corrections

Remark 32.148. To avoid overfitting, the correction terms are penalized:

$$J_\delta = \int_0^L (\delta\kappa'(s))^2 ds + \int_0^L (\delta\phi'(s))^2 ds.$$

Remark 32.149. This enforces smooth, small deviations from the parametric model.

—

32.20.6. Residual Structure

Remark 32.150. The residual now consists of:

- parametric contributions,
 - correction contributions,
 - data fitting terms.
-

32.20.7. Block Structure

Remark 32.151. The parameter vector decomposes as:

$$(\mathbf{T}, \delta\boldsymbol{\kappa}) \quad \text{and} \quad (\mathbf{T}_\phi, \delta\boldsymbol{\phi}).$$

Remark 32.152. This yields a multi-level block structure:

- coarse scale (parameters),
 - fine scale (corrections).
-

32.20.8. Interpretation

Remark 32.153. The parametric model captures:

- geometric intent,
- transition structure,
- engineering semantics.

Remark 32.154. The correction term captures:

- measurement noise,
 - local deviations,
 - modelling imperfections.
-

32.20.9. Connection to Measurement Data

Remark 32.155. Measured polylines can be fitted by minimizing:

$$\sum_i \|\gamma(s_i) - p_i\|^2,$$

with the hybrid representation.

Remark 32.156. The parametric part explains the global structure, while the correction absorbs residual errors.

32.20.10. Computational Strategy

- first optimize parametric variables,
 - then refine using correction terms,
 - optionally alternate between both levels.
-

32.20.11. Relation to LOD

Remark 32.157. The hybrid model naturally supports level-of-detail concepts:

- coarse LOD: parametric model only,
 - fine LOD: parametric + corrections.
-

32.20.12. Connection to AIM

Summary 32.158. The hybrid representation combines structured modelling and local flexibility.

It enables robust alignment reconstruction from imperfect data while preserving the interpretability of parametric models.

This approach bridges engineering design and data-driven fitting within the AIM framework.

32.21. Hybrid Representation and Block Structure

32.21.1. Motivation

Remark 32.159. Purely parametric models provide a low-dimensional and interpretable representation of alignments, but lack flexibility when fitting real-world data.

32. Optimization of Alignments

Remark 32.160. Purely sampled models provide maximum flexibility, but lead to high-dimensional and often ill-conditioned optimization problems.

Remark 32.161. A hybrid formulation combines both advantages.

32.21.2. Hybrid Representation

Definition 32.162 (Hybrid curvature model).

$$\kappa(s) = \kappa_{\text{param}}(s; \mathbf{T}) + \delta\kappa(s),$$

where:

- κ_{param} is a structured parametric model,
- $\delta\kappa(s)$ is a correction term.

Definition 32.163 (Hybrid cant model).

$$\phi(s) = \phi_{\text{param}}(s; \mathbf{T}_\phi) + \delta\phi(s).$$

32.21.3. Discretization of Corrections

Remark 32.164. The correction terms are represented discretely:

$$\delta\kappa_i = \delta\kappa(s_i), \quad \delta\phi_i = \delta\phi(s_i).$$

Definition 32.165 (Extended parameter vector).

$$\mathbf{x} = (\mathbf{T}, \mathbf{T}_\phi, \delta\kappa, \delta\phi).$$

32.21.4. Regularization

Remark 32.166. To avoid overfitting, correction terms are regularized:

$$J_\delta = \int_0^L (\delta\kappa'(s))^2 ds + \int_0^L (\delta\phi'(s))^2 ds.$$

Remark 32.167. Thus, the correction captures only smooth, physically meaningful deviations.

32.21.5. Residual Structure

Remark 32.168. The residual vector consists of:

- parametric contributions,
- correction contributions,
- physical constraints,

32. Optimization of Alignments

- data fitting terms.

32.21.6. Block Structure of Variables

Remark 32.169. The parameter vector decomposes naturally as:

$$(\mathbf{T}, \delta\boldsymbol{\kappa}) \quad \text{and} \quad (\mathbf{T}_\phi, \delta\boldsymbol{\phi}).$$

Remark 32.170. This yields a two-level structure:

- coarse scale: parametric variables,
- fine scale: correction variables.

32.21.7. Jacobian Block Structure

Definition 32.171 (Block Jacobian). The Jacobian takes the form:

$$J_r = \begin{pmatrix} J_{\kappa\kappa}^{\text{param}} & J_{\kappa\kappa}^{\text{corr}} & 0 & 0 \\ 0 & 0 & J_{\phi\phi}^{\text{param}} & J_{\phi\phi}^{\text{corr}} \\ J_{\text{dyn},\kappa} & J_{\text{dyn},\kappa}^\delta & J_{\text{dyn},\phi} & J_{\text{dyn},\phi}^\delta \end{pmatrix}.$$

Remark 32.172. Key properties:

- parametric and correction terms are weakly coupled,
- curvature and cant are largely separable,
- coupling occurs mainly through dynamics.

32.21.8. Hierarchical Interpretation

Remark 32.173. The hybrid model introduces a hierarchical structure:

- parametric layer defines global alignment structure,
- correction layer refines local deviations.

Remark 32.174. This reflects engineering practice:

- design defines intent,
- construction introduces deviations.

32.21.9. Computational Strategy

- optimize parametric variables first,
- introduce correction terms for refinement,
- optionally alternate between both levels.

32.21.10. Connection to Level-of-Detail

Remark 32.175. The hybrid representation naturally supports LOD concepts:

- coarse LOD: parametric model only,
- fine LOD: parametric + corrections.

32.21.11. Key Insight

Proposition 32.176. *The hybrid formulation separates structural modelling from data fitting while preserving a sparse and well-conditioned optimization problem.*

32.21.12. Connection to AIM

Summary 32.177. The combination of hybrid modelling and block structure is a central feature of the AIM framework.

It enables:

- interpretable parametric modelling,
- robust handling of imperfect data,
- efficient optimization due to sparsity and locality.

Thus, it bridges the gap between engineering design and data-driven reconstruction.

33. Optimizer Architecture

33.1. Overview

Remark 33.1. The previous chapters introduced the mathematical and physical foundations of alignment modelling and optimization.

Remark 33.2. In this chapter, these concepts are combined into a concrete system architecture that enables the implementation of a full alignment optimization engine.

Remark 33.3. The architecture consists of the following main components:

- data ingestion (import and preprocessing),
 - initial guess construction,
 - constraint library,
 - objective engine,
 - numerical solver,
 - live visualization.
-

33.2. Data Flow

Remark 33.4. The overall data flow can be summarized as:

Input Data $\rightarrow$ Initial Guess $\rightarrow$ Optimization $\rightarrow$ Result

Remark 33.5. In the implemented system, this flow is realized through a sequence of well-defined transformations.

Drop $\rightarrow$ Parser $\rightarrow$ Grabbeltisch $\rightarrow$ Initial Guess $\rightarrow$ SQP $\rightarrow$ Model $\rightarrow$ View

33.3. Optimization Session

Definition 33.6 (Optimization session). An optimization session encapsulates all data and operations required for a single optimization run.

```
class OptimizationSession {
  constructor({ input, constraints, objectives }) {
    this.input = input
    this.constraints = constraints
    this.objectives = objectives
    this.initialGuess = null
    this.result = null
  }

  buildInitialGuess() {}
  runOptimization() {}
  commitToModel(projectModel) {}
}
```

Remark 33.7. This abstraction separates:

- data preparation,
- numerical optimization,
- integration into the system model.

33.4. Initial Guess Construction

Remark 33.8. The initial guess transforms raw geometric data into optimization variables.

```
function buildInitialGuess(polyline) {
  const s = computeArcLength(polyline)

  const theta = computeDirections(polyline)
  const kappa = diff(theta) / ds

  const kappaSmooth = smooth(kappa)

  const phi = kappaSmooth.map(k =>
    Math.asin(clamp(v*v*k/g, -1, 1))
  )
}
```

```

    return { kappa: kappaSmooth, phi }
}

```

Remark 33.9. This step is essential for robust convergence of the optimization algorithm.

33.5. Constraint Library

Definition 33.10 (Constraint). A constraint is a function mapping the current state to a residual.

```

function minRadius(kappa, Rmin) {
    return Math.abs(kappa) - 1/Rmin
}

function parallelConstraint(p1, p2, d) {
    const dx = p1.x - p2.x
    const dy = p1.y - p2.y
    return Math.sqrt(dx*dx + dy*dy) - d
}

```

Remark 33.11. All constraints are expressed uniformly as residuals and integrated into the optimization problem.

33.6. Objective Engine

Definition 33.12 (Objective function). The objective function is constructed as a weighted sum of residual terms.

```

function objectiveJerk(kappa, phi, ds, v, g) {
    const r = []
    for (let i = 0; i < kappa.length - 1; i++) {
        const dk = (kappa[i + 1] - kappa[i]) / ds
        const dphi = (phi[i + 1] - phi[i]) / ds
        const j = v*v*v*dk - g*v*Math.cos(phi[i]) * dphi
        r.push(j)
    }
    return r
}

```

```
function buildObjectiveResiduals(ctx, objectives) {
  const r = []

  for (const obj of objectives) {
    const ri = obj.evaluate(ctx)
    const w = Math.sqrt(obj.weight ?? 1)
    for (const v of ri) r.push(w * v)
  }

  return r
}
```

Remark 33.13. This formulation enables flexible adaptation to different design criteria.

33.7. Residual Assembly

Remark 33.14. The optimization problem is expressed entirely in terms of residuals.

```
function buildResidualVector(ctx, constraints, objectives) {
  return [
    ...buildConstraintResiduals(ctx, constraints),
    ...buildObjectiveResiduals(ctx, objectives)
  ]
}
```

33.8. Numerical Solver

33.8.1. Gauss–Newton / SQP Core

```
async function runSQPInteractive(x0, onStep) {
  let x = x0

  for (let k = 0; k < maxIter; k++) {

    const r = residuals(x)
    const J = jacobian(x)

    const p = solve(J, r)
```

```

    x = add(x, scale(p, alpha))

    onStep({ x, k, r })

    await nextFrame()
  }

  return x
}

```

Remark 33.15. The solver operates on the residual formulation and exploits sparsity.

33.9. Live Visualization

Remark 33.16. A key feature of the system is the live visualization of the optimization process.

```

optSession.runOptimization({
  onStep: ({ x, k }) => {
    const geom = reconstruct(x)

    geoRender.updateAlignment(geom)
    plotKappa(x.kappa)
    plotPhi(x.phi)
  }
})

```

Remark 33.17. This enables interactive exploration and debugging of the optimization.

33.10. Network Extension

Remark 33.18. The architecture extends naturally to networks of alignments.

```

function residuals(network) {

  const r = []

  for (edge of network.edges) {
    r.push(...smoothness(edge))
  }
}

```

```

for (node of network.nodes) {
  r.push(...nodeConstraints(node))
}

return r
}

```

Remark 33.19. This allows simultaneous optimization of multiple connected tracks.

33.11. System Integration

Remark 33.20. The optimization engine is integrated into the user workflow through the interactive selection interface.

```

onUserFinalizeSelection(selection) {

  const alignmentData =
    buildAlignmentFromSelection(selection)

  const optSession = new OptimizationSession({
    alignmentData
  })

  optSession.buildInitialGuess()
  optSession.runOptimization()
}

```

33.12. Summary

Summary 33.21. The presented architecture combines mathematical modelling, numerical optimization, and interactive visualization into a unified system.

By representing all constraints and objectives as residuals, the system achieves a high degree of modularity and extensibility.

This architecture enables the transition from static alignment modelling to dynamic, optimization-driven design of railway infrastructure.

34. Realization-Aware Optimization

34.1. Motivation

Remark 34.1. Classical alignment optimization operates directly on analytical target geometry.

Remark 34.2. Real railway tracks, however, differ from their analytical target states.

Remark 34.3. The physically realized alignment is influenced by:

- construction procedures,
- machine behaviour,
- ballast and track mechanics,
- local correction limits,
- discretization,
- and measurement feedback.

Remark 34.4. Thus, optimization should minimize:

$$J(\mathcal{R}(\kappa, \phi)),$$

rather than:

$$J(\kappa, \phi).$$

Remark 34.5. Within AIM, optimization is therefore interpreted as optimization of expected realized operational behaviour rather than of purely analytical geometry.

34.2. Realization Operator

Definition 34.6 (Realization operator). The realization operator is:

$$\mathcal{R} : \text{pose3}_{\text{target}} \mapsto \text{pose3}_{\text{real}}.$$

Remark 34.7. The operator represents the transformation from analytical target state to physically realized railway state.

Remark 34.8. The realization operator may include:

- sampling,
- machine action,
- smoothing,
- local correction kernels,
- and structural response of the track.

34.3. Optimization on Realized States

Remark 34.9. Objective functions should be evaluated on realized states rather than on analytical target states.

Definition 34.10 (Realization-aware optimization). The general optimization problem is:

$$\min_{\kappa, \phi} J(\mathcal{R}(\kappa, \phi)).$$

Remark 34.11. This formulation explicitly integrates realization effects into the optimization process.

34.4. Pose3 Optimization

Remark 34.12. Optimization operates on the coupled pose3 state:

$$(\gamma, \theta, \phi).$$

Remark 34.13. Curvature and cant must therefore be optimized jointly rather than independently.

Remark 34.14. This reflects the dynamic coupling between geometry and vehicle behaviour.

34.5. Dynamic Objectives

34.5.1. Effective Lateral Acceleration

Definition 34.15 (Effective lateral acceleration). The effective lateral acceleration is:

$$a_{\text{eff}} = v^2 \kappa - g \sin \phi.$$

Remark 34.16. This quantity represents the dynamically relevant acceleration acting on the vehicle.

34.5.2. Jerk

Definition 34.17 (Jerk). The jerk is:

$$j = v^3 \kappa' - gv \cos \phi \phi'.$$

Remark 34.18. Jerk constraints strongly influence transition design and passenger comfort.

34.5.3. Typical Objective Components

Remark 34.19. Typical optimization objectives include:

- curvature smoothness,
- cant smoothness,
- jerk minimization,
- cant deficiency control,
- vehicle comfort,
- energy efficiency,
- realization robustness.

34.6. Realization Constraints

Remark 34.20. Optimization must respect realization constraints such as:

- machine limits,
- tamping limits,
- ballast behaviour,
- local correction ranges,
- and measurement resolution.

Remark 34.21. Thus, the feasible design space is constrained not only by geometry and standards, but also by realizability.

34.7. Inverse Realization

Remark 34.22. The optimization problem may be interpreted as an inverse realization problem.

Definition 34.23 (Inverse realization formulation). The objective is:

$$\mathcal{R}(\kappa_{\text{target}}, \phi_{\text{target}}) \approx (\kappa_{\text{desired}}, \phi_{\text{desired}}).$$

Remark 34.24. The analytical target state therefore acts as a pre-image under the realization operator.

34.8. Frequency Interpretation

Remark 34.25. The realization operator acts approximately as a low-pass filter on curvature and cant.

Remark 34.26. High-frequency components are attenuated during realization.

Remark 34.27. Optimization must therefore avoid introducing geometrically meaningless high-frequency corrections that cannot survive realization.

34.9. Sparse and Hybrid Optimization

Remark 34.28. Sparse and hybrid models are particularly suitable for realization-aware optimization.

Remark 34.29. Sparse structures provide:

- numerical stability,
- low parameter dimension,
- and interpretable geometry.

Remark 34.30. Hybrid models additionally permit:

- local corrections,
- realization compensation,
- and adaptive refinement.

34.10. Runtime Optimization

Remark 34.31. Optimization may operate continuously during:

- construction,
- maintenance,
- measurement feedback,
- and operational monitoring.

Remark 34.32. Thus, optimization becomes a runtime process rather than a purely offline planning step.

34.11. Vehicle-Space Interpretation

Remark 34.33. The physically relevant object is not necessarily the track centerline, but the motion of the railway vehicle within space.

Remark 34.34. This includes:

- vehicle center of gravity,
- bogie motion,
- vehicle envelope,
- platform interaction,
- and clearance behaviour.

Remark 34.35. Such interpretations connect realization-aware optimization to Schwerpunkt-oriented alignment concepts [12].

34.12. Ellipsoid-Aware Optimization

Remark 34.36. For large-scale or high-precision systems, optimization may be performed directly on the Earth reference surface.

Remark 34.37. In this case:

- curvature becomes geodesic curvature,
- frames become surface-attached frames,
- and world-to-track projection depends on ellipsoid geometry.

34.13. Key Insight

Proposition 34.38. *The relevant optimization target is not the analytical alignment itself, but the expected realized operational behaviour.*

Remark 34.39. This fundamentally changes the interpretation of railway alignment optimization.

34.14. Connection to AIM

Summary 34.40. Realization-aware optimization extends classical alignment optimization toward physically realizable railway states.

Within AIM, optimization is formulated on:

- pose3 states,

34. *Realization-Aware Optimization*

- realization operators,
- runtime projection structures,
- and dynamically relevant vehicle-space behaviour.

This provides a unified framework linking geometry, realization, dynamics, optimization, and operational interpretation.

Teil VIII.

Engineering and Standards

Remark 34.41. This part places the AIM framework into the context of railway engineering practice. Mathematical models alone are insufficient unless they can be related to engineering constraints, standards, and operational rules.

Remark 34.42. The purpose of this part is therefore to connect formal modelling with practical admissibility conditions arising from railway design and operation.

Remark 34.43. The following chapters discuss standards, rule systems, and structured constraint formulations relevant to alignment engineering.

35. Standards and Data Models

35.1. Motivation

Remark 35.1. Railway alignment data does not exist in isolation. It is embedded in a landscape of established standards and engineering data models.

Remark 35.2. Relevant formats include:

- LandXML,
- IFC (Industry Foundation Classes),
- railML,
- and various proprietary or legacy formats.

Remark 35.3. The AIM framework does not replace these standards. Instead, it provides a consistent internal representation that can interpret and integrate them.

35.2. General Perspective

Remark 35.4. Most existing standards describe railway geometry in a layered or fragmented way.

- horizontal alignment,
- vertical profile,
- cant (superelevation),
- additional metadata.

Remark 35.5. In contrast, the AIM framework proposes a unified state-based representation:

$$(\gamma(s), \theta(s), \kappa(s), \phi(s)).$$

Remark 35.6. This representation serves as a canonical internal model.

35.3. LandXML

35.3.1. Overview

Remark 35.7. LandXML is one of the most widely used formats for alignment data exchange.

Remark 35.8. It typically represents:

- horizontal geometry (CoordGeom),
- vertical profiles,
- cant definitions.

35.3.2. Characteristics

- structured XML format,
- hierarchical organization,
- multiple alignments per file,
- explicit naming of geometric elements.

35.3.3. Limitations

Remark 35.9. Despite its structure, LandXML exhibits several limitations:

- separation of horizontal and vertical geometry,
- lack of a unified dynamic interpretation,
- ambiguity in coordinate reference systems,
- inconsistent usage across software systems.

35.3.4. AIM Interpretation

Remark 35.10. Within the AIM framework, LandXML is treated as a *fat container*:

- all available data is parsed,
- relevant components are extracted,
- the result is transformed into a unified state model.

Remark 35.11. The conversion step

$$\text{LandXML} \rightarrow \text{AIM (sparse)}$$

is essential for further processing.

—

35.4. IFC

35.4.1. Overview

Remark 35.12. IFC is a general BIM standard designed for interdisciplinary data exchange.

Remark 35.13. In the context of infrastructure, IFC is extended by infrastructure schemas (IFC Rail, IFC Alignment).

35.4.2. Characteristics

- object-oriented data model,
- rich semantic structure,
- integration of geometry and metadata,
- support for relationships and hierarchies.

35.4.3. Challenges

Remark 35.14. For alignment modelling, IFC presents several challenges:

- complex schema structure,
- multiple representation alternatives,
- inconsistent implementation across tools,
- limited support for advanced alignment concepts.

35.4.4. AIM Interpretation

Remark 35.15. The AIM framework interprets IFC data by extracting:

- alignment geometry,
- reference systems,
- semantic relations.

Remark 35.16. AIM acts as a normalization layer that reduces IFC complexity to a computationally usable form.

—

35.5. railML

35.5.1. Overview

Remark 35.17. railML is a domain-specific XML standard for railway systems.

Remark 35.18. It focuses on:

- infrastructure,
- timetable data,
- rolling stock.

35.5.2. Relevance for Alignment

Remark 35.19. railML provides a more explicit representation of railway topology than most other formats.

Remark 35.20. However, detailed geometric modelling is often secondary to network and operational aspects.

35.5.3. AIM Interpretation

Remark 35.21. Within AIM, railML is particularly relevant for:

- network topology,
 - connections between tracks,
 - system-level structure.
-

35.6. Legacy and Proprietary Formats

Remark 35.22. In practice, a large amount of alignment data exists in proprietary or legacy formats.

Remark 35.23. Examples include:

- TRA / GRA (Technet, Vermessung),
- spreadsheet-based formats,
- CAD exports.

35.6.1. Characteristics

- incomplete or implicit structure,
- missing type information,
- reliance on conventions rather than formal schema.

35.6.2. AIM Strategy

Remark 35.24. These formats are interpreted using:

- sniffing and classification,
- format-specific parsers,
- reconstruction of implicit structure.

—

35.7. Canonical Internal Model

Definition 35.25 (AIM core model). The AIM framework uses a canonical internal representation based on:

$$(\gamma(s), \theta(s), \kappa(s), \phi(s)).$$

Remark 35.26. All external formats are mapped to this representation.

35.7.1. Advantages

- unified handling of geometry and dynamics,
- compatibility with numerical optimization,
- independence from specific file formats,
- extensibility toward new standards.

—

35.8. Transformation Pipeline

Remark 35.27. The transformation from external data to the AIM model follows a pipeline structure.

```
external file
→ parser
→ fat representation
→ validation
→ sparse conversion
→ AIM state model
```

Remark 35.28. This pipeline ensures robustness and traceability of data processing.

—

35.9. Relation to BIM

35.9.1. Conceptual Position

Remark 35.29. BIM (Building Information Modelling) aims to integrate geometry, semantics, and lifecycle information.

Remark 35.30. However, alignment modelling is often underrepresented or treated as a secondary aspect.

35.9.2. AIM Contribution

Remark 35.31. The AIM framework contributes to BIM by:

- providing a rigorous alignment model,
- enabling dynamic interpretation,
- supporting optimization-driven design.

Remark 35.32. In this sense, AIM can be interpreted as a specialization of BIM for alignment-centric infrastructure modelling.

35.10. Interoperability

Remark 35.33. A key requirement for practical application is interoperability with existing tools and workflows.

Remark 35.34. The AIM framework supports interoperability through:

- import adapters,
 - export converters,
 - intermediate canonical representation.
-

35.11. Discussion

Remark 35.35. Existing standards focus primarily on data exchange and documentation.

Remark 35.36. The AIM framework complements these standards by introducing:

- a computationally consistent model,
- a dynamic interpretation,
- integration with optimization methods.

Remark 35.37. Thus, AIM bridges the gap between static data representation and active engineering design.

35.12. Summary

Summary 35.38. Railway alignment data is governed by a variety of standards, each with its own strengths and limitations.

The AIM framework provides a unified internal representation that allows these heterogeneous sources to be interpreted consistently.

By combining standard-based data exchange with a state-based, optimization-ready model, AIM enables a transition from static data handling to dynamic and computational alignment design.

36. Constraint Code Architecture

36.1. Motivation

Remark 36.1. Railway alignment rules must be represented in a form that is both mathematically meaningful and computationally tractable.

Remark 36.2. Within the AIM framework, this is achieved by expressing all rules as constraint residuals that can be evaluated uniformly inside an optimization loop.

Remark 36.3. The implementation goal is therefore not a collection of isolated rule checks, but a modular constraint library.

36.2. General Principle

Definition 36.4 (Constraint residual). A constraint is represented as a function

$$c(\mathbf{x}) \in \mathbb{R}^m$$

whose value vanishes in the ideal case and whose magnitude measures violation.

Remark 36.5. This unified representation allows the same mathematical object to be used for:

- feasibility checks,
- penalty terms,
- direct inclusion into SQP solvers.

```
function evaluateConstraint(ctx) {  
    return [residual1, residual2, ...]  
}
```

36.3. Constraint Registry

Remark 36.6. A practical implementation should organize constraints in a registry.

```
const ConstraintRegistry = {  
    geometric: {},  
    dynamic: {},  
}
```

```

    topologic: {},
    regulatory: {}
}

```

Remark 36.7. This keeps domain logic extensible and avoids hard-coded monolithic solver structures.

36.4. Constraint Object Model

```

const constraint = {
  id: "min_radius",
  group: "regulatory",
  appliesTo: "edge",
  weight: 1.0,
  evaluate: (ctx) => []
}

```

Remark 36.8. The fields have the following meaning:

- `id`: unique name,
- `group`: semantic class,
- `appliesTo`: edge, node, pair, network,
- `weight`: optional scaling,
- `evaluate`: residual callback.

36.5. Context Object

Remark 36.9. Constraints should not operate directly on raw global arrays. Instead, they receive a normalized context object.

```

const ctx = {
  edge,
  node,
  pair,
  i,
  kappa,
  phi,
  geometry,
  speed,
  params,
}

```



```

    metadata
}

```

Remark 36.10. This provides a stable interface between model, solver, and rule system.

36.6. Geometric Constraints

36.6.1. Position Continuity

```

function continuityConstraint(p1, p2) {
    return [
        p1.x - p2.x,
        p1.y - p2.y
    ]
}

```

Remark 36.11. This enforces C^0 -continuity between adjacent geometric elements or connected alignments.

36.6.2. Direction Continuity

```

function directionConstraint(t1, t2) {
    return [
        t1.dx - t2.dx,
        t1.dy - t2.dy
    ]
}

```

Remark 36.12. This corresponds to tangent continuity and therefore to C^1 -style behaviour in geometric terms.

36.6.3. Curvature Continuity

```

function curvatureConstraint(k1, k2) {
    return [k1 - k2]
}

```

Remark 36.13. This enforces continuity of curvature across boundaries.

36.7. Dynamic Constraints

36.7.1. Equilibrium Constraint

```

function equilibriumConstraint(kappa, phi, v, g) {
    return [v * v * kappa - g * Math.sin(phi)]
}

```

Remark 36.14. This residual measures cant deficiency or excess in analytic form.

36.7.2. Jerk Constraint

```
function jerkConstraint(kappa0, kappa1, phi0, phi1, ds, v, g) {
  const dk = (kappa1 - kappa0) / ds
  const dphi = (phi1 - phi0) / ds
  const j = v * v * v * dk - g * v * Math.cos(phi0) * dphi
  return [j]
}
```

Remark 36.15. This constraint reflects dynamic smoothness and comfort-related limits.

36.8. Regulatory Constraints

36.8.1. Minimum Radius

```
function minRadiusConstraint(kappa, Rmin) {
  return [Math.abs(kappa) - 1 / Rmin]
}
```

Remark 36.16. A non-positive value indicates compliance with the radius limit.

36.8.2. Maximum Cant

```
function maxCantConstraint(phi, phiMax) {
  return [Math.abs(phi) - phiMax]
}
```

36.8.3. Cant Gradient

```
function cantGradientConstraint(phi0, phi1, ds, limit) {
  return [Math.abs((phi1 - phi0) / ds) - limit]
}
```

36.9. Topological Constraints

36.9.1. Switch Node Constraint

```
function switchConstraint(main, branch) {
  return [
    main.x - branch.x,
    main.y - branch.y,
    main.dx - branch.dx,
  ]
}
```

```

    main.dy - branch.dy
  ]
}

```

Remark 36.17. This expresses the basic coupling at branching nodes.

36.9.2. Parallel Track Spacing

```

function parallelConstraint(p1, p2, d) {
  const dx = p1.x - p2.x
  const dy = p1.y - p2.y
  return [Math.sqrt(dx * dx + dy * dy) - d]
}

```

36.10. Constraint Assembly

Remark 36.18. The full residual vector is assembled from all active constraints.

```

function buildConstraintResiduals(ctx, constraints) {
  const r = []

  for (const c of constraints) {
    const ri = c.evaluate(ctx)
    const w = Math.sqrt(c.weight ?? 1)
    for (const v of ri) r.push(w * v)
  }

  return r
}

```

Remark 36.19. This produces a single solver-compatible vector independent of the semantic source of the constraints.

36.11. Factory Functions

Remark 36.20. Constraints should be created through factories rather than by writing anonymous closures throughout the code base.

```

function makeMinRadiusConstraint(Rmin, weight = 1) {
  return {
    id: "min_radius",
    group: "regulatory",
    appliesTo: "edge",
  }
}

```

```

    weight,
    evaluate: ({ kappa, i }) => [
      Math.abs(kappa[i]) - 1 / Rmin
    ]
  }
}

function makeJerkConstraint(v, g, ds, weight = 1) {
  return {
    id: "jerk",
    group: "dynamic",
    appliesTo: "edge",
    weight,
    evaluate: ({ kappa, phi, i }) => {
      const dk = (kappa[i + 1] - kappa[i]) / ds
      const dphi = (phi[i + 1] - phi[i]) / ds
      return [v * v * v * dk - g * v * Math.cos(phi[i]) * dphi]
    }
  }
}

```

36.12. Constraint Presets

Remark 36.21. In practical railway design, different projects require different constraint configurations.

```

const presetHighSpeed = [
  makeMinRadiusConstraint(4000, 10),
  makeJerkConstraint(v, g, ds, 8),
  makeMaxCantConstraint(phiMax, 6)
]

const presetExistingLine = [
  makeMinRadiusConstraint(800, 3),
  makeJerkConstraint(v, g, ds, 2),
  makeParallelConstraint(trackDistance, 5)
]

```

36.13. Integration into the Solver

```

function buildResidualVector(ctx, constraints, objectives) {
  return [

```

```

        ...buildConstraintResiduals(ctx, constraints),
        ...buildObjectiveResiduals(ctx, objectives)
    ]
}
```

Remark 36.22. This shows the central design principle of the architecture: constraints and objectives share the same residual-based interface.

36.14. Discussion

Remark 36.23. The proposed architecture has several advantages:

- modularity,
- extensibility,
- compatibility with least-squares solvers,
- transparent mapping between engineering rules and code.

Remark 36.24. Instead of scattering rule checks throughout the application, the constraint library concentrates domain logic in a single mathematical layer.

36.15. Summary

Summary 36.25. The constraint library translates railway rules, topology, and dynamic requirements into a unified residual-based architecture.

This provides the technical bridge between theoretical modelling, engineering regulations, and computational optimization.

As a result, the AIM framework becomes not only mathematically coherent, but also directly implementable in software.

37. Railway Design Rules

37.1. Motivation

Remark 37.1. Railway alignment design is not solely a geometric or mathematical task. It is governed by engineering rules, safety requirements, and operational constraints.

Remark 37.2. These rules originate from:

- national regulations,
- international standards (e.g. UIC),
- operator-specific guidelines (e.g. Deutsche Bahn),
- empirical engineering practice.

Remark 37.3. Within the AIM framework, such rules are not treated as external constraints only, but are integrated into the modelling and optimization process.

37.2. Categories of Rules

37.2.1. Geometric Rules

- minimum radius,
- continuity of alignment,
- transition length requirements,
- curvature monotonicity in transitions.

37.2.2. Kinematic and Dynamic Rules

- limits on lateral acceleration,
- limits on jerk,
- compatibility of curvature and cant,
- speed-dependent requirements.

37.2.3. Operational Rules

- allowable speeds,
- safety margins,
- maintenance considerations,
- interoperability constraints.

37.2.4. Topological Rules

- constraints at switches and crossings,
 - parallel track spacing,
 - network connectivity conditions.
-

37.3. Geometric Constraints

37.3.1. Minimum Radius

Definition 37.4 (Minimum radius constraint). The curvature must satisfy

$$|\kappa(s)| \leq \kappa_{\max}, \quad \kappa_{\max} = \frac{1}{R_{\min}}.$$

Remark 37.5. The value of $R_{\min}$ depends on:

- design speed,
 - vehicle characteristics,
 - regulatory standards.
-

37.3.2. Continuity Conditions

Remark 37.6. Railway alignments must satisfy at least:

- C^0 continuity (position),
- C^1 continuity (direction),
- typically C^2 continuity (curvature).

Remark 37.7. Higher-order smoothness may be required for high-speed lines.

37.3.3. Transition Length

Definition 37.8 (Minimum transition length). A transition between curvatures must satisfy

$$L \geq L_{\min}(v, \kappa_1, \kappa_2).$$

Remark 37.9. Typical formulations relate transition length to:

- allowable rate of change of lateral acceleration,
- cant development limits.

—

37.4. Dynamic Constraints

37.4.1. Lateral Acceleration

Definition 37.10 (Lateral acceleration). The lateral acceleration is given by

$$a_y(s) = v^2 \kappa(s).$$

Constraint 37.11.

$$|a_y(s)| \leq a_{\max}.$$

—

37.4.2. Cant and Cant Deficiency

Definition 37.12 (Cant angle). Let $\phi(s)$ denote the cant angle.

Definition 37.13 (Effective acceleration). The effective lateral acceleration is

$$a_{\text{eff}}(s) = v^2 \kappa(s) - g \tan(\phi(s)).$$

Constraint 37.14.

$$|a_{\text{eff}}(s)| \leq a_{\text{eff},\max}.$$

—

37.4.3. Jerk

Definition 37.15 (Jerk). The lateral jerk is defined as

$$j(s) = \frac{d}{dt} a_y(s).$$

Remark 37.16. Using s as parameter, one obtains

$$j(s) = v^3 \kappa'(s).$$

Constraint 37.17.

$$|j(s)| \leq j_{\max}.$$

—

37.5. Coupling of Curvature and Cant

Remark 37.18. Curvature and cant must not be treated independently.

Constraint 37.19.

$$|\kappa'(s)| \leq f_1(v, \phi), \quad |\phi'(s)| \leq f_2(v, \kappa).$$

Remark 37.20. These relations express that:

- curvature changes require corresponding cant development,
- cant changes must respect dynamic limits.

—

37.6. Speed-Dependent Constraints

Remark 37.21. All relevant constraints depend on speed.

Definition 37.22 (Speed profile). A speed profile is a function

$$v : [0, L] \rightarrow \mathbb{R}_+.$$

Remark 37.23. Constraints must be evaluated pointwise with respect to $v(s)$.

—

37.7. Constraint Representation in AIM

37.7.1. Abstract Form

Definition 37.24 (Constraint functional). A constraint is represented as a functional

$$C_i[\kappa, \phi, v] \leq 0.$$

Remark 37.25. This allows constraints to be:

- evaluated continuously,
- discretized for numerical optimization,
- combined into a unified constraint system.

—

37.7.2. Discrete Representation

Remark 37.26. In numerical implementation, constraints are evaluated on a grid

$$s_0, \dots, s_N.$$

Definition 37.27 (Discrete constraint).

$$C_i(\mathbf{x}) = \max_k C_i(s_k).$$

—

37.8. Integration into Optimization

Remark 37.28. Constraints enter the optimization problem as:

$$g_i(\mathbf{x}) \leq 0.$$

Remark 37.29. They may be handled via:

- hard constraints (feasibility),
- penalty terms,
- barrier methods.

—

37.9. Relation to Standards

Remark 37.30. The presented constraints correspond to rules found in:

- UIC codes,
- national railway regulations,
- operator guidelines.

Remark 37.31. The AIM framework provides a mathematical representation that allows these rules to be applied consistently across different data sources.

—

37.10. Discussion

Remark 37.32. Traditional workflows treat rules as external checks applied after design.

Remark 37.33. In contrast, AIM integrates rules directly into:

- modelling,
- reconstruction,
- optimization.

Remark 37.34. This leads to a shift from validation to constraint-driven design.

37.11. Summary

Summary 37.35. Railway design rules define the feasible set of alignment solutions.

Within the AIM framework, these rules are formalized as mathematical constraints and integrated into the optimization process.

This allows alignment design to be treated as a constrained optimization problem that respects both geometric and dynamic requirements.

Teil IX.

Applications

Remark 37.36. This part illustrates how the AIM framework can be applied to concrete railway and modelling problems. Its purpose is not only to demonstrate feasibility, but also to show how the theoretical concepts developed earlier become operational in practice.

Remark 37.37. The applications range from classical railway alignment questions to broader perspectives such as BIMalignment and use-case-driven interpretation.

Remark 37.38. Thus, this part serves as the bridge from formal framework to practical engineering relevance.

38. Railway Alignment

38.1. Motivation

Remark 38.1. Railway alignment is one of the most demanding application domains for curvature-based geometric modelling.

Remark 38.2. In railway engineering, alignment is not merely a planar curve design problem. It is a coupled system involving horizontal geometry, vertical profile, cant, operating speed, dynamic vehicle response, safety, and maintenance.

Remark 38.3. This makes railway alignment a natural and severe test case for alignment-based information modelling.

38.2. Railway Alignment as a Multi-Layer Object

Definition 38.4 (Railway alignment). A railway alignment is the spatial path associated with a railway track, typically represented through coordinated horizontal, vertical, and cant-related descriptions.

Remark 38.5. In practical standards and data models, railway alignment is usually not represented as a single monolithic object, but as an ordered system of segments and auxiliary alignment layers.

38.2.1. Horizontal alignment

Remark 38.6. The horizontal alignment is primarily governed by curvature. In its classical engineering decomposition it consists of straight segments, circular arcs, and transition curves.

38.2.2. Vertical alignment

Remark 38.7. The vertical alignment describes elevation and gradient behaviour along the route. In practical data models it is segmented into line, arc, and transition primitives.

OPEN: Unified mathematical treatment of horizontal and vertical alignment.

38.2.3. Cant alignment

Definition 38.8 (Cant). Cant is the elevation difference between the rails, introduced to compensate lateral acceleration in curves.

Remark 38.9. Cant is not an accessory parameter. It forms an alignment layer of its own, with segment structure, continuity rules, and possible smoothing requirements.

Remark 38.10. Recent railway conceptual models explicitly represent cant as an ordered sequence of segments, including line and transition segments, linked to the horizontal alignment by chainage or equivalent position reference.

OPEN: Integrate cant into a unified geometric and kinematic state model.

38.3. Parameter Coupling

Remark 38.11. Railway alignment is characterized by strong coupling between geometric and operational variables.

- curvature,
- cant,
- speed,
- transition length,
- gradient,
- and boundary conditions imposed by rolling stock and operation

Remark 38.12. These quantities are not independent. Increased speed requires either larger radii, larger cant, longer transition lengths, or a different trade-off between these variables.

Remark 38.13. Likewise, high cant or severe curvature may improve one operational aspect while worsening another, for example when stopped trains or restricted sight distances are involved.

38.4. Why Transition Curves Matter in Railways

Remark 38.14. The transition curve is not merely a geometric connector between a straight line and a circular arc.

Remark 38.15. Its primary role is to avoid abrupt changes in lateral forces, improve ride comfort, reduce dynamic excitation, and limit wear on both vehicle and infrastructure.

Remark 38.16. The state-of-the-art review literature explicitly emphasizes that transition curves are crucial for safe and comfortable railway travel and for reducing maintenance demand.

Remark 38.17. This is also why modern railway research increasingly evaluates transition curves not only geometrically, but in terms of lateral change of acceleration, force development, wear, and vehicle dynamics.

38.5. From Classical Geometry to Dynamic Interpretation

Remark 38.18. Classical alignment design often idealizes the influence of curvature, cant, and transition shape through simplified or quasi-static considerations.

Remark 38.19. However, practical evidence and later engineering work show that the vehicle-track system exhibits dynamic responses which depend strongly on the detailed shape of the transition curve.

Remark 38.20. In particular, non-linear transition curves were introduced to mitigate the abrupt onset and disappearance of the variation of unbalanced lateral acceleration that arises with linear transitions at higher speeds.

Remark 38.21. This gives railway transition design a systems character: the geometric curve acts as a control function for dynamic behaviour.

38.6. Curvature, Cant, and Speed

Remark 38.22. Curvature alone does not determine railway behaviour. The relevant quantity is the combined system of curvature, cant, speed, and their rates of change.

Remark 38.23. Cant and cant variation influence lateral acceleration compensation, while transition shape influences how these effects develop along the track.

Remark 38.24. Practical railway design manuals also show that excessive superelevation, insufficient tangent lengths, or poorly located curves can create operational and derailment risks even when the pure geometry is formally traversable.

OPEN: Formal coupled model of curvature, cant, speed, and dynamic response.

38.7. Operational and Maintenance Constraints

Remark 38.25. Railway alignment is constrained not only by construction feasibility, but also by long-term operation and maintenance.

- minimum radii,
- maximum cant,
- limits on cant change,
- curvature and transition limits,
- sight distance restrictions,
- turnout and crossing constraints,
- wear and maintenance considerations

Remark 38.26. Practical design guides repeatedly emphasize that sharper curvature may be physically negotiable, yet still be undesirable because of higher maintenance cost, lower operating efficiency, or increased derailment risk.

38.8. Freight, Passenger, High-Speed, and Special Systems

Remark 38.27. Railway alignment is not uniform across operational domains. Freight, passenger, high-speed, light rail, and special systems such as maglev may lead to different admissible geometry and different transition preferences.

Remark 38.28. This means that an alignment model must be flexible enough to represent domain-specific constraints without losing mathematical coherence.

OPEN: Taxonomy of railway application classes and their geometric consequences.

38.9. Chainage and Engineering Referencing

Definition 38.29 (Chainage). Chainage is the engineering reference system used to identify positions along an alignment.

Remark 38.30. In practice, alignment data is often anchored more robustly by chainage than by raw Cartesian coordinates alone, especially when the alignment is revised or partially reconstructed.

Remark 38.31. Modern railway information models therefore include dedicated chainage mechanisms in order to preserve engineering references despite local alignment changes.

OPEN: Relation between arc-length parameterization and engineering chainage.

38.10. Railway Alignment in Data Models and BIM

Remark 38.32. Railway alignment is increasingly represented within interoperable data models. These models typically distinguish horizontal alignment, vertical alignment, and cant as related but separate geometric layers.

Remark 38.33. Current IFC-related railway modelling efforts explicitly represent transition curves, vertical transition segments, and cant segments, and already reveal that infrastructure alignment cannot be reduced to a single curve primitive.

Remark 38.34. This strongly supports the view adopted in the present manuscript: alignment should be understood as a structured object rather than as an undifferentiated line.

OPEN: Formal docking of the sparse alignment model to IFC/railway data models.

38.11. Why Railway Alignment is a Privileged Use Case

Remark 38.35. Railway alignment combines

- strict geometric structure,
- strong parameter coupling,
- dynamic sensitivity,
- safety implications,
- heavy maintenance consequences,
- and demanding interoperability requirements.

Remark 38.36. For that reason, railway alignment is not merely one application among many, but an especially revealing use case for a general theory of curvature-based structured curve modelling.

38.12. Outlook

Remark 38.37. A mature railway-alignment framework should therefore support:

- structured geometric representation,
- multiple transition families,
- coupled cant and profile modelling,
- numerical reconstruction,
- dynamic and safety-oriented evaluation,
- and interoperability with engineering standards and BIM models.

Summary 38.38. Railway alignment is a multi-layer object rather than a single planar curve. It couples geometry, cant, speed, vehicle dynamics, operational constraints, maintenance, and engineering referencing.

This makes railway alignment a privileged and highly demanding domain for alignment-based information modelling and provides the main application context for the concepts developed in this manuscript.

39. Examples

39.1. Motivation

Remark 39.1. Theoretical concepts become meaningful only when illustrated by concrete examples.

Remark 39.2. The following examples demonstrate how different transition concepts can be interpreted within the curvature-based framework.

39.2. Example 1: Clothoid Transition

39.2.1. Definition

Definition 39.3. The clothoid is defined by linear curvature:

$$\kappa(s) = as.$$

39.2.2. Properties

- $\kappa'(s) = \text{const}$
- $\kappa''(s) = 0$

Remark 39.4. The clothoid introduces curvature at a constant rate.

39.2.3. Interpretation

Remark 39.5. In the curvature-based framework, the clothoid represents a linear control function.

Remark 39.6. Its simplicity explains its historical dominance in railway design.

39.3. Example 2: Polynomial Transition

39.3.1. Definition

Definition 39.7. A polynomial transition may be defined as

$$\kappa(s) = a_1s + a_2s^2 + a_3s^3.$$

39.3.2. Properties

- $\kappa'(s)$ is variable
- $\kappa''(s)$ is controllable

39.3.3. Interpretation

Remark 39.8. Polynomial transitions allow shaping of higher-order behaviour.

Remark 39.9. They can be used to reduce dynamic excitation by smoothing curvature variation.

—

39.4. Example 3: Comparison of Curvature Profiles

Remark 39.10. Consider two transitions:

- a clothoid,
- a cubic polynomial.

Remark 39.11. Both achieve the same boundary conditions:

$$\kappa(0) = 0, \quad \kappa(L) = \kappa_0.$$

39.4.1. Observation

Remark 39.12. Although geometrically equivalent at the endpoints, the internal behaviour differs.

- clothoid: constant curvature growth,
- polynomial: variable curvature growth.

Remark 39.13. This leads to different dynamic responses.

—

39.5. Example 4: Influence of Speed

Remark 39.14. Lateral acceleration is given by

$$a_{\text{lat}} = v^2 \kappa(s).$$

Remark 39.15. For identical geometry, increasing speed amplifies dynamic effects.

Remark 39.16. This highlights that transition design cannot be separated from operational conditions.

—

39.6. Example 5: Cant Interaction

Remark 39.17. Cant modifies the effective lateral acceleration experienced by the vehicle.

Remark 39.18. The combination of curvature and cant determines the system behaviour.

Remark 39.19. Transition design must therefore consider both geometric and cant evolution.

—

39.7. Example 6: Interpretation as Control Problem

Remark 39.20. Consider $\kappa(s)$ as an input function.

Remark 39.21. The system produces:

- forces,
- vehicle response,
- wear.

Remark 39.22. Different transition curves correspond to different control strategies.

—

39.8. Example 7: Numerical Comparison of Two Transition Laws

39.8.1. Setting

Remark 39.23. Consider two transition curves of equal length

$$L = 100 \text{ m}$$

connecting a straight segment to a circular arc with final curvature

$$\kappa_1 = \frac{1}{1000} \text{ m}^{-1}.$$

Remark 39.24. Thus, both transitions satisfy

$$\kappa(0) = 0, \quad \kappa(L) = 0.001.$$

39.8.2. Case A: Clothoid

Definition 39.25. The clothoid is given by

$$\kappa_A(s) = \frac{\kappa_1}{L} s = \frac{0.001}{100} s = 10^{-5} s.$$

Remark 39.26. Its curvature derivative is constant:

$$\kappa'_A(s) = 10^{-5} \text{ m}^{-2}.$$

39.8.3. Case B: Cubic Polynomial Transition

Definition 39.27. Consider the normalized cubic law

$$\widehat{\kappa}(u) = 3u^2 - 2u^3, \quad u = \frac{s}{L}.$$

The corresponding physical curvature law is

$$\kappa_B(s) = \kappa_1 \left(3 \left(\frac{s}{L} \right)^2 - 2 \left(\frac{s}{L} \right)^3 \right).$$

Remark 39.28. This transition satisfies

$$\kappa_B(0) = 0, \quad \kappa_B(L) = 0.001,$$

but its curvature derivative is not constant.

39.8.4. Comparison of Curvature Growth

Remark 39.29. At the midpoint $s = 50$ m, the clothoid gives

$$\kappa_A(50) = 10^{-5} \cdot 50 = 0.0005.$$

Remark 39.30. For the cubic transition, with $u = 0.5$,

$$\kappa_B(50) = 0.001 \left(3 \cdot 0.5^2 - 2 \cdot 0.5^3 \right) = 0.001(0.75 - 0.25) = 0.0005.$$

Remark 39.31. Hence both transitions have the same midpoint curvature, but they differ in the way curvature is introduced.

39.8.5. Comparison of Curvature Derivatives

Remark 39.32. For the clothoid,

$$\kappa'_A(s) = 10^{-5} \text{ m}^{-2}$$

is constant over the whole interval.

Remark 39.33. For the cubic transition,

$$\kappa'_B(s) = \kappa_1 \left(\frac{6s}{L^2} - \frac{6s^2}{L^3} \right).$$

With $L = 100$ and $\kappa_1 = 0.001$, this becomes

$$\kappa'_B(s) = 0.001 \left(\frac{6s}{10000} - \frac{6s^2}{1000000} \right).$$

39. Examples

Remark 39.34. At the start and end points:

$$\kappa'_B(0) = 0, \quad \kappa'_B(100) = 0.$$

At the midpoint $s = 50$:

$$\kappa'_B(50) = 0.001 \left(\frac{300}{10000} - \frac{15000}{1000000} \right) = 0.001(0.03 - 0.015) = 1.5 \times 10^{-5} \text{ m}^{-2}.$$

39.8.6. Interpretation

Remark 39.35. The clothoid introduces curvature with constant rate from the very beginning. The cubic transition starts more gently, reaches a higher curvature growth rate in the middle, and then tapers off smoothly toward the end.

Remark 39.36. This shows that identical boundary conditions do not imply identical dynamic behaviour. Even when two transitions connect the same geometric states, their internal curvature evolution may differ substantially.

39.8.7. Dynamic Implication

Remark 39.37. For a vehicle speed v , the lateral acceleration is

$$a_{\text{lat}}(s) = v^2 \kappa(s).$$

Hence the rate of change of lateral acceleration is proportional to

$$v^2 \kappa'(s).$$

Remark 39.38. The numerical example therefore illustrates a central point of this manuscript: transition curves should be compared not only by endpoint conditions, but by their full curvature evolution and its derivatives.

—

39.9. Example 8: Speed and Dynamic Effects

39.9.1. Setting

Remark 39.39. We consider the same two transition curves as before:

- clothoid (linear curvature),
- cubic transition.

Remark 39.40. Let the vehicle speed be

$$v = 200 \text{ km/h} = 55.56 \text{ m/s}.$$

39.9.2. Lateral Acceleration

Proposition 39.41. *The lateral acceleration is given by*

$$a_{\text{lat}}(s) = v^2 \kappa(s).$$

Remark 39.42. With

$$v^2 \approx 3086 \text{ m}^2/\text{s}^2,$$

we obtain for the final curvature

$$a_{\text{lat,max}} = 3086 \cdot 0.001 = 3.086 \text{ m/s}^2.$$

39.9.3. Rate of Change of Acceleration

Definition 39.43. The rate of change of lateral acceleration is

$$\frac{da_{\text{lat}}}{ds} = v^2 \kappa'(s).$$

39.9.4. Clothoid

Remark 39.44. For the clothoid:

$$\kappa'(s) = 10^{-5}.$$

Remark 39.45. Thus:

$$\frac{da_{\text{lat}}}{ds} = 3086 \cdot 10^{-5} = 0.0309 \text{ m/s}^2/\text{m}.$$

Remark 39.46. The acceleration increases at a constant rate along the transition.

39.9.5. Cubic Transition

Remark 39.47. At the midpoint $s = 50 \text{ m}$:

$$\kappa'_B(50) = 1.5 \times 10^{-5}.$$

Remark 39.48. Thus:

$$\frac{da_{\text{lat}}}{ds} = 3086 \cdot 1.5 \cdot 10^{-5} = 0.0463 \text{ m/s}^2/\text{m}.$$

39. Examples

Remark 39.49. At the start and end:

$$\kappa'_B(0) = \kappa'_B(L) = 0$$

and therefore

$$\frac{da_{\text{lat}}}{ds} = 0.$$

—

39.9.6. Interpretation

Remark 39.50. The clothoid introduces acceleration immediately with a constant rate.

Remark 39.51. The cubic transition starts smoothly (zero rate), increases the rate in the middle, and returns smoothly to zero.

—

39.9.7. Engineering Implication

Remark 39.52. From a passenger comfort perspective:

- clothoid: predictable but abrupt start,
- cubic: smoother entry and exit, but stronger variation internally.

Remark 39.53. From a system perspective:

- clothoid: simple control,
- cubic: distributed control with reduced boundary effects.

—

39.9.8. Key Observation

Proposition 39.54. *Even for identical geometry and speed, different curvature laws produce different acceleration profiles.*

—

Summary 39.55. Dynamic behaviour depends not only on curvature magnitude, but on its rate of change.

The numerical example shows that transition curves act as control functions shaping the evolution of forces in the system.

This supports the central thesis that alignment design is fundamentally a problem of curvature control.

—

39.10. Example 9: Optimization Sketch

39.10.1. Motivation

Remark 39.56. Traditional transition design selects predefined curve families.

Remark 39.57. In contrast, the curvature-based framework allows direct optimization of the transition function.

—

39.10.2. Problem Formulation

Definition 39.58. We seek a curvature function

$$\kappa(s), \quad s \in [0, L],$$

such that:

$$\kappa(0) = 0, \quad \kappa(L) = \kappa_1.$$

Remark 39.59. Additionally, we may impose smoothness conditions such as

$$\kappa'(0) = 0, \quad \kappa'(L) = 0.$$

—

39.10.3. Objective

Definition 39.60. A simple objective is to minimize the maximum rate of curvature change:

$$\min_{\kappa} \max_{s \in [0, L]} |\kappa'(s)|.$$

Remark 39.61. This corresponds to minimizing peak dynamic excitation.

—

39.10.4. Candidate Solutions

Remark 39.62. Two candidate solutions are:

- clothoid: constant κ' ,
- cubic transition: zero boundary derivatives.

—

39.10.5. Qualitative Comparison

Remark 39.63. The clothoid distributes curvature growth evenly over the interval.

Remark 39.64. The cubic transition reduces boundary effects but increases curvature variation in the interior.

39.10.6. Optimality Consideration

Proposition 39.65. *An optimal transition balances boundary smoothness and internal variation of curvature.*

Remark 39.66. The optimization problem therefore involves trade-offs between:

- peak curvature derivative,
 - smoothness at endpoints,
 - distribution of curvature change.
-

39.10.7. General Optimization Framework

Remark 39.67. More general formulations may include objectives such as:

- minimizing jerk,
- minimizing energy,
- minimizing wear-related forces.

Remark 39.68. These objectives can be expressed as functionals of $\kappa(s)$ and its derivatives.

39.10.8. Interpretation

Remark 39.69. The optimization problem is defined directly in curvature space.

Remark 39.70. This eliminates the need to select a predefined transition family.

Remark 39.71. Instead, transition curves emerge as solutions of an optimization problem.

39.10.9. Connection to AIM

Remark 39.72. Within the AIM framework, the curvature function becomes a parameterized object subject to constraints and optimization.

Remark 39.73. This enables integration with numerical methods and automated design processes.

39.10.10. Key Result

Theorem 39.74. *Transition curves can be derived as solutions of optimization problems in curvature space rather than selected from predefined families.*

39.11. Summary

Summary 39.75. The present example is formalized in the optimization chapter, where classical transition families are derived as minimizers of curvature-based functionals.

The optimization sketch demonstrates that transition design can be formulated as a mathematical optimization problem.

This shifts alignment design from rule-based construction to model-based optimization and provides a natural link to numerical methods and automated engineering workflows.

39.12. Summary

Summary 39.76. The examples demonstrate that:

- transition curves can be compared via $\kappa(s)$,
- different functional forms lead to different system behaviour,
- geometry alone does not determine performance,
- transition design is inherently a control problem.

Two transitions may connect the same straight and circular arc and still exhibit different internal behaviour.

The clothoid is characterized by constant curvature growth, whereas the cubic transition provides a smoother start and end, but stronger variation in the interior.

This confirms that transition design is fundamentally a problem of curvature control rather than mere endpoint matching.

40. Use Cases

40.1. Motivation

Remark 40.1. A theoretical modelling framework becomes substantially more convincing when it is connected to concrete engineering scenarios.

Remark 40.2. The following use cases illustrate how the proposed AIM framework can be applied to realistic railway alignment problems.

Remark 40.3. The examples are not meant as software demonstrations only. They are intended to clarify the interaction between:

- data,
- structured alignment models,
- dynamic interpretation,
- and optimization.

40.2. Use Case 1: Reconstruction of an Existing Alignment

40.2.1. Scenario

Remark 40.4. An existing railway alignment is available only through measured points, legacy file formats, or partially structured engineering data.

Remark 40.5. Typical input may consist of:

- a measured polyline,
- imported TRA/GRA data,
- LandXML fragments,
- or mixed data of uncertain quality.

40.2.2. Task

Remark 40.6. The task is to reconstruct a mathematically coherent alignment model from heterogeneous data.

40.2.3. Pipeline

raw data
→ parsing
→ grabbeltisch sorting
→ sparse / pose3 state reconstruction
→ initial guess
→ numerical refinement

40.2.4. Interpretation

Remark 40.7. This use case emphasizes that railway geometry is not simply “read from a file”. Instead, it is reconstructed as a structured state object.

Remark 40.8. The AIM framework interprets imported data as evidence for an alignment, not as unquestionable truth.

40.3. Use Case 2: Smoothing of an Existing Track

40.3.1. Scenario

Remark 40.9. A measured or existing alignment is geometrically feasible, but exhibits undesirable local irregularities.

Remark 40.10. Examples include:

- noisy curvature estimates,
- abrupt cant changes,
- locally degraded transition zones.

40.3.2. Task

Remark 40.11. The task is to derive an improved alignment preserving the overall route while reducing local dynamic excitation.

40.3.3. Objective Structure

Remark 40.12. Typical objective terms include:

- fitting to the original track,
- smoothness of curvature,
- smoothness of cant,
- reduction of jerk.

40.3.4. Interpretation

Remark 40.13. This is one of the clearest examples showing that alignment should be treated as an optimization problem rather than as a purely geometric description.

40.4. Use Case 3: Design of a New Transition Zone

40.4.1. Scenario

Remark 40.14. A new transition between straight track and circular arc is required.

Remark 40.15. Classically, one would choose a predefined transition family such as a clothoid. In the present framework, the transition may instead be derived through optimization.

40.4.2. Task

Remark 40.16. The task is to construct a transition satisfying:

- boundary curvature conditions,
- cant conditions,
- dynamic smoothness criteria.

40.4.3. AIM Perspective

Remark 40.17. In the AIM framework, the transition is not primarily selected as a named curve family. Rather, it is derived as a controlled curvature evolution.

Remark 40.18. This use case directly illustrates the central thesis that transition curves are control functions rather than merely geometric connectors.

40.5. Use Case 4: Coupled Optimization of Curvature and Cant

40.5.1. Scenario

Remark 40.19. A railway section is to be improved with respect to comfort and dynamic performance.

40.5.2. Task

Remark 40.20. The task is to optimize both:

$$\kappa(s) \quad \text{and} \quad \phi(s),$$

instead of treating cant as a secondary afterthought.

40.5.3. Objective Terms

Remark 40.21. Relevant objective terms include:

- effective lateral acceleration,
- jerk,
- curvature smoothness,
- cant smoothness.

40.5.4. Interpretation

Remark 40.22. This use case demonstrates the transition from pose2-based geometry to pose3-based railway state modelling.

Remark 40.23. It also shows that railway alignment design is inherently a coupled problem.

40.6. Use Case 5: Schwerpunkt-Trassierung

40.6.1. Scenario

Remark 40.24. Instead of designing only the track centerline, one seeks to design the motion of a representative point of the vehicle system.

40.6.2. Task

Remark 40.25. The task is to optimize the trajectory of the center of mass, or of another dynamically relevant reference point, by adjusting curvature and cant.

40.6.3. Model

Remark 40.26. The design variables are:

$$\kappa(s), \quad \phi(s),$$

while the induced trajectory

$$\gamma_c(s)$$

becomes the primary object of evaluation.

40.6.4. Interpretation

Remark 40.27. This use case represents the conceptual extension from classical alignment design toward the new idea of Schwerpunkt-Trassierung.

Remark 40.28. It is one of the clearest expressions of the originality of the present framework.

40.7. Use Case 6: Multi-Track and Network Optimization

40.7.1. Scenario

Remark 40.29. A railway system rarely consists of a single isolated alignment. Instead, one must often deal with:

- parallel tracks,
- switches,
- branching structures,
- coupled corridor geometries.

40.7.2. Task

Remark 40.30. The task is to optimize several alignments simultaneously under topological and geometric coupling constraints.

40.7.3. Typical Constraints

- coincident switch nodes,
- tangent continuity,
- parallel track spacing,
- compatible cant and geometry transitions.

40.7.4. Interpretation

Remark 40.31. This use case shows that the AIM framework extends naturally from single curves to entire railway systems.

40.8. Use Case 7: Interactive Optimization from the Grabbeltisch

40.8.1. Scenario

Remark 40.32. A user imports multiple heterogeneous data sources and arranges them in the grabbeltisch interface according to their semantic role.

40.8.2. Task

Remark 40.33. The system must transform this user-guided sorting into a mathematically meaningful optimization input.

40.8.3. Pipeline

drop files
→ detect / parse
→ sort on grabbeltisch
→ build alignment input
→ construct initial guess
→ optimize
→ preview
→ commit to model

40.8.4. Interpretation

Remark 40.34. This use case is particularly important because it connects the abstract framework to actual engineering workflow.

Remark 40.35. It also demonstrates that user interaction and mathematical modelling are not opposites, but complementary parts of the same system.

40.9. Use Case 8: Standards and BIM Integration

40.9.1. Scenario

Remark 40.36. An optimized alignment must be exchanged with other systems through standardized or semi-standardized formats.

40.9.2. Task

Remark 40.37. The task is to translate the internal alignment state into layered representations suitable for engineering and BIM workflows.

40.9.3. Interpretation

Remark 40.38. This use case highlights the value of a structured internal model. Only such a model can be mapped consistently to multi-layer data models containing horizontal alignment, profile, and cant.

40.10. Comparative Summary of Use Cases

Use Case	Primary Input	Primary Output
Reconstruction	raw / imported geometry	structured alignment state
Smoothing	measured / degraded track	improved alignment
Transition design	boundary conditions	optimized transition law
Curvature–cant coupling	route + speed + cant	pose3-consistent state
Schwerpunkt	vehicle-based reference	optimized reference trajectory
Network optimization	multi-track graph	globally consistent corridor
Grabbeltisch workflow	user-sorted imports	optimization-ready input
BIM integration	structured state model	exportable layered model

40.11. Conclusion

Summary 40.39. The presented use cases demonstrate that the AIM framework is not limited to theoretical alignment descriptions.

It supports:

- reconstruction,
- optimization,
- dynamic interpretation,
- user-guided engineering workflows,
- and system-level railway design.

Together, these use cases show that alignment can be treated as a structured, dynamic, and optimization-ready object.

41. BIM Alignment Modelling

41.1. Motivation

Remark 41.1. While BIM provides a powerful framework for infrastructure data integration, the modelling of railway alignment within BIM remains structurally fragmented and computationally limited.

Remark 41.2. The AIM framework suggests that alignment should not be treated as a secondary geometric component, but as a primary organizing structure.

Remark 41.3. This chapter introduces the concept of BIMalignment as a synthesis of BIM data modelling and alignment-centred computational representation.

41.2. Concept of BIMalignment

Definition 41.4 (BIMalignment). BIMalignment is an alignment-centred modelling approach in which infrastructure geometry, semantics, and dynamics are organized around a canonical alignment state representation.

Remark 41.5. It combines:

- BIM data structures (IFC, LandXML, etc.),
 - AIM state representation,
 - computational and optimization capabilities.
-

41.3. Core Representation

Remark 41.6. The central object in BIMalignment is the alignment state:

$$(\gamma(s), \theta(s), \kappa(s), \phi(s)).$$

Remark 41.7. This representation:

- unifies horizontal and vertical geometry,

- integrates cant,
 - enables dynamic interpretation,
 - supports optimization.
-

41.4. Alignment as Reference Structure

Remark 41.8. In BIMalignment, infrastructure elements are defined relative to the alignment.

- track geometry follows alignment,
- platforms reference stationing,
- signals are positioned along chainage,
- structures are aligned to geometry.

Remark 41.9. This establishes alignment as the primary coordinate and reference system.

41.5. Transformation Pipeline

IFC / LandXML / legacy formats
→ parsing
→ normalization
→ AIM state model
→ BIMalignment model
→ analysis / optimization
→ export to BIM

Remark 41.10. This pipeline separates:

- data exchange (BIM),
 - computation (AIM),
 - application (BIMalignment).
-

41.6. Bidirectional Integration

41.6.1. Import

Remark 41.11. BIM data is interpreted and transformed into the AIM state model.

Remark 41.12. This includes:

- reconstruction of alignment geometry,
 - interpretation of stationing,
 - extraction of cant and profile information.
-

41.6.2. Export

Remark 41.13. Optimized or modified alignments are exported back into BIM formats.

Remark 41.14. This ensures compatibility with:

- existing design tools,
 - project workflows,
 - lifecycle data systems.
-

41.7. Advantages of BIMalignment

41.7.1. Unified model

Remark 41.15. BIMalignment replaces fragmented representations with a unified alignment-centred model.

41.7.2. Computational capability

Remark 41.16. Unlike traditional BIM, the model is directly usable for:

- dynamic evaluation,
 - constraint checking,
 - optimization.
-

41.7.3. Consistency

Remark 41.17. A single state representation ensures consistency between:

- geometry,
 - dynamics,
 - constraints,
 - and derived quantities.
-

41.7.4. Extensibility

Remark 41.18. The framework allows integration of:

- new transition models,
 - advanced vehicle dynamics,
 - additional constraint libraries,
 - network-level interactions.
-

41.8. Relation to Existing BIM Workflows

Remark 41.19. BIMalignment does not replace existing BIM processes, but augments them.

Remark 41.20. It introduces a computational layer that enables:

- deeper analysis,
 - design optimization,
 - improved data interpretation.
-

41.9. Role of the Grabbeltisch

Remark 41.21. The interpretation of BIM data often requires human input.

Remark 41.22. The grabbeltisch serves as an intermediate layer where:

- imported data is classified,
- semantic roles are assigned,
- alignment components are structured.

Remark 41.23. This enables robust handling of heterogeneous and imperfect data.

41.10. Challenges

41.10.1. Standard complexity

Remark 41.24. BIM standards such as IFC are complex and require careful mapping to alignment-centred models.

41.10.2. Data quality

Remark 41.25. Real-world data is often incomplete, inconsistent, or noisy.

41.10.3. Adoption

Remark 41.26. Integration into existing workflows requires:

- tool support,
 - user acceptance,
 - compatibility with established practices.
-

41.11. Outlook

Remark 41.27. BIMalignment provides a foundation for future systems in which:

- alignment is the central modelling object,
 - geometry and dynamics are unified,
 - optimization becomes an integral part of design,
 - BIM evolves from static representation to computational modelling.
-

41.12. Summary

Summary 41.28. BIMalignment extends BIM by introducing an alignment-centred, computationally enabled modelling approach.

It bridges the gap between data representation and engineering computation, enabling consistent, dynamic, and optimization-ready handling of railway alignment within BIM environments.

42. Contribution

42.1. Motivation

Remark 42.1. Railway alignment has traditionally been approached as a geometric design problem, supplemented by engineering rules and dynamic considerations.

Remark 42.2. However, these aspects are often treated separately, leading to fragmented models and limited theoretical coherence.

42.2. Central Hypothesis

Theorem 42.3 (Curvature as System Variable). *A railway alignment can be represented as a curvature-driven system in which*

$$\kappa(s)$$

acts as the primary state variable.

Remark 42.4. All geometric, kinematic, and dynamic properties of the alignment follow from $\kappa(s)$ and its derivatives.

42.3. System Interpretation of Alignment

Proposition 42.5. *The behaviour of a railway alignment can be described by the mapping*

$$\kappa(s) \longrightarrow \text{forces} \longrightarrow \text{system response} \longrightarrow \text{performance}.$$

Remark 42.6. This establishes alignment as a system rather than a static geometric object.

42.4. Unification of Transition Curves

Theorem 42.7 (Unified Transition Representation). *All transition curves can be expressed as curvature functions*

$$\kappa(s)$$

within a normalized domain.

Remark 42.8. This allows comparison of historically distinct transition concepts within a single mathematical framework.

42.5. Transition Curves as Control Functions

Theorem 42.9 (Control Interpretation). *Transition curves act as control functions governing the evolution of curvature and thus the dynamic behaviour of the system.*

Remark 42.10. This interpretation replaces the traditional view of transition curves as purely geometric connectors.

42.6. Structured Alignment Model

Proposition 42.11. *An alignment can be represented as a structured sequence of elements:*

- *fixed curvature elements,*
- *transition elements,*
- *auxiliary alignment layers such as cant.*

Remark 42.12. This structure separates representation, geometry, and system behaviour.

42.7. Coupling with Engineering Variables

Remark 42.13. Alignment design is governed by coupled variables including:

- curvature,
- cant,
- speed,
- and their rates of change.

Proposition 42.14. *Dynamic constraints can be expressed as conditions on*

$$\kappa(s), \quad \kappa'(s), \quad \kappa''(s).$$

42.8. Numerical and Optimization Perspective

Remark 42.15. The curvature-based representation allows direct application of numerical integration and optimization methods.

Proposition 42.16. *Alignment design can be formulated as a parameterized optimization problem in curvature space.*

42.9. Integration with Data Models

Remark 42.17. Modern railway data models represent alignment as a multi-layer structure consisting of horizontal, vertical, and cant components.

Proposition 42.18. *The structured alignment model proposed here is compatible with such multi-layer representations.*

42.10. Main Contributions

Summary 42.19. The contributions of this work can be summarized as follows:

- A curvature-based system interpretation of alignment.
- A unified representation of transition curves.
- The interpretation of transition curves as control functions.
- A structured alignment model separating geometry and behaviour.
- Integration of geometry, dynamics, and engineering constraints.
- A foundation for alignment-based information modelling (AIM).

The numerical comparison of transition laws shows that identical boundary conditions may still lead to substantially different internal curvature evolution and therefore different dynamic behaviour.

42.11. Scientific Positioning

Remark 42.20. The presented approach does not introduce a new transition curve, but a new conceptual framework for understanding alignment.

Remark 42.21. It connects previously separated domains:

- geometry,
 - dynamics,
 - engineering design,
 - and data modelling.
-

42.12. Outlook

Remark 42.22. The framework provides a basis for future work in:

- dynamic simulation,
- alignment optimization,
- network-level modelling,
- and BIM integration.

43. Discussion

43.1. Purpose of the Discussion

Remark 43.1. The previous chapters introduced the mathematical foundations, the state-based model, the optimization framework, the software-oriented architecture, and the intended engineering use cases of the AIM approach.

Remark 43.2. The present chapter discusses the significance, scope, and limitations of this framework.

Remark 43.3. Its purpose is not only to summarize previous results, but to position the AIM approach with respect to classical railway alignment design, modern computational methods, and infrastructure data modelling.

43.2. From Geometry to System Modelling

Remark 43.4. A central shift introduced by the AIM framework is the transition from pure geometry to system modelling.

Remark 43.5. Classical alignment theory typically treats the railway path as a curve to be constructed, checked, and documented.

Remark 43.6. In contrast, the AIM framework interprets alignment as a structured state object whose behaviour is governed by curvature, cant, and their derivatives.

Remark 43.7. This leads to a more general view: alignment is not merely a shape, but a functional object with geometric, dynamic, and operational meaning.

43.3. Transition Curves Reinterpreted

Remark 43.8. One of the most important conceptual consequences of the AIM framework is the reinterpretation of transition curves.

Remark 43.9. In many traditional presentations, transition curves appear mainly as named geometric objects, such as clothoids or engineering variants.

Remark 43.10. Within AIM, transition curves are instead understood as control functions for the evolution of curvature and therefore for the dynamic behaviour of the railway system.

Remark 43.11. This reinterpretation is significant because it unifies geometric, dynamic, and optimization-based viewpoints.

43.4. Advantages of the AIM Perspective

43.4.1. Unified internal representation

Remark 43.12. A major strength of AIM lies in the use of a canonical internal model. External data sources may differ substantially in structure and semantics, but can still be mapped into a common state-based representation.

Remark 43.13. This reduces dependence on file formats and enables consistent processing across heterogeneous sources.

43.4.2. Optimization readiness

Remark 43.14. The AIM representation is designed not only for description, but also for computation.

Remark 43.15. Since curvature, cant, and derived quantities are represented explicitly, the model is naturally suited for least-squares methods, SQP, and related optimization approaches.

43.4.3. Dynamic interpretation

Remark 43.16. The framework incorporates dynamics directly through effective lateral acceleration, jerk, cant deficiency, and related quantities.

Remark 43.17. This strengthens the engineering relevance of the model, especially for high-speed and comfort-sensitive applications.

43.4.4. Extensibility

Remark 43.18. The architecture is modular:

- new transition families may be added,
- new constraints may be introduced,
- objective functionals may be reweighted,
- network-level coupling may be extended.

Remark 43.19. This makes AIM suitable not only for one fixed tool, but as a long-term framework.

43.5. Comparison with Classical Approaches

Remark 43.20. Classical railway alignment design often follows a decomposition into:

- horizontal alignment,

- vertical profile,
- cant,
- subsequent rule checking.

Remark 43.21. This decomposition is practical and historically well established. However, it tends to separate modelling from evaluation and often treats dynamics only in a downstream manner.

Remark 43.22. The AIM approach differs by integrating:

- state representation,
- dynamic evaluation,
- and optimization

into a single conceptual structure.

Remark 43.23. This is not intended to invalidate classical methods, but to provide a more general framework into which they can be embedded as special cases.

43.6. Relation to BIM and Data Standards

Remark 43.24. AIM is not designed as a competitor to standards such as LandXML, IFC, or railML.

Remark 43.25. Instead, it serves as a normalization and computational layer between heterogeneous data models and actual engineering reasoning.

Remark 43.26. This is especially important because most standards are primarily designed for exchange and documentation, whereas AIM is designed for interpretation, analysis, and optimization.

Remark 43.27. In that sense, AIM may be viewed as a computational core that complements BIM-style information structures.

43.7. Role of the Grabbeltisch

Remark 43.28. An important practical insight of the present work is that fully automatic interpretation of imported data is neither realistic nor always desirable.

Remark 43.29. The grabbeltisch concept acknowledges that engineering understanding is often needed to assign semantic roles to imported objects.

Remark 43.30. This user-guided intermediate layer is not a weakness of the framework, but a deliberate design choice.

Remark 43.31. It allows the system to combine:

- machine-based parsing,
- human interpretation,
- mathematical reconstruction.

43.8. Limits of the Present Formulation

43.8.1. Physical simplifications

Remark 43.32. The dynamic model introduced in the present manuscript is intentionally simplified.

Remark 43.33. Although effective acceleration and jerk provide meaningful engineering proxies, they do not yet replace full multibody simulation, detailed wheel–rail contact models, or full vehicle-specific dynamics.

43.8.2. Standards formalization

Remark 43.34. While many railway rules can be written as constraint residuals, the full formalization of national and operator-specific regulations remains an open and demanding task.

43.8.3. Topological complexity

Remark 43.35. The extension from single alignments to complete railway networks is one of the strongest features of the framework, but also one of its most challenging aspects.

Remark 43.36. Switches, crossings, branching structures, and corridor-wide coupling introduce substantial mathematical and computational complexity.

43.8.4. Industrial robustness

Remark 43.37. The framework is conceptually strong, but practical deployment requires robust implementations, extensive validation, and careful interface design.

Remark 43.38. In particular, optimization quality depends strongly on:

- good initial guesses,
- stable numerical methods,
- suitable weighting of objectives and constraints.

43.9. Scientific Positioning

Remark 43.39. The contribution of the AIM framework is not the invention of yet another transition curve.

43. Discussion

Remark 43.40. Its real contribution lies in the proposal of a broader conceptual framework in which:

- alignment is a state-based object,
- transition curves are control functions,
- geometry and dynamics are treated jointly,
- imported data is interpreted rather than merely accepted,
- and optimization becomes a natural component of design.

Remark 43.41. This positioning places AIM between several established fields:

- railway engineering,
- differential geometry,
- optimization,
- BIM and infrastructure data modelling,
- and interactive engineering software.

43.10. Philosophical Perspective

Remark 43.42. At a deeper level, AIM reflects a shift in engineering modelling: from static representation toward functional interpretation.

Remark 43.43. In this sense, the track is not only a line in space, but a rule-bound, state-dependent object whose meaning arises from its behaviour.

Remark 43.44. This is particularly visible in the move from pose2 to pose3 and from simple curve design to Schwerpunkt-Trassierung.

43.11. Outlook of the Discussion

Remark 43.45. The present discussion suggests that AIM should be understood not as a finished final system, but as a foundational framework.

Remark 43.46. Its long-term value lies in its ability to host:

- richer vehicle models,
- deeper optimization strategies,
- stronger network formulations,
- tighter BIM integration,
- and increasingly automated engineering workflows.

43.12. Summary

Summary 43.47. The AIM framework reinterprets railway alignment as a structured, dynamic, and optimization-ready object.

Its main strengths are the unification of geometry and dynamics, the reinterpretation of transition curves as control functions, and the ability to connect data interpretation, engineering constraints, and numerical optimization.

Its current limitations lie mainly in the simplifications of the dynamic model, the partial formalization of standards, and the remaining challenge of full industrial robustness.

Nevertheless, the framework provides a strong foundation for a new alignment-centred perspective on railway infrastructure modelling.

44. Vehicle Space and Platform Interaction

44.1. Motivation

Remark 44.1. Classical railway alignment modelling is primarily based on the track centerline.

Remark 44.2. In this view, the railway vehicle is treated indirectly through:

- clearance envelopes,
- cant rules,
- safety margins,
- and operational standards.

Remark 44.3. This abstraction has proven remarkably successful and enabled the development of reliable railway systems over many decades.

Remark 44.4. However, the physically relevant object is not the centerline itself, but the spatial motion of the railway vehicle along the track.

Remark 44.5. This becomes particularly visible in:

- tight urban rail systems,
- platform-edge design,
- articulated vehicles,
- narrow tunnel geometries,
- and highly optimized clearance situations.

Remark 44.6. This perspective is closely related to approaches that interpret alignment geometry from the viewpoint of vehicle motion rather than purely from the track centerline.

Such concepts have been explored in advanced railway engineering, including Schwerpunkt-oriented alignment approaches [12].

44.2. Centerline as Reduced Representation

Remark 44.7. The classical alignment model represents the railway track by:

$$\gamma(s), \quad \kappa(s), \quad \phi(s).$$

Remark 44.8. This defines:

- position,
- curvature,
- and cant,

along the alignment.

Remark 44.9. The vehicle itself is not explicitly represented.

Remark 44.10. Instead, its behaviour is incorporated indirectly through engineering rules and geometric tolerances.

Proposition 44.11. *The centerline representation is a reduced-order approximation of the actual spatial motion of railway vehicles.*

Remark 44.12. This approximation is generally sufficient for conventional railway engineering.

44.3. Vehicle-Induced Geometry

Remark 44.13. A railway vehicle does not follow the centerline as a rigid point.

Remark 44.14. Its occupied space depends on:

- bogie geometry,
- wheelbase,
- car-body length,
- articulation,
- cant,
- suspension,
- and dynamic motion.

Remark 44.15. As a consequence:

- vehicle ends may swing outward,
- the center of the car body may move inward,
- and the effective occupied space varies along the track.

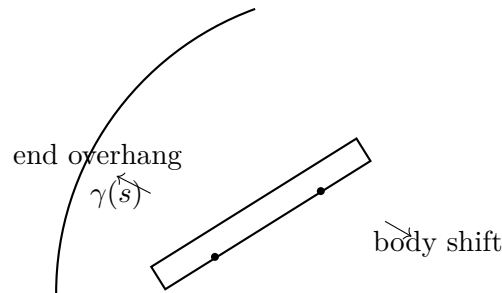


Abbildung 44.1.: Vehicle motion differs from the centerline geometry. Car-body overhang and bogie-constrained motion generate occupied space that cannot be represented by the centerline alone.

44.4. Platform Interaction

Remark 44.16. Platform-edge design is one of the most sensitive examples of vehicle-space interaction.

Remark 44.17. The effective gap between vehicle and platform depends on:

- curvature,
- cant,
- vehicle geometry,
- suspension state,
- and operational conditions.

Remark 44.18. In some urban railway systems, such as light rail and metro systems, vehicle overhang is explicitly considered in platform-edge geometry.

Remark 44.19. This demonstrates that the practically relevant geometry is not merely the track centerline, but the dynamically occupied vehicle space.

44.5. Standards and Conservative Approximation

Remark 44.20. Conventional railway standards often address vehicle motion indirectly through conservative clearance envelopes.

Remark 44.21. This approach increases robustness and simplifies engineering workflows.

Remark 44.22. However, it may also lead to:

- increased platform gaps,
- reduced passenger comfort,
- larger-than-necessary clearance reserves,

- and loss of geometric precision.

Remark 44.23. Thus, many existing systems compensate modelling limitations through additional safety margins.

44.6. Towards Vehicle-Centric Alignment Modelling

Remark 44.24. Future alignment systems may extend the classical centerline model towards vehicle-centric representations.

Remark 44.25. In such a framework, the relevant object would no longer be only:

$$\gamma(s),$$

but the occupied vehicle space evolving along the alignment.

Remark 44.26. Possible extensions include:

- dynamic vehicle envelopes,
- bogie-constrained motion models,
- passenger-space optimization,
- platform interaction models,
- and runtime vehicle-state evaluation.

Remark 44.27. Within the AIM framework, such concepts are regarded as potential future extensions rather than immediate replacement of existing engineering practice.

44.7. Connection to Schwerpunkt-Trassierung

Remark 44.28. The concept of Schwerpunkt-Trassierung already shifts the interpretation of alignment design from purely geometric construction toward the motion of a dynamically relevant state.

Remark 44.29. Vehicle-space modelling can be interpreted as a continuation of this idea.

Remark 44.30. Instead of optimizing only a centerline curve, one may eventually seek to optimize the spatial behaviour of vehicles and passengers along the track.

44.8. Connection to AIM

Summary 44.31. Classical railway alignment modelling represents the railway track by a centerline-based geometric abstraction.

This abstraction is highly successful, but does not explicitly represent the spatial behaviour of railway vehicles.

44. Vehicle Space and Platform Interaction

Practical situations such as platform interaction reveal the limitations of purely centerline-based modelling and motivate future vehicle-centric extensions.

Within the AIM framework, this is regarded as an important long-term research direction.

Teil X.

Outlook

Remark 44.32. This part concludes the manuscript by placing the AIM framework into a broader perspective. It discusses how the presented ideas may be extended, generalized, and embedded into future modelling and engineering workflows.

Remark 44.33. The goal is not to close the framework, but to identify its open ends, its possible evolution, and its relevance beyond the present document.

Remark 44.34. The following chapters therefore address BIM-related perspectives and future research directions.

45. BIM and Alignment Modelling

45.1. Motivation

Remark 45.1. Building Information Modelling (BIM) has become a central paradigm in modern infrastructure projects.

Remark 45.2. Its goal is the integration of geometry, semantics, and lifecycle information into a unified digital representation.

Remark 45.3. However, despite its broad scope, the representation of railway alignments within BIM remains conceptually and technically limited.

45.2. BIM as a Data Integration Paradigm

Remark 45.4. BIM is fundamentally concerned with:

- object-based modelling,
- semantic enrichment of geometry,
- interoperability across disciplines,
- lifecycle management of infrastructure assets.

Remark 45.5. In this context, geometry is typically associated with objects such as:

- buildings,
- bridges,
- track elements,
- terrain models.

Remark 45.6. Alignment geometry is often embedded implicitly within these objects, rather than treated as a primary modelling entity.

45.3. Alignment in BIM Standards

45.3.1. IFC Alignment

Remark 45.7. Recent extensions of IFC introduce explicit alignment entities.

Remark 45.8. These aim to represent:

- horizontal alignment,
- vertical alignment,
- cant (superelevation),
- stationing.

Remark 45.9. While this is a significant step forward, several issues remain:

- separation of components,
 - lack of unified state representation,
 - limited support for dynamic interpretation,
 - complexity of schema usage.
-

45.3.2. Layered Representation

Remark 45.10. In most BIM-related formats, alignment is decomposed into layers:

- horizontal geometry,
- vertical profile,
- cant definition.

Remark 45.11. This decomposition is practical for data exchange, but does not directly support computation or optimization.

45.4. Limitations of Current BIM Alignment Modelling

45.4.1. Static representation

Remark 45.12. BIM primarily represents geometry as static data.

Remark 45.13. Dynamic properties such as acceleration, jerk, or vehicle response are not part of the core model.

45.4.2. Fragmentation

Remark 45.14. The separation of horizontal, vertical, and cant components leads to a fragmented representation.

Remark 45.15. This fragmentation complicates:

- consistency checks,
 - dynamic evaluation,
 - optimization-based design.
-

45.4.3. Lack of computational core

Remark 45.16. BIM standards focus on data exchange and documentation.

Remark 45.17. They do not provide a unified computational model for alignment analysis and optimization.

45.5. AIM as a Computational Layer for BIM

45.5.1. Conceptual Role

Remark 45.18. The AIM framework can be understood as a computational core that complements BIM data models.

Remark 45.19. Instead of replacing BIM, AIM operates between:

- external data formats,
 - and internal engineering reasoning.
-

45.5.2. Canonical Representation

Remark 45.20. AIM introduces a unified representation:

$$(\gamma(s), \theta(s), \kappa(s), \phi(s)).$$

Remark 45.21. This representation:

- integrates horizontal geometry,
- integrates vertical and cant behaviour,
- supports dynamic evaluation,

- is directly usable in optimization.
-

45.5.3. Transformation Pipeline

BIM / IFC / LandXML

- parsing
- fat representation
- normalization
- AIM state model
- analysis / optimization
- export

Remark 45.22. This pipeline separates:

- data exchange concerns,
 - computational modelling,
 - engineering reasoning.
-

45.6. Alignment-Centred BIM

45.6.1. Shift of perspective

Remark 45.23. The AIM framework suggests a shift toward alignment-centred modelling.

Remark 45.24. Instead of treating alignment as one component among many, it becomes a primary organizing structure.

45.6.2. Implications

- infrastructure elements are referenced to alignment,
 - geometry is derived from alignment state,
 - dynamic behaviour becomes part of the model,
 - optimization becomes a native operation.
-

45.7. Integration with BIM Workflows

45.7.1. Import

Remark 45.25. Existing BIM data is imported via:

- IFC parsing,
 - LandXML parsing,
 - legacy format interpretation.
-

45.7.2. Processing

Remark 45.26. Within AIM, the imported data is:

- normalized,
 - reconstructed,
 - enriched with dynamic interpretation,
 - potentially optimized.
-

45.7.3. Export

Remark 45.27. Results may be exported back into:

- IFC alignment structures,
- LandXML files,
- other engineering formats.

Remark 45.28. This ensures compatibility with existing BIM ecosystems.

45.8. Relation to BIMinfra

Remark 45.29. Infrastructure BIM (often referred to as BIMinfra) extends BIM concepts to linear infrastructure such as railways and roads.

Remark 45.30. AIM can be interpreted as a specialization within BIMinfra focusing on:

- alignment-centric modelling,
 - dynamic interpretation,
 - optimization-driven design.
-

45.9. Discussion

Remark 45.31. BIM provides a powerful framework for data integration, but lacks a strong computational alignment model.

Remark 45.32. AIM addresses this gap by:

- introducing a unified state representation,
- enabling dynamic evaluation,
- supporting optimization,
- and integrating heterogeneous data sources.

Remark 45.33. Together, BIM and AIM can be seen as complementary:

- BIM for data integration and lifecycle,
 - AIM for computation and alignment reasoning.
-

45.10. Summary

Summary 45.34. Current BIM standards provide important structures for data exchange and semantic modelling, but remain limited in their treatment of railway alignment.

The AIM framework complements BIM by introducing a unified, state-based, and optimization-ready representation of alignment.

This enables a transition from static data handling toward dynamic, computational, and alignment-centred infrastructure modelling.

46. Future Work

46.1. Purpose of this Chapter

Remark 46.1. The AIM framework presented in this manuscript establishes a mathematical and conceptual foundation for alignment-centred modelling.

Remark 46.2. However, many aspects remain open and provide opportunities for further research, development, and practical implementation.

Remark 46.3. This chapter outlines potential directions for future work, ranging from mathematical extensions to software architecture and engineering applications.

46.2. Extension of the Dynamic Model

46.2.1. From simplified dynamics to vehicle models

Remark 46.4. The current formulation relies on simplified dynamic quantities such as effective lateral acceleration and jerk.

Remark 46.5. Future work should extend this toward more realistic vehicle models, including:

- multi-body vehicle dynamics,
- wheel–rail interaction,
- suspension behaviour,
- track elasticity.

46.2.2. Speed-dependent modelling

Remark 46.6. The integration of variable speed profiles

$$v(s)$$

opens the possibility for more realistic evaluation and optimization of alignments under operational conditions.

46.3. Schwerpunkt-Trassierung

46.3.1. Conceptual development

Remark 46.7. The idea of designing the trajectory of a dynamically relevant point, such as the center of mass, represents a fundamental extension of classical alignment design.

Remark 46.8. Further work is required to:

- formalize the mapping from track geometry to vehicle trajectory,
- incorporate full dynamic models,
- define appropriate objective functions.

46.3.2. Integration into optimization

Remark 46.9. A key research direction is the coupling of Schwerpunkt-Trassierung with optimization methods, enabling:

- comfort-driven design,
 - dynamic optimization,
 - direct evaluation of vehicle response.
-

46.4. Advanced Optimization Methods

46.4.1. SQP extensions

Remark 46.10. While Sequential Quadratic Programming provides a powerful baseline, future work may include:

- tailored SQP formulations for sparse alignment structures,
- improved constraint handling,
- adaptive weighting strategies.

46.4.2. Global optimization

Remark 46.11. Alignment optimization problems are often non-convex.

Remark 46.12. Future research may explore:

- global optimization techniques,
 - multi-start strategies,
 - hybrid deterministic–stochastic methods.
-

46.5. Network-Level Modelling

46.5.1. From single alignment to network

Remark 46.13. Real railway systems consist of interconnected alignments forming complex networks.

Remark 46.14. Future work should extend the AIM framework to:

- branching structures,
- switches and crossings,
- corridor-wide optimization.

46.5.2. Topological constraints

Remark 46.15. A systematic treatment of topological constraints remains a major challenge.

Remark 46.16. This includes:

- node consistency,
 - multi-track coupling,
 - network-wide feasibility conditions.
-

46.6. Constraint and Objective Libraries

46.6.1. Constraint formalization

Remark 46.17. Although many railway rules can be expressed as constraints, a complete formalization of regulatory frameworks remains an open task.

Remark 46.18. Future work should aim at:

- systematic encoding of standards,
- validation against real projects,
- parameterization for different railway systems.

46.6.2. Objective design

Remark 46.19. The design of objective functions is crucial for optimization-based alignment design.

Remark 46.20. Potential extensions include:

- cost models,

- maintenance optimization,
 - energy efficiency,
 - lifecycle considerations.
-

46.7. Integration with BIM and Data Standards

46.7.1. Deeper BIM integration

Remark 46.21. The integration of AIM with BIM workflows can be further developed by:

- tighter coupling with IFC alignment models,
- bidirectional synchronization,
- embedding AIM states within BIM objects.

46.7.2. Standard extensions

Remark 46.22. The AIM framework may also inspire extensions to existing standards, for example:

- inclusion of dynamic properties,
 - support for optimization metadata,
 - representation of control-based alignment models.
-

46.8. Interactive and Real-Time Systems

46.8.1. Live optimization

Remark 46.23. One of the most promising directions is the development of systems that allow real-time feedback during alignment design.

Remark 46.24. This includes:

- live preview during optimization,
- interactive constraint adjustment,
- immediate visualization of dynamic effects.

46.8.2. User interaction

Remark 46.25. The grabbeltisch concept may be extended to:

- interactive constraint definition,
 - visual debugging of alignment states,
 - intuitive editing of transition parameters.
-

46.9. AI and Assisted Design

46.9.1. Data-driven approaches

Remark 46.26. Machine learning techniques may complement the AIM framework by:

- learning from existing alignments,
- suggesting initial solutions,
- identifying patterns in design choices.

46.9.2. Hybrid systems

Remark 46.27. A promising direction is the combination of:

- physics-based modelling (AIM),
 - data-driven methods (AI),
 - interactive user control.
-

46.10. AXTRAN Revival and Beyond

Remark 46.28. Classical systems such as AXTRAN demonstrated the feasibility of optimization-based alignment design.

Remark 46.29. Future work may aim to:

- reconstruct and generalize such systems,
 - extend them to network-level problems,
 - integrate modern numerical and computational techniques.
-

46.11. Software Architecture and Deployment

46.11.1. Scalability

Remark 46.30. Practical applications require scalable implementations capable of handling large datasets and complex networks.

46.11.2. Multi-view systems

Remark 46.31. Future systems may include:

- synchronized multi-window views,
 - combined 2D/3D representations,
 - integration of GIS and CAD functionality.
-

46.12. Long-Term Vision

Remark 46.32. In the long term, the AIM framework may contribute to a new paradigm of infrastructure modelling in which:

- alignment becomes the central organizing structure,
 - geometry, dynamics, and optimization are unified,
 - engineering design becomes an interactive computational process.
-

46.13. Summary

Summary 46.33. The AIM framework opens a wide range of research and development directions.

These include extensions of the dynamic model, integration with BIM, network-level optimization, real-time systems, and AI-assisted design.

Together, these directions point toward a future in which railway alignment is no longer treated as static geometry, but as a dynamic, computational, and optimizable system at the core of infrastructure engineering.

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